**SAIGON UNIVERSITY**
**FACULTY OF INFORMATION TECHNOLOGY**

# COURSE PROJECT REPORT

## TOURVOICE

Course: Software Engineering

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Class: DCT124C5

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# CHAPTER 1: INTRODUCTION AND PROJECT OBJECTIVES

## 1.1. Background and Rationale

This chapter explains why TourVoice is worth building, what it must achieve and where it stops. It moves from the context of western Chợ Lớn to four concrete problems, turns those problems into six measurable objectives, and then fixes the scope, the working method and the vocabulary that the remaining five chapters rely on. Every later chapter refers back to the problem codes (P1–P4), objective codes (O1–O6) and subsystem codes (PH1–PH5) introduced here.

### *1.1.1. Tourism demand in Ho Chi Minh City*

In the first ten months of 2025, Ho Chi Minh City received about 6.5 million international visitors, 17.7% more than in the same period of 2024, together with more than 33 million domestic visitors; tourism revenue reached VND 208.066 trillion, an increase of 22% [1]. Two observations follow. The audience is large, so a tool that improves the visit can reach many people. The audience is also linguistically mixed: a visitor from China, Korea, Japan or an English-speaking country arrives with a different language, and a domestic visitor may share the language but not the history of the place. A guide that speaks only one language therefore serves only part of this audience.

### *1.1.2. Western Chợ Lớn and the four destinations*

Chợ Lớn is described as the largest Chinatown in Vietnam. The Hoa (Chinese-Vietnamese) community now accounts for about 5% of the city's population, yet its influence remains visible in temples, guildhalls and food, and Bình Tây Market is both a major wholesale market and a place for sightseeing and shopping [2]. TourVoice concentrates on the western part of Chợ Lớn, the former District 6, and on four representative destinations: Bình Tiên Communal House (Đình Bình Tiên), a heritage site; Bình Tây Market; the Phú Lâm food street; and Bình Phú Park. The market and the food street each lead to a group of food spots, so the content covers both places and what to eat around them.

Figure 1.1 places the four destinations on a common scale. The point to notice is their spacing: every destination lies within about 1.5 km in a straight line of another, so the four form one walkable cluster that a single, area-scoped system can serve.

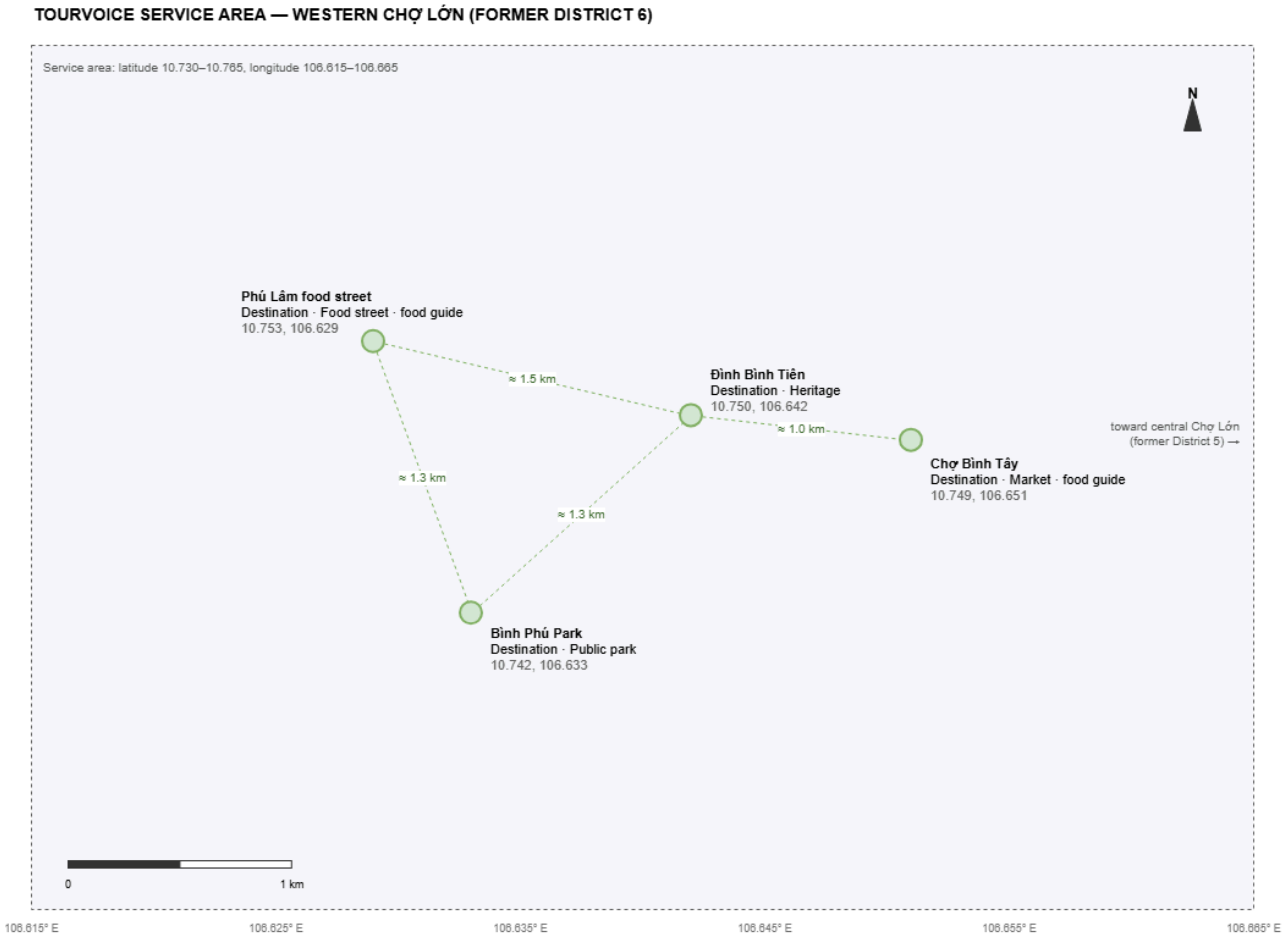



*Figure 1.1. The four destinations in western Chợ Lớn, plotted from the approximate seed coordinates of the project*

The report keeps the name “District 6” because visitors and residents still use it. Administratively, the district level no longer exists since 1 July 2025: under Resolution 1685/NQ-UBTVQH15, for example, the new Bình Tây Ward was formed from former Wards 2 and 9 of District 6 [3]. The area is now covered by the Bình Tây, Bình Tiên, Bình Phú and Phú Lâm wards, and the system stores addresses under these new ward names. Bình Tiên Communal House, despite its name, lies in Bình Tây Ward.

### *1.1.3. Problem statement*

The difficulty visitors face is not a lack of places to see or food to eat, but a lack of explanation they can understand at the moment they stand in front of a place. The team's requirements analysis of the four destinations identified four concrete problems. They are numbered because each objective in Section 1.2 answers at least one of them.

| Code | Problem | Who is affected |
|---|---|---|
| P1 | Visitors from China, Korea, Japan and English-speaking countries have no trustworthy narration in their own language for the four destinations. | Guests |
| P2 | Standing in a market or a food street, visitors do not know which dishes to try, where the stalls are or how far they must walk. | Guests |
| P3 | Recording human voices in five languages for every revision of a text is not feasible for a student team with no budget; machine translation and synthetic speech are affordable but must be reviewed by a person before visitors see them. | Content team |
| P4 | The manager cannot see how many visitors are using the system at a given moment, and therefore cannot judge whether the QR codes placed at each destination work. | Administrator |

*Table 1.1. Problems that motivate TourVoice*

### *1.1.4. Rationale for the chosen approach*

The usual ways of explaining a place each fail at least one of these problems. A human guide gives rich explanations but costs money and cannot stand at every stall when a visitor arrives. Printed signs are fixed in content and space, so they rarely carry more than two languages. A general translation app converts words but knows nothing about a specific communal house or dish unless someone writes that knowledge first.

A pre-recorded audio guide needs a separate recording for every language and a new recording for every correction, which is exactly the cost described in P3.

These observations lead to three constraints that shape the whole project: no paid service may be required, every text a visitor hears must have been checked by a person, and the visitor must not have to install anything. TourVoice meets them with one chain of choices. The team writes each narration once in Vietnamese, a free machine translation service drafts the other four languages, and an administrator reviews every draft before it is published. The visitor's own browser then reads the reviewed text aloud with the voice already installed on the phone, so no audio file is produced, stored or downloaded. A QR code at each destination opens the page for that place directly, and an anonymous signal from each open page lets the administrator count the devices in use. Chapter 2 examines the concepts and existing systems behind these choices, and Chapter 4 justifies each one against its alternatives.

## 1.2. Project Objectives

### *1.2.1. General objective*

The general objective is to design and build TourVoice, a mobile-first web system that lets visitors to four destinations in western Chợ Lớn hear and read reviewed narration about the places and their food in five languages (Vietnamese, English, Chinese, Korean and Japanese), without paying for any external service, and that gives the administrator a live view of how many devices are using it.

### *1.2.2. Specific objectives*

The general objective is broken into six specific objectives, each with a target that can be measured at the end of the project. Table 1.2 states the target, the problem it answers and the evidence that will decide whether it is met. The targets are deliberately numeric: “fast” or “reliable” cannot be tested, while “at most 2 seconds” can.

| Code | Objective | Measurable target | Answers | Evidence (Chapter 5) |
|---|---|---|---|---|
| O1 | Complete multilingual content | 4 destinations and at least 12 food spots, each with Reviewed narration in all 5 languages | P1, P2 | Count of Reviewed translations per place and language |
| O2 | Zero external cost | No paid service; characters sent to the translation service stay at or below 45,000 per day, 90% of the free 50,000 [4] | P3 | Translation usage log; test of the quota guard |
| O3 | Quick access to narration | At most 2 s from pressing Listen to hearing speech on a Fast 4G connection; at most 3 taps from the home page to a narration | P1 | Timed tests on real phones; task walkthrough |
| O4 | Shared access through one QR code | At least 4 real phones scanning the same code can open the page and listen; load test with at least 50 simulated devices | P1, P4 | Field test; load test report |
| O5 | Live usage count | Dashboard count changes within 5 s when a page opens or closes, and within 60 s when a device loses its connection | P4 | Timed tests with real devices |

| | | | | |
|---|---|---|---|---|
| O6 | Maintainable backend | Three-layer rule checked automatically on every pull request; at least 70% unit-test coverage of the business layer; every change to the main branch merged through a pull request with a passing pipeline | All | Pipeline records; coverage report |

*Table 1.2. Specific objectives, targets and evidence*

Objectives O1 to O5 describe what visitors and the administrator experience, while O6 describes the quality of the software that delivers it, which the Software Engineering course requires and which protects O1 to O5 as the code grows. The same six codes return three times in the report: Chapter 3 turns them into requirements, Chapter 4 names the design decision that serves each one, and Chapter 5 reports the measured value against each target. An objective that cannot be traced along this chain does not belong in the project.

### *1.2.3. Expected outcomes*

At the end of the project the team expects to deliver four results. The first is a requirements and use case model of the system (Chapter 3). The second is a design that covers the architecture, the data model and the key flows, with the reason for every major choice (Chapter 4). The third is a working, deployed system loaded with the reviewed content of the four destinations, together with test evidence for O1 to O6 (Chapter 5). The fourth is an honest assessment of what was and was not achieved, with directions for further work (Chapter 6). These are intended outcomes rather than claims: a function is reported as complete in Chapter 5 only when evidence supports it.

### *1.2.4. Extended objective*

A further objective is to use the visitor's GPS position to start the narration of a nearby place automatically. This feature depends on geospatial queries in MongoDB and on browser location permission. It is treated as an optional extension: it will be

designed so that the core system does not depend on it, and it will be implemented only if time allows. If it is not implemented, it is reported as future work in Chapter 6.

## 1.3. Scope and Target Users

This section defines who the system is for and where the project stops. A clear boundary keeps the work achievable for a three-person group within one course.

### *1.3.1. Target users*

TourVoice has three kinds of users. A Guest is any visitor, foreign or domestic, who opens the site on their own phone, usually by scanning the QR code at a destination; a Guest needs no account and is the main user of the system. A Customer is a Guest who has registered with an email address in order to keep favorites, a listening history and a record of visited destinations, including progress such as “3 of 4 destinations visited”. Because a Customer can do everything a Guest can, the Customer role specialises the Guest role. An Admin is a team member or an appointed content owner who manages places and food spots, reviews translations, prints QR codes and monitors usage. Chapter 3 specifies these actors and their use cases in detail.

### *1.3.2. System boundary and subsystems*

Figure 1.2 draws the boundary of the system. Three actors exchange data with TourVoice, and three external systems sit outside it. The point to notice is the lower left corner: the system sends reviewed text and a language code to the speech engine of the visitor's own device, and the speech itself passes from that device to the visitor without touching the server. This is what allows narration in five languages without producing or storing any audio. The translation service is the only external service the server calls, and map tiles and directions come from free map services.

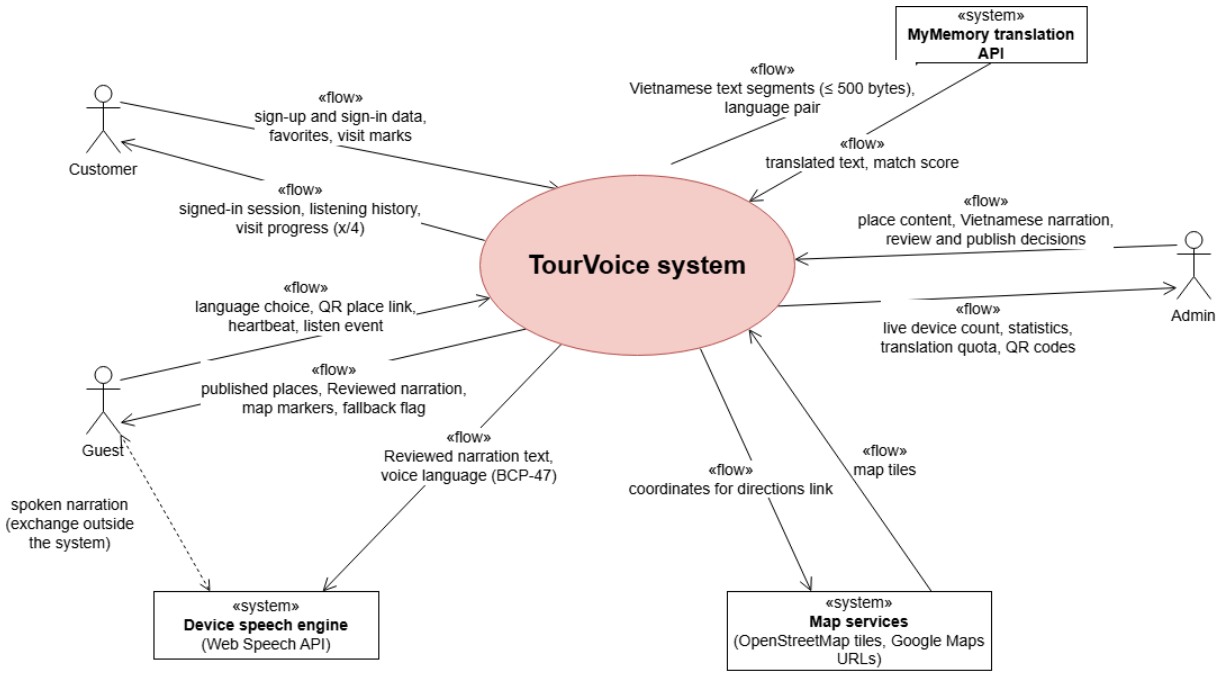



*Figure 1.2. Context diagram of the TourVoice system*

Inside the boundary, the work is divided into five subsystems, listed in Table 1.3. The division follows the business concepts rather than the technical layers, so that each subsystem can be analysed, designed and tested as one unit. The subsystem codes and colours are kept unchanged in every later figure, including the conceptual model, the use case diagrams and the entity-relationship diagram.

| Code | Subsystem | Responsibility | Main data |
|---|---|---|---|
| PH1 | Places and map | Destinations, food spots and their categories, coordinates, map, directions link, QR entry to a place | Places |
| PH2 | Multilingual narration | Vietnamese source text, machine translation, human review, language fallback, reading aloud with the device voice, listening records, translation quota | Place translations, languages, listen events, usage logs |

| | | | |
|---|---|---|---|
| PH3 | Customer accounts | Registration, favorites, listening history, visited destinations | Favorites, visits |
| PH4 | Presence and statistics | Anonymous heartbeat from open pages, live device count, access sessions, usage statistics | Access sessions, in-memory presence |
| PH5 | Administration and access control | Sign-in, roles, user management, administrator overview | Users |

*Table 1.3. Subsystems of TourVoice*

### *1.3.3. Out of scope*

A project of eight weeks and three members succeeds only if it states clearly what it will not do. Table 1.4 lists the exclusions and the reason for each; most of them protect one of the objectives in Table 1.2.

| **Excluded** | **Reason** |
|---|---|
| Native Android or iOS applications; offline listening | A web page opened from a QR code needs no installation, which serves O3 and O4 |
| Recorded human voices; translation in real time while a visitor reads | Recording breaks O2; unreviewed real-time text breaks the review constraint behind O1 |
| Narration triggered automatically by GPS; route drawing inside the application | Directions are delegated to Google Maps through a link; GPS triggering adds permission, accuracy and battery problems outside the eight-week plan |
| Tour booking, payment and visitor reviews | Not part of the four problems in Table 1.1 |

| | |
|---|---|
| Speech recognition and chatbots | Not needed to deliver narration |
| More than one tourist area | The content and its review are bounded to the four destinations |

*Table 1.4. Items deliberately left out of scope*

## 1.4. Methodology and Tools

### *1.4.1. Development approach: iterations with release gates*

The team develops TourVoice iteratively over eight weeks, in five increments that end at three release gates. Setup (week 1) ends at gate G1; content core (weeks 2–3) and narration (weeks 4–5) end at gate G2; accounts (week 6) and the final integration and testing (weeks 7–8) end at gate G3. The order is core-first: the content model and narration carry the purpose of the system, so they are built and tested before accounts and live statistics. Requirements are still being refined, for example the list of food spots and the reviewer for three languages, so short increments let the team adjust the plan without rework.

Figure 1.3 shows the cycle that every increment follows. The point to notice is the inner loop on the right: a change reaches the main branch only after the automated pipeline has built it, tested it and checked the layer rule, and a failing pipeline returns the change to the team. The outer loop shows that a gate is reviewed only when its increments are complete. The criteria of gate G3 are the targets of O1 to O6, so the method ends exactly where the evaluation in Chapter 5 begins.

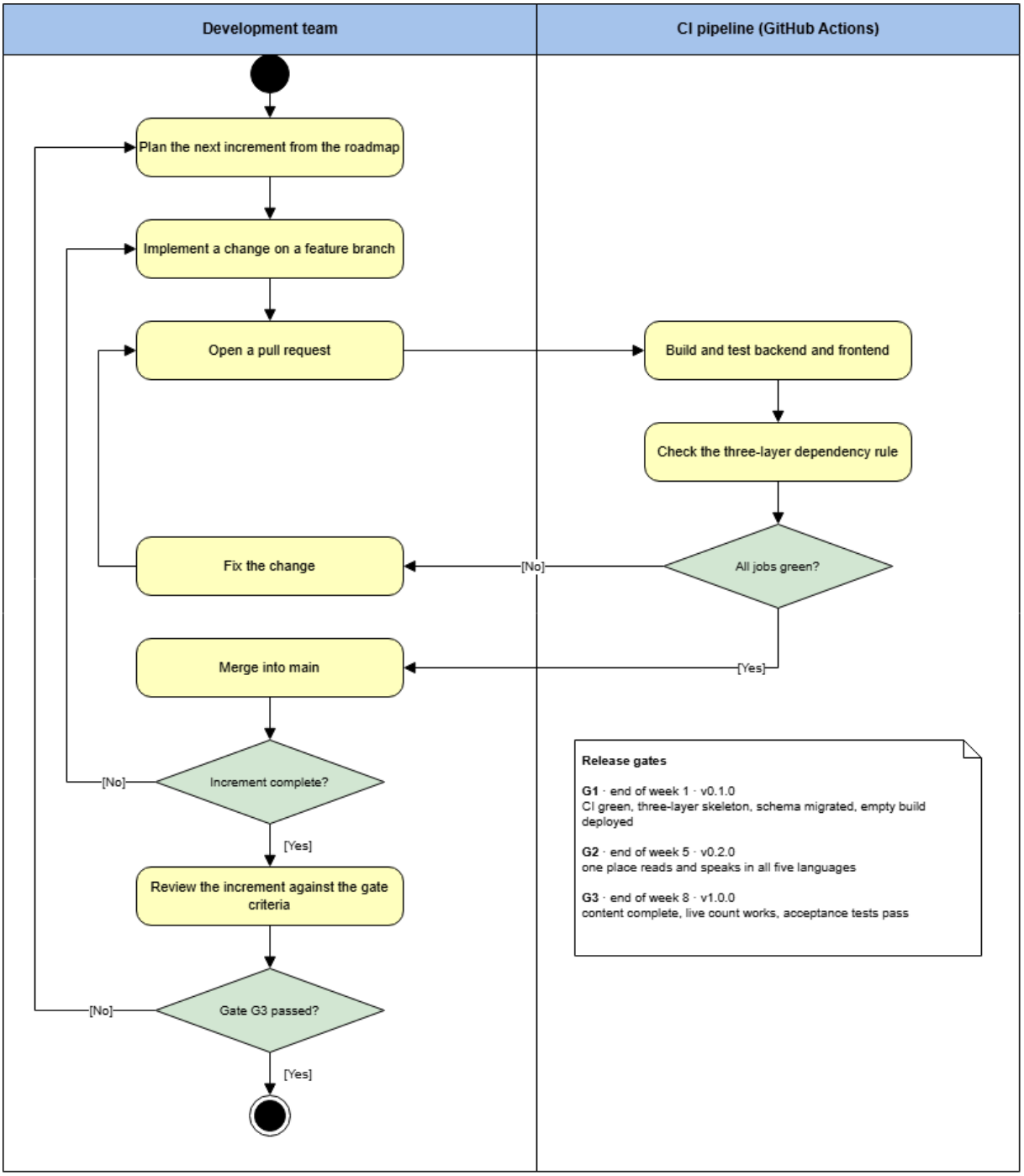



*Figure 1.3. Iterative development cycle with release gates*

### *1.4.2. Analysis and design method*

The analysis and design follow a fixed sequence of models in which each model is derived from the one before it, so that no element appears without a reason. The sequence starts from how visitors obtain information today, identifies the actors and the system boundary, groups the business concepts into subsystems, and then describes

what each actor can do through use cases and activity diagrams. Design continues with sequence diagrams for the key flows, state machines for objects with a lifecycle such as a translation moving from Draft to Reviewed, the database in three levels, and finally the screens. Names are kept identical across models: an actor in a use case diagram carries the same name as the corresponding swimlane and lifeline, and a state in a state machine carries the same name as the value stored in the database. Chapter 1 already contributes the context diagram (Figure 1.2); Chapter 2 describes current practice, Chapter 3 contains the analysis models and Chapter 4 the design models.

### *1.4.3. Quality assurance*

Quality is built into the workflow rather than checked at the end. The backend is organised in three layers, presentation, business logic and data access, and an automated architecture test fails the pipeline if a layer depends on a layer it must not use. The business layer, which holds the rules such as language fallback and the translation quota, is covered by unit tests. Every pull request runs separate jobs for the backend, the frontend and the container build. These practices are the means to objective O6 and are reported with their results in Chapter 5.

### *1.4.4. Zero-cost principle and tools*

The zero-cost principle means that no paid subscription or paid API is required for TourVoice to work. It does not mean that the system needs no resources: it still needs a server to host the API and the database, which can be a low-cost virtual server or, for the demonstration, a team member's machine. Table 1.5 lists the tools and the role of each; Chapter 2 explains each technology and why it was chosen over its alternatives.

| Area | Technology | Role in TourVoice | Cost |
|---|---|---|---|
| Backend | ASP.NET Core Web API (.NET 10), Entity Framework Core, SignalR | Three-layer API; pushes the live count to the dashboard | Free, open source |

| | | | |
|---|---|---|---|
| Database | MySQL 8.4 in Docker | Places, translations, accounts, listening and usage records | Free, open source |
| Frontend | React, Vite, TypeScript | Visitor web pages and administrator dashboard | Free |
| Narration | Web Speech API of the browser | Reads reviewed text aloud with the device voice [5] | Free, no quota |
| Translation | MyMemory translation API | Drafts English, Chinese, Korean and Japanese from Vietnamese | Free up to 50,000 characters per day [4] |
| Map | Leaflet with OpenStreetMap; Google Maps URLs | Shows destinations and food spots; opens walking directions | Free, no API key |
| Delivery | Docker Compose, GitHub Actions, GitHub Container Registry, Cloudflare Tunnel | Packages, tests and deploys every change | Free within usage limits |

*Table 1.5. Technologies and their roles*

### 1.5. Key Terms and Conventions

One concept keeps one name throughout the report. Table 1.6 fixes the terms that later chapters use without redefining them.

| Term | Meaning in this report |
|---|---|
| Place | Any item the system narrates: a destination or a food spot |

| | |
|---|---|
| Destination | One of the four main sites; a destination may own a group of food spots |
| Food spot | A dish or stall linked to a destination, classified as noodles, rice, snack, drink or dessert |
| Narration | The text about a place in one language, read aloud by the device voice |
| Translation status | Draft (machine output or edited, not visible to visitors) or Reviewed (approved and visible) |
| Translation source | Machine (drafted by the translation service) or Manual (written or edited by a person) |
| Place status | Draft, Published or Hidden; visitors see only Published places |
| Language fallback | Showing the Vietnamese text, with a notice, when the requested language is not yet Reviewed |
| Device voice | Speech produced by the visitor's own browser and phone, not by the server |
| Quota guard | The rule that blocks translation requests that would exceed 45,000 characters in a day |
| Presence | The set of devices that currently have a TourVoice page open |
| Heartbeat | A small anonymous signal an open page sends every 15 seconds to stay counted |
| Access session | One continuous visit by a device, stored for statistics |
| Release gate | A milestone (G1, G2, G3) with explicit criteria that must be met before the next stage |

*Table 1.6. Key terms*

Identifiers follow one scheme so that every element can be traced across chapters. Problems are numbered P1–P4, objectives O1–O6 and subsystems PH1–PH5. Functional requirements (FR-xx) and business rules (BR-xx) keep the numbers of the project's requirements document. Use cases are numbered UCnn, user stories USnn with acceptance criteria ACn, screens SCRnn and sequence diagrams SD-nn. Figures and tables are numbered by chapter, for example Figure 3.2 or Table 4.1.

### 1.6. Report Structure

The report has six chapters, and each one answers a single question whose answer raises the question of the next.

- **Chapter 1, Introduction and Project Objectives,** has asked why the project is needed and answered it with four problems, six objectives and a defined scope.
- **Chapter 2, Theoretical Background and Related Work,** asks what the project can build on: the concepts behind web systems, machine translation and speech synthesis, the selected technologies, how visitors obtain information today and what existing guide systems already offer.
- **Chapter 3, System Analysis,** asks what the system must do and states the requirements, actors, use cases and screen layout.
- **Chapter 4, System Design,** asks how the system is built and why, presenting the database, the architecture, the classes and the detailed interface together with the alternatives that were rejected.
- **Chapter 5, Results,** asks whether the system meets O1 to O6 and reports the implementation, the evaluation and the tests.
- **Chapter 6, Conclusion and Future Work,** compares the results with the objectives of this chapter and sets out what remains.

The objectives above say what TourVoice must achieve, but not yet what is already known about achieving it. Before the requirements can be written, the next chapter examines the concepts, technologies and existing systems on which a multilingual audio guide can be built, and identifies the gaps that TourVoice must fill.

# CHAPTER 2: THEORETICAL BACKGROUND AND RELATED WORK

Chapter 1 ended with six objectives but no account of what is already known about reaching them. This chapter supplies that account in four steps. Section 2.1 explains the concepts that each objective depends on, Section 2.2 selects the technologies that put those concepts into practice, Section 2.3 examines how visitors are informed today and what existing guide systems offer, and Section 2.4 turns the shortcomings found there into the gaps that TourVoice must fill. The chapter is therefore not a survey for its own sake: every concept, tool and system discussed here is kept because a later decision in Chapter 3 or Chapter 4 relies on it.

## 2.1. Theoretical Foundations

Six concepts underlie the objectives of Chapter 1. Each subsection below states the concept, the property of it that matters for TourVoice and the objective that depends on that property. Table 2.1 at the end of the section summarises these links.

### *2.1.1. Mobile-first web systems and the client–server model*

In a web information system the client, a browser, presents the interface, while a server applies the business rules and keeps the data. The two communicate through an application programming interface (API). In the REST style, the API exposes resources such as a place or a translation through addresses and standard HTTP operations, and each request carries everything the server needs to answer it. This statelessness is what allows many independent phones to use the same server at the same time without coordinating with one another, which is the situation described by objective O4.

A web client has a second advantage over a native application: it runs in the browser that every smartphone already has. A visitor who scans a code reaches the content in one step, with nothing to download or install. "Mobile-first" means that the pages are designed for a phone screen and then widened for larger screens, not the reverse. Both properties serve objective O3, which limits the path from the home page to a narration to three taps.

### *2.1.2. Layered architecture*

A layered architecture divides the server into layers with one responsibility each and allows a layer to depend only on the layer below it. TourVoice uses the common three-layer form. The presentation layer receives HTTP requests and returns responses, the business logic layer applies rules such as language fallback and the translation quota, and the data access layer reads and writes the database and calls external services. Because the business layer does not depend on HTTP or on a particular database, its rules can be tested in isolation. The dependency rule is valuable only if it is enforced, so the rule is checked by an automated test on every change. These two properties, isolated rules and an enforced dependency direction, are the substance of objective O6.

### *2.1.3. Machine translation with human review*

Machine translation (MT) produces a draft in a target language from a source text. Its quality varies with the language pair and with the type of text. Proper names, historical terms and food names, which are frequent in a guide about Chợ Lớn, are the parts most likely to be mistranslated. The translation industry handles this with post-editing: a person checks and corrects machine output before it is used, and the standard ISO 18587 defines this process and the competence expected of the post-editor [6].

TourVoice applies the same principle as a content lifecycle. Every translation is either Draft, visible only to the administrator, or Reviewed, visible to visitors. A machine output always starts as Draft, and any edit to a Reviewed text returns it to Draft. The MyMemory service used to produce the drafts combines a translation memory, a store of earlier human translations, with machine translation. Its free tier allows 50,000 characters per day when a contact address is supplied [4]. Two consequences follow for the design: translation must be requested only when the Vietnamese source has changed, and the daily volume must be guarded below the limit. These are the conditions behind objectives O1 and O2 and the response to problem P3.

### *2.1.4. Speech synthesis on the server and on the device*

Text-to-speech (TTS) converts text into spoken audio. It can be performed in two places, and the choice changes the cost structure of the whole system. Figure 2.1 compares the two approaches. The point to notice is the dashed loop in each row. In approach A a single correction forces new audio files to be produced, stored and delivered for every language; in approach B it changes only text, which is reviewed again.

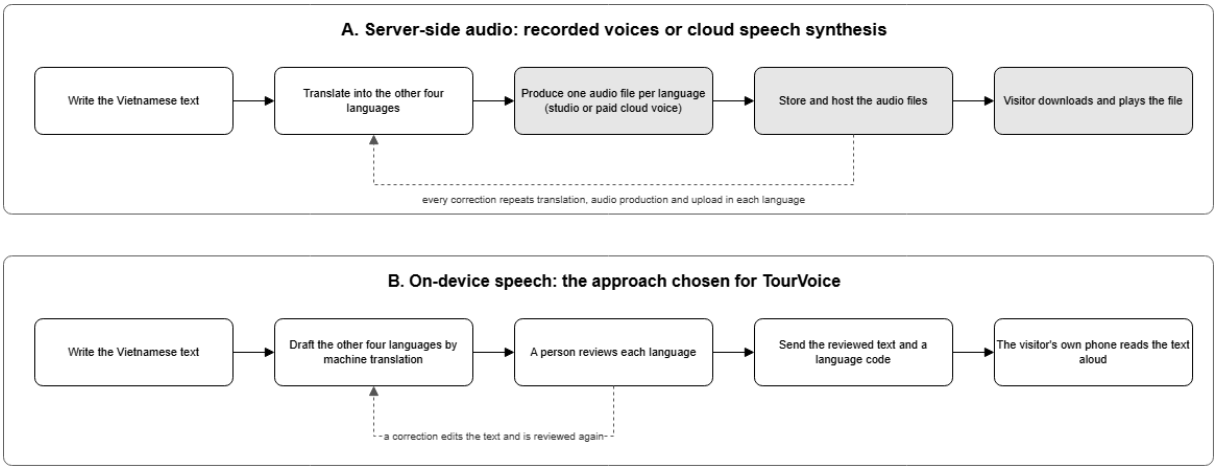



*Figure 2.1. Two ways to deliver multilingual narration*

In approach A the audio is created once on a server, either by recording a human voice or by a cloud TTS service, and each visitor downloads a file. The voice is consistent, but cost and storage grow with the number of languages and with every revision, and a paid cloud service conflicts with objective O2. In approach B the browser's Web Speech API reads the text aloud with a voice installed on the device [5]. No audio file is produced, stored or transferred, and speech begins as soon as the text is on the phone, which serves objective O3. The price is variability: the voice differs between phones, and a phone without a voice for Chinese, Korean or Japanese cannot speak that language at all. A design that adopts approach B must therefore always show the text and treat a missing voice as a planned fallback rather than an error.

### *2.1.5. Presence detection with heartbeats*

To count the devices that are using the system at a given moment, the server must know when a device arrives and when it leaves. Arrival is easy to observe; departure is not, because a phone may close the page, lose its signal or run out of battery without saying so. The usual solution is a heartbeat. While the page is open, it sends a small signal at a fixed interval, and the server keeps each device in its count for a time-to-live (TTL) after the last signal. A device that has stopped signalling is removed when its TTL expires. A page that closes normally can also send an explicit leave signal, so that the count falls at once.

The interval and the TTL determine how quickly the count reacts. With a heartbeat every 15 seconds, a TTL of 45 seconds and a removal sweep every 5 seconds, a silent device disappears from the count within 45 + 5 = 50 seconds. This is inside the 60-second target of objective O5 while tolerating two lost heartbeats. To reach the administrator without being polled, each change of the count is pushed to the dashboard over a persistent connection. SignalR provides this push over WebSockets and falls back to other transports when WebSockets are unavailable.

### *2.1.6. QR codes and map links as entry points*

A QR code is a two-dimensional barcode, standardised in ISO/IEC 18004, that any current phone camera can read. When it encodes a web address, it connects a physical place to a specific page without typing. Because the address is printed, it cannot be changed after the code is placed on site, so the address of each place must remain stable once published. For the map, OpenStreetMap offers free map tiles under a usage policy that requires attribution and moderate use, and a Google Maps URL opens walking directions in the visitor's own map application without an API key. Together, the code and the link answer the two halves of problem P2: knowing what is here, and knowing how to walk to the next place.

| **Concept** | **Property that matters** | **Objective served** |
|---|---|---|

| | | |
|---|---|---|
| Stateless client–server web system | Many phones served independently; no installation | O3, O4 |
| Three-layer architecture | Rules testable in isolation; dependency rule enforceable | O6 |
| Machine translation with human review | Cheap drafts, published only after review | O1, O2 |
| On-device speech synthesis | No audio files, no quota; voice varies by device | O2, O3 |
| Heartbeat presence with TTL | Departure detected within a bounded time; count pushed | O5 |
| QR codes and map links | One-step entry to a place; directions without a key | O3, O4 |

*Table 2.1. Concepts and the objectives that depend on them*

## 2.2. Technologies and Tools

The technologies were selected with four criteria taken directly from Chapter 1, applied in this order:

1. No paid service may be required for the system to work (O2).
2. No installation may be required from the visitor (O3, O4).
3. Enforceable structure: the tool must support the three-layer rule, automated tests and automated delivery (O6).
4. Team fit: the tool must be learnable within eight weeks by three second-year students, which favours the C# and .NET stack taught in the course.

A candidate that failed an earlier criterion was rejected even if it was stronger on a later one. Table 2.2 records the outcome for each need, with the alternatives that were considered. Version numbers are those used in the project repository at the time of writing.

| Need | Chosen (version) | Alternatives considered | Reason for the choice |
|---|---|---|---|
| Server and data mapping | ASP.NET Core Web API on .NET 10, Entity Framework Core 10 | Node.js with Express; Java Spring Boot | Matches the course language; built-in dependency injection makes the three layers explicit; an architecture-test library exists for .NET |
| Database | MySQL 8.4 in a Docker container | SQL Server; MongoDB; SQLite | Free and open source; relational keys and unique constraints fit one translation per place and language; SQLite handles many concurrent writers poorly; MongoDB gives up the constraints that protect the review lifecycle |
| Visitor pages and dashboard | React 19, Vite 8, TypeScript 6 | Native Android and iOS applications; server-rendered pages | Opens from a QR code with no installation; one code base for both interfaces; types shared with the API contract catch errors early |
| Translation drafts | MyMemory API | Google Cloud Translation; Azure Translator | Usable without an account or payment details, 50,000 characters per day with a contact email [4]; the cloud services require a billing account |
| Narration voice | Web Speech API of the browser | Cloud TTS services; recorded voices | No audio files, no quota and no cost (Section 2.1.4) [5]; accepted |

| | | | |
|---|---|---|---|
| | | | trade-off: the voice depends on the device |
| Live count to the dashboard | ASP.NET Core SignalR | Polling the server every few seconds; raw WebSockets | Pushes each change as it happens and falls back automatically when WebSockets are blocked; polling by every dashboard adds load without improving freshness |
| Map and directions | Leaflet with OpenStreetMap tiles; Google Maps URLs | Google Maps JavaScript API | No API key or billing account; directions are delegated to the visitor's own map application |
| Tests and delivery | xUnit 2.9, Moq, NetArchTest 1.3, GitHub Actions, Docker Compose, GitHub Container Registry, Cloudflare Tunnel | Manual testing and manual deployment | Turns the layer rule, unit tests and deployment into checks that run on every pull request, which O6 requires |

*Table 2.2. Technology choices, alternatives and reasons*

Two choices in Table 2.2 accept a known weakness. MyMemory is a community-grade service whose quality must be raised by review, and the Web Speech API gives up control over the voice. Both weaknesses are acceptable only because the design contains a counter-measure: the Draft and Reviewed lifecycle for the first, and the text fallback for the second. Chapter 4 shows where these counter-measures sit in the architecture.

## 2.3 Survey of Existing Systems

The survey looks at two things: how visitors to the four destinations are informed today, without any dedicated system, and what existing guide systems already offer. The first is modelled as business processes, which is the first step of the analysis method in Section 1.4.2; the second is compared against criteria drawn from P1 to P4.

### *2.3.1. Current practice at the four destinations*

The current processes were reconstructed from the team's analysis of the four destinations and from published descriptions of guide services in Vietnam [9], [10]. They describe typical practice rather than measured behaviour; the field survey that will confirm the final list of food spots (Section 1.3.4) will also be used to check them. Two processes matter: how a visitor looks for information, and how a site would produce a multilingual guide with the usual means.

Figure 2.2 follows a foreign visitor through a destination. The point to notice is that every decision of the visitor ends in a lane outside their own. When the sign is not in the visitor's language, a general translation app produces a word-by-word reading of a short, fixed text. When the explanation is not enough, the visitor turns to a vendor who usually speaks only Vietnamese. When choosing food, the visitor relies on generic listings that know nothing about the dishes of this market. None of these sources is specific, complete and in the visitor's language at once, which is problem P1, and the food decision is problem P2. The final object, a visit that is never recorded,

is the root of problem P4.

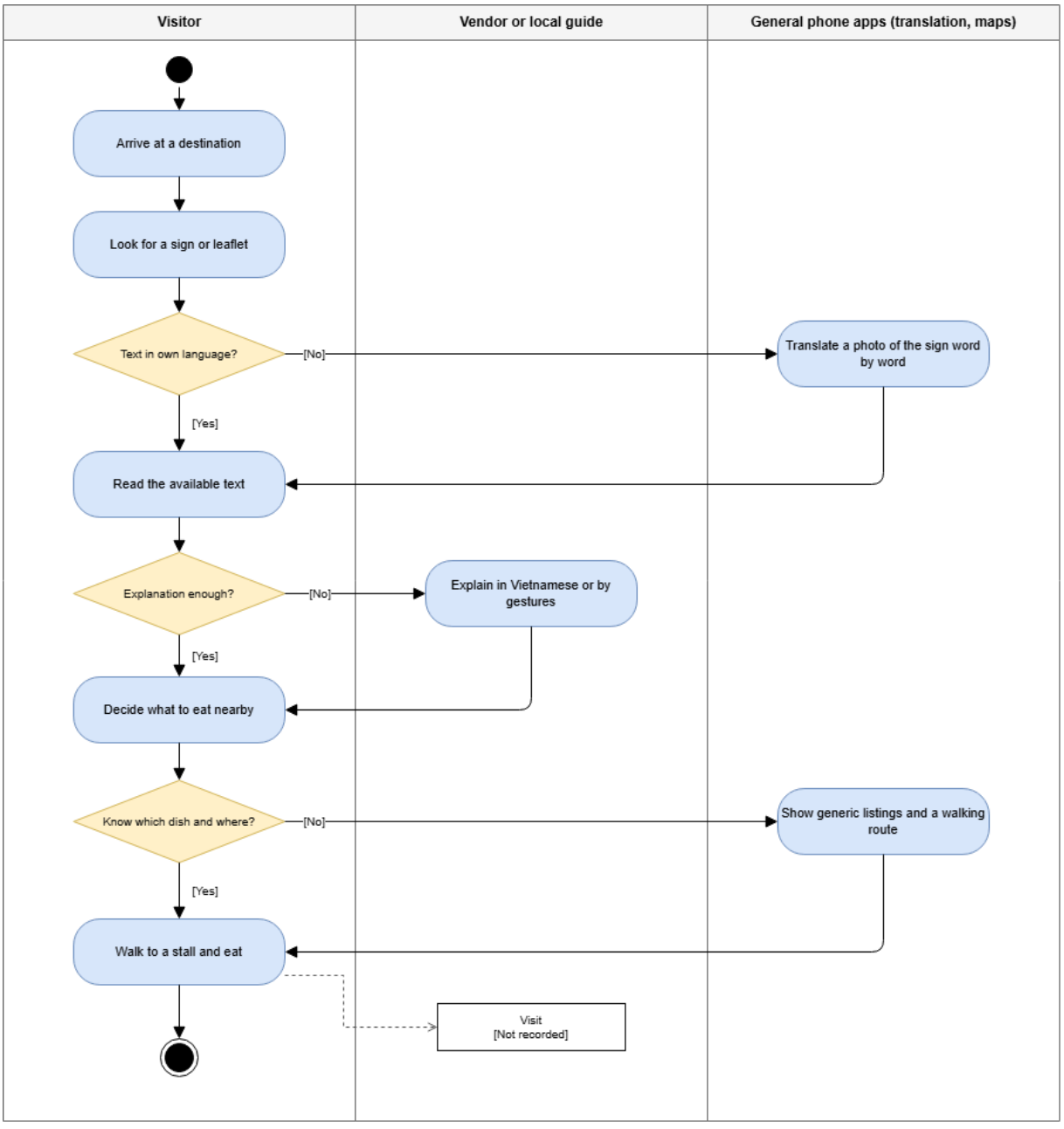



*Figure 2.2. Current process of a visitor seeking information at a destination*

Figure 2.3 shows how a site produces a recorded multilingual guide today. The point to notice is the long loop on the left. A correction of a single sentence, for example a wrong date or a stall that has moved, returns to the writer and passes again through approval, translation and studio recording for every language. This is the cost described in problem P3, and it explains why most signs at small sites carry one or two

languages and are rarely updated. The manager's only view of use is informal feedback, so the usage figures remain unknown.

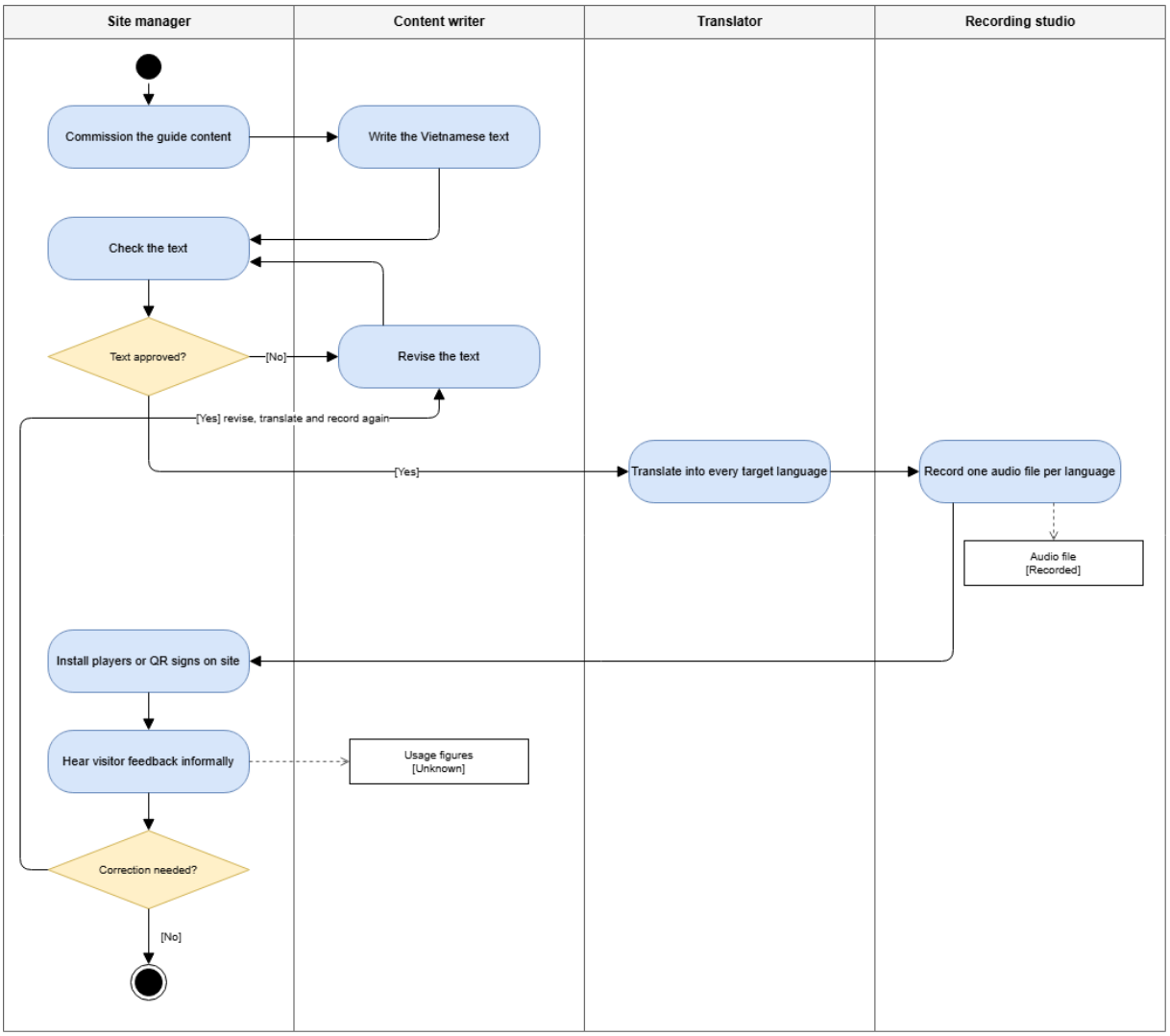



*Figure 2.3. Current process of producing a multilingual audio guide*

The two processes also supply the vocabulary for the analysis in Chapter 3. The lanes become candidate actors: the visitor becomes the Guest, and the site manager and the content writer merge into the Admin. The translator and the recording studio disappear as people, because their work is taken over by the translation service and the device voice, with the Admin's review in place of the translator's judgement. The business objects, a narration and its status, become the conceptual model.

### *2.3.2. Existing guide systems*

Four kinds of existing solutions address parts of the problem. izi.TRAVEL is a free platform with about 25,000 audio tours in 2,500 cities and guides from more than 3,000 museums; tours are written by local authors and museums, can be downloaded for offline use and can be opened from QR codes in museums, but require the app [7]. SmartGuide offers GPS-triggered audio guides for more than 1,000 destinations in thirteen languages, with offline maps as a paid option; its language list does not include Vietnamese, Korean or Japanese [8]. Google Translate can translate text in a camera image into many languages, which helps with signs but carries no knowledge of the place. Finally, Vietnamese sites have begun to use QR-based guides: the Ho Chi Minh Museum in Huế opened an automated guide with 43 QR stops in Vietnamese and English that works with any QR scanner [9]. A 2024 review of QR use at heritage sites warns that a code without well-maintained content becomes a formality [10]. Table 2.3 compares these solutions with the proposed system.

| **Criterion** | **izi.TRAVEL [7]** | **SmartGuide [8]** | **Google Translate camera** | **QR guide, Ho Chi Minh Museum Huế [9]** | **TourVoice (proposed)** |
|---|---|---|---|---|---|
| Opens without installing an app | No | No | No | Yes | Yes |
| Content specific to the four destinations | Only if a local author creates a tour | Not documented | No | No (other site) | Yes |
| Vietnamese, English, Chinese, Korean and Japanese | Depends on each tour | No Vietnamese, Korean or Japanese | Yes, text only | Vietnamese and English | Yes |

| | | | | | |
|---|---|---|---|---|---|
| Spoken narration | Yes, recorded | Yes | No | Yes | Yes, device voice |
| Food guidance with walking directions | Depends on each tour | Sights and GPS routes | No | No | Yes |
| Content reviewed by a person before publication | Depends on the author | Editorial | No, raw machine output | Yes | Yes, Draft to Reviewed |
| Cost to the visitor | Free app, paid premium options | Free app, paid guides and offline use | Free | Free | Free |
| Live usage visible to the site manager | Not documented | Not documented | No | Not documented | Yes |

*Table 2.3. Comparison of existing solutions and the proposed system*

Table 2.3 shows that no single solution meets all criteria. The international platforms are strong on scale and audio quality, but they depend on an app and on a local author, and none is known to cover the four destinations in the five required languages. General translation tools are available everywhere but know nothing about a place and publish unreviewed output. The Vietnamese QR guides prove that visitors accept a scan-and-listen model without an app, but they cover few languages, rely on recorded audio and do not report usage. TourVoice does not compete with these systems on scale; it combines the no-installation entry of the local QR guides with reviewed content in five languages and a live view of use.

### 2.4. Identified Gaps and Requirements for the New System

The survey reduces to five gaps between what visitors and managers need and what current practice or existing systems provide. Table 2.4 states each gap, the evidence for it in Section 2.3, the problem and objective it relates to, and the group of functional requirements (FR) and business rules (BR) of the project requirements document that will close it. Chapter 3 specifies these requirements in full; this table is the bridge that explains why each group exists.

| Gap | Evidence | Problem | Objective | Requirements that close it |
|---|---|---|---|---|
| 1. No reviewed narration about the four destinations in the visitor's language | Figure 2.2: fixed signs, word-by-word translation, vendors who speak Vietnamese; Table 2.3: no solution covers the five languages for these sites | P1 | O1, O3 | Multilingual visitor pages and narration player (FR-01 to FR-04, FR-08, FR-09, FR-17, FR-18); language fallback (BR-01) |
| 2. Food choice and walking route are not linked to the destination | Figure 2.2: generic listings and routes; Table 2.3: only tour-dependent coverage | P2 | O1 | Food guide, map and directions link (FR-04 to FR-07) |
| 3. Multilingual content is costly to produce and to correct | Figure 2.3: every correction repeats translation and recording; Figure 2.1, approach A | P3 | O1, O2 | Translation, review and quota pipeline (FR-23 to FR-26; BR-04 to BR-07) |

| | | | | |
|---|---|---|---|---|
| 4. Entry depends on an app or on several steps | Table 2.3: international platforms require installation | P1, P4 | O3, O4 | QR entry to each place and printable codes (FR-16, FR-32) |
| 5. The manager cannot see whether the site is used | Figure 2.3: usage figures unknown; Table 2.3: no solution reports live usage | P4 | O5 | Heartbeat presence, live count and access statistics (FR-19, FR-31, FR-33) |

*Table 2.4. Gaps, their evidence and the requirements that close them*

Two further points complete the picture. First, objective O6 does not close a gap visible to visitors; it is a quality requirement that keeps the five closures above reliable as the code grows, and Chapter 3 states it as a non-functional requirement. Second, customer accounts (FR-10 to FR-14) do not answer any gap in Table 2.4. They are a convenience for returning visitors, such as keeping favourites and showing progress across the four destinations, and this lower priority is why the plan in Section 1.4.1 builds them after the content and narration core.

# CHAPTER 3: SYSTEM ANALYSIS

Chapter 2 ended with five gaps between what visitors and the manager need and what current practice provides. A gap says what is missing, but not who will use the system, which business objects it must keep or which rules those objects obey. This chapter answers those questions in the order of the analysis method of Section 1.4.2, where each model is derived from the one before it. Section 3.1 turns the lanes of the current processes into actors. Section 3.2 turns the objects of those processes into business concepts and groups them into the five subsystems. Section 3.3 states the requirements, rules and quality targets that the concepts must satisfy. Sections 3.4 and 3.5 express the requirements as use cases and follow the three most important ones step by step. Section 3.6 derives the screens from the use cases. The result is a description of what TourVoice must do, without yet deciding how; the design decisions are the subject of Chapter 4.

## 3.1. Actor

An actor is a role that exchanges information with the system, not a particular person. The candidates come from the lanes of the two current processes in Section 2.3.1, because a lane already records who does what before any system exists. Table 3.1 shows what happens to each lane once TourVoice is introduced. Some lanes become actors directly, some merge, and some disappear because the system or an external service takes over their work.

| Lane in Figure 2.2 or 2.3 | Role after TourVoice | Reason |
|---|---|---|
| Visitor | Guest (actor) | Opens pages, chooses a language and listens without an account |
| Visitor who wants to keep a record | Customer (actor, specialises Guest) | Needs favorites, history and visited destinations, which require an identity |

| | | |
|---|---|---|
| Site manager, Content writer | Admin (actor) | One role writes the Vietnamese text, reviews translations, publishes places and watches use |
| Translator | MyMemory translation API (indirect) together with the Admin's review | The service drafts; the human judgement of the translator becomes the review step |
| Recording studio | Device speech engine (indirect) | The visitor's own browser reads the reviewed text aloud, so no recording exists |
| Vendor or local guide | Not an actor | Their explanation is replaced by the narration and the food guide |
| General phone apps | Map services (indirect) | Only walking directions and map tiles remain outside the system |

*Table 3.1. From the lanes of the current processes to the actors of TourVoice*

Figure 3.1 shows the result. The point to notice is the hollow triangle between Customer and Guest. It states that a Customer is a Guest, so every function open to a Guest is also open to a Customer, and the Customer adds only the account functions. This keeps the main path of the system, reaching a narration, free of any registration, which objective O3 requires. The three systems in the dashed frame are indirect actors: they never start a use case, but a use case cannot complete without them. Naming them here makes the dependence on external services visible early, and Chapter 4 designs a fallback for each one.

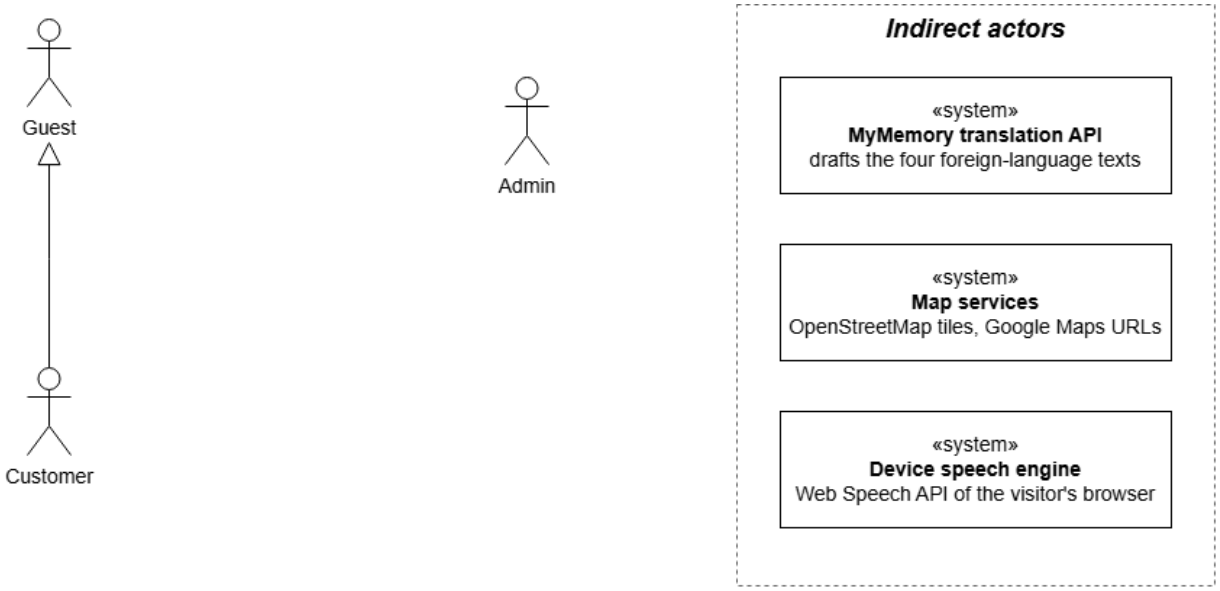



*Figure 3.1. Actors of the TourVoice system*

### 3.2. Business Concepts and Subsystems

The business objects of the current processes supply the concepts that the system must keep. Figure 2.3 contains a narration whose state changes as it moves through writing, translation and recording; Figure 2.2 ends with a visit that is never recorded. Following these objects, and the nouns of the requirements document, gives ten concepts. Each one is kept only because a requirement in Section 3.3 reads or changes it, and attributes are left to the database design in Chapter 4.

Figure 3.2 groups the concepts into the five subsystems introduced in Table 1.3. The point to notice is where the relations cross the colour boundaries. Every crossing starts at Place or at User, the two objects that the other subsystems refer to, while the concepts inside one region change together for the same reason. This is the test used to draw the boundaries: a concept belongs to the subsystem whose rules change it.

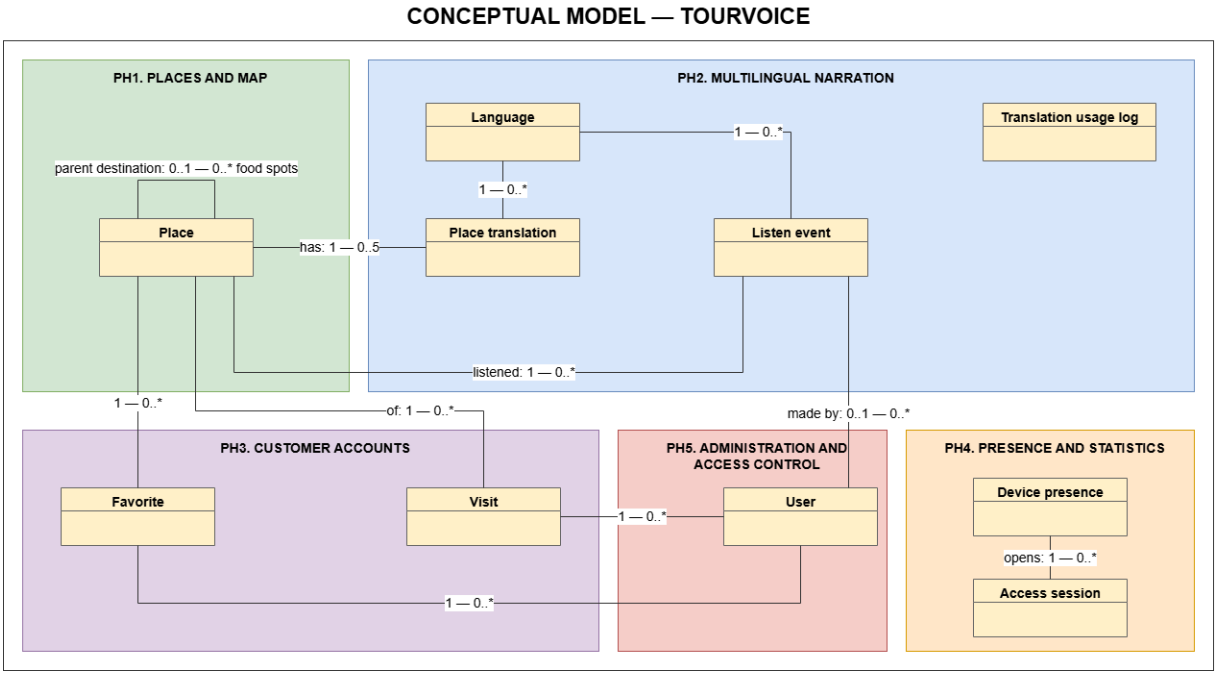



*Figure 3.2. Conceptual model of TourVoice*

Three decisions in the model shape the rest of the analysis:

- Narration is a property of a Place translation, not of a Place. A place has at most five translations, one per language, and each carries its own status, Draft or Reviewed. This is the business object "narration [status]" of Figure 2.3, and it is where the review requirement of problem P3 is enforced.
- Destinations and food spots are one concept. A food spot is a Place whose parent is a destination with a food guide. Both are narrated, published and opened by QR code in the same way, so one concept avoids duplicating every function.
- Presence and access sessions are separate. Device presence answers "how many devices now" and is kept only in memory, because it changes every few seconds; an Access session is the stored trace used later for statistics. Problem P4 needs both views, and Chapter 4 explains why they are stored differently.

## 3.3. System Requirements

The requirements are stated at three levels. Functional requirements say what the system must do, business rules say under which conditions it may do it, and non-functional requirements say how well it must do it. Each level is traced backwards to a gap of Table 2.4 or an objective of Table 1.2, so that no requirement exists without a reason.

### *3.3.1. Functional requirements*

The functional requirements keep the codes of the project requirements document (PRD). Where the PRD groups several requirements under one feature, Table 3.2 lists the group as a range. The groups are ordered by subsystem.

| Subsystem | Requirements | The system shall | Closes gap (Table 2.4) |
|---|---|---|---|
| PH1 | FR-01 to FR-04 | Provide the Home, Explore, Food and Culture and Map pages for published destinations and food spots | 1, 2 |
| PH1 | FR-04 to FR-07 | Show the food guide of a destination by five dish types, place every place on a map and open walking directions in the visitor's map application | 2 |
| PH1 | FR-16, FR-32 | Open the page of a place directly from its QR code, and let the Admin download each code for printing | 4 |
| PH2 | FR-17, FR-18 | Offer five languages, suggest one from the browser setting and remember the visitor's choice | 1 |

| | | | |
|---|---|---|---|
| PH2 | FR-08, FR-09 | Read reviewed narration aloud with the device voice, sentence by sentence, with pause, skip and three speeds; show the text when no voice exists; record anonymous listens | 1 |
| PH2 | FR-23 to FR-26 | Let the Admin write the Vietnamese text, draft the other four languages by machine translation, preview, edit and approve each language, and see the daily translation quota | 3 |
| PH3 | FR-10 to FR-14 | Let a Guest register, and let a Customer keep favorites, a listening history and visited destinations with progress across the four destinations | None (convenience) |
| PH4 | FR-19, FR-31, FR-33 | Count open devices by heartbeat and push the live count and access statistics to the dashboard | 5 |
| PH5 | Other requirements in FR-20 to FR-33 | Provide sign-in with roles, the dashboard overview and the management of places, languages, users and quotas | Supports all |

*Table 3.2. Functional requirements by subsystem*

Table 3.2 confirms the observation at the end of Chapter 2: every group except the accounts of PH3 closes at least one gap. This is why PH3 is planned after the content and narration core in Section 1.4.1.

### *3.3.2. Business rules*

A business rule is a condition that must hold whatever screen or function triggers it. Stating the rules separately from the functions matters for TourVoice, because the same rule is often reached from several places. The language fallback, for example, applies to a place page, a food spot card and a QR entry alike. Table 3.3 lists the eighteen rules of the PRD.

| Code | Rule | Subsystem |
|---|---|---|
| BR-01 | When the requested language is not Reviewed, the Vietnamese text is shown with a notice (language fallback) | PH2 |
| BR-02 | A food spot must belong to a destination that has a food guide | PH1 |
| BR-03 | Coordinates must lie inside the service area (latitude 10.730 to 10.765, longitude 106.615 to 106.665) | PH1 |
| BR-04 | Machine translation may start only after the Vietnamese text, at most 4,000 characters, is Reviewed | PH2 |
| BR-05 | A language is not translated again while the reviewed Vietnamese text is unchanged; a changed source marks the translation outdated | PH2 |
| BR-06 | A translation request is refused when the characters sent today plus the request would exceed 45,000, 90% of the free 50,000 (quota guard) | PH2 |

| | | |
|---|---|---|
| BR-07 | Only Reviewed translations are shown to visitors; editing a Reviewed translation returns it to Draft | PH2 |
| BR-08 | A place can be published only with a Reviewed Vietnamese text, valid coordinates and a cover image; after the first publication its address (slug) is locked | PH1 |
| BR-09 | A listen is recorded only after 10 seconds of speech, without IP address or location | PH2 |
| BR-10 | Progress equals visited destinations divided by published destinations | PH3 |
| BR-11 | A locked account cannot sign in, and at least one active Admin must always remain | PH5 |
| BR-12 | A password has at least 8 characters and is stored only as a hash | PH5 |
| BR-13 | Only a Draft place without food spots can be deleted | PH1 |
| BR-14 | A device identifier is stored only as a hash; the dashboard receives counts, never identifiers | PH4 |
| BR-15 | An open page sends a heartbeat every 15 s; a device is removed 45 s after its last heartbeat | PH4 |

| | | |
|---|---|---|
| BR-16 | A device that returns after more than 30 minutes opens a new access session | PH4 |
| BR-17 | Every change of the live count reaches the dashboard within 5 s | PH4 |
| BR-18 | A QR code encodes only the address of a place, never a language | PH1 |

*Table 3.3. Business rules*

Two groups of rules carry most of the weight. BR-04 to BR-07 together turn the review principle of Section 2.1.3 into enforceable conditions, and BR-14 to BR-17 turn the heartbeat model of Section 2.1.5 into the numbers behind objective O5.

### *3.3.3. Non-functional requirements*

The non-functional requirements translate the measurable targets of Table 1.2 into qualities of the software. Table 3.4 keeps the codes of the PRD where they exist and lists only the qualities that a later chapter designs for or measures.

| **Quality** | **Requirement** | **Objective** |
|---|---|---|
| Response time | At most 2 s from Listen to speech on Fast 4G and at most 3 taps from the home page to a narration; public reads at the 95th percentile within 300 ms (NFR-01); largest page element shown within 2.5 s and visitor scripts at most 250 KB compressed (NFR-02) | O3 |

| | | |
|---|---|---|
| Concurrency | At least 4 real phones on one QR code; heartbeats of 50 simulated devices at the 95th percentile within 200 ms, with no server errors and no rate-limit refusals (NFR-13) | O4 |
| Counting accuracy | Exact live count in the four-phone scenario: open, close a tab, cut the network, two tabs on one phone (NFR-14) | O5 |
| Cost | No paid service; translation volume held by the quota guard (BR-06) | O2 |
| Privacy | No GPS position or IP address stored; device identifier hashed (NFR-05) | P4, BR-14 |
| Security | HTTPS, cross-origin access limited to the site, no internal error details returned, images at most 2 MB with type check (NFR-04); at most 120 requests per minute per device and 1,200 per IP address | O4 |
| Accessibility | Contrast at least 4.5:1, touch targets at least 44 px, text always shown under the player, pages usable from 360 to 1,440 px wide (NFR-06) | O3 |
| Maintainability | Three-layer rule checked on every pull request, business-layer test coverage at least 70%, every change merged through a pull request with a passing pipeline | O6 |

*Table 3.4. Non-functional requirements*

The last row is the requirement that Chapter 2 deferred: objective O6 closes no visible gap, but without it the other rows cannot be kept as the code grows.

### 3.4. List of Functions

Each requirement group of Table 3.2 is now expressed as one or more use cases, a use case being one goal that an actor reaches through the system. A group becomes several use cases when its parts are done by different actors or at different times. The narration pipeline FR-23 to FR-26 is the clearest case: writing, translating and reviewing are separate goals of the Admin, often done on different days, and each is guarded by its own rule. Use cases are numbered in the order of the subsystems, so that the number alone tells the reader where a function belongs. A primary actor starts the use case; a secondary actor takes part in it without starting it. Table 3.5 is the resulting list.

| ID | Use case | Primary actor | Secondary actor | Subsystem | Requirements and rules |
|---|---|---|---|---|---|
| UC01 | Browse destinations | Guest | N/A | PH1 | FR-01 to FR-04 |
| UC02 | Open a place by QR code | Guest | N/A | PH1 | FR-16; BR-18 |
| UC03 | View food spots | Guest | N/A | PH1 | FR-04 to FR-07; BR-02 |

| | | | | | |
|---|---|---|---|---|---|
| UC04 | View map and directions | Guest | Map services | PH1 | FR-04 to FR-07; BR-03 |
| UC05 | Manage places and food spots | Admin | N/A | PH1 | FR-20 to FR-33; BR-02, BR-03, BR-08, BR-13 |
| UC06 | Print QR codes | Admin | N/A | PH1 | FR-32; BR-08, BR-18 |
| UC07 | Choose a language | Guest | N/A | PH2 | FR-17, FR-18; BR-01 |
| UC08 | Listen to a narration | Guest | Device speech engine | PH2 | FR-08, FR-09; BR-01, BR-07, BR-09 |
| UC09 | Write the Vietnamese narration | Admin | N/A | PH2 | FR-23 to FR-26; BR-04, BR-05 |
| UC10 | Translate missing languages | Admin | MyMemory translation API | PH2 | FR-23 to FR-26; BR-04, BR-05, BR-06 |

| | | | | | |
|---|---|---|---|---|---|
| UC11 | Review a translation | Admin | N/A | PH2 | FR-23 to FR-26; BR-05, BR-07 |
| UC12 | Preview with the device voice | Admin | Device speech engine | PH2 | FR-23 to FR-26 |
| UC13 | Monitor the translation quota | Admin | N/A | PH2 | FR-23 to FR-26; BR-06 |
| UC14 | Manage languages | Admin | N/A | PH2 | FR-20 to FR-33; BR-01 |
| UC15 | Register an account | Guest | N/A | PH3 | FR-10 to FR-14; BR-12 |
| UC16 | Manage favorites | Customer | N/A | PH3 | FR-10 to FR-14 |
| UC17 | View listening history | Customer | N/A | PH3 | FR-10 to FR-14; BR-09 |

| | | | | | |
|---|---|---|---|---|---|
| UC18 | Mark a destination as visited | Customer | N/A | PH3 | FR-10 to FR-14; BR-10 |
| UC19 | Report presence | Guest | N/A | PH4 | FR-19; BR-14, BR-15, BR-16 |
| UC20 | Monitor live usage | Admin | N/A | PH4 | FR-31, FR-33; BR-14 to BR-17 |
| UC21 | View access statistics | Admin | N/A | PH4 | FR-31; BR-14 |
| UC22 | Sign in | Customer, Admin | N/A | PH5 | FR-20 to FR-33; BR-11, BR-12 |
| UC23 | Manage users | Admin | N/A | PH5 | FR-20 to FR-33; BR-11, BR-12 |

*Table 3.5. List of use cases*

UC19 deserves a remark. The Guest does not press anything to report presence; the open page does it on the Guest's behalf. It is still modelled as a use case of the Guest, because the goal, being counted while the page is open, belongs to the visitor's session and not to the Admin who reads the count.

### 3.5. Use Case Model

The model is presented in the way the actors meet the system: one overall diagram, then one overview diagram per actor, each followed by the specifications of that actor's use cases. Every specification uses the same template. The normal flow is the expected path. An alternative flow such as 3a is another way of reaching the same goal that departs from step 3 and rejoins the normal flow. An exception such as UC08.E1 is a condition that prevents a step; it states the message the actor sees and whether the use case stops or continues. Three activity diagrams are placed after the use cases whose flows they draw, because these flows carry the measured objectives: listening (P1, O3), producing a narration (P3, O1 and O2) and counting live use (P4, O5).

### *3.5.1. Overall use case diagram*

Figure 3.3 places the 23 use cases of Table 3.5 in five packages, one per subsystem. The point to notice is how unevenly the actors are spread. The Guest reaches all visitor functions of PH1, PH2 and PH4 without an account, the Customer adds only the four account functions of PH3 and Sign in, and the Admin's functions are concentrated in the back office of PH1 and PH2. The model therefore puts the burden of work on the Admin and keeps the visitor's path short, which is the intention of objectives O1 and O3.

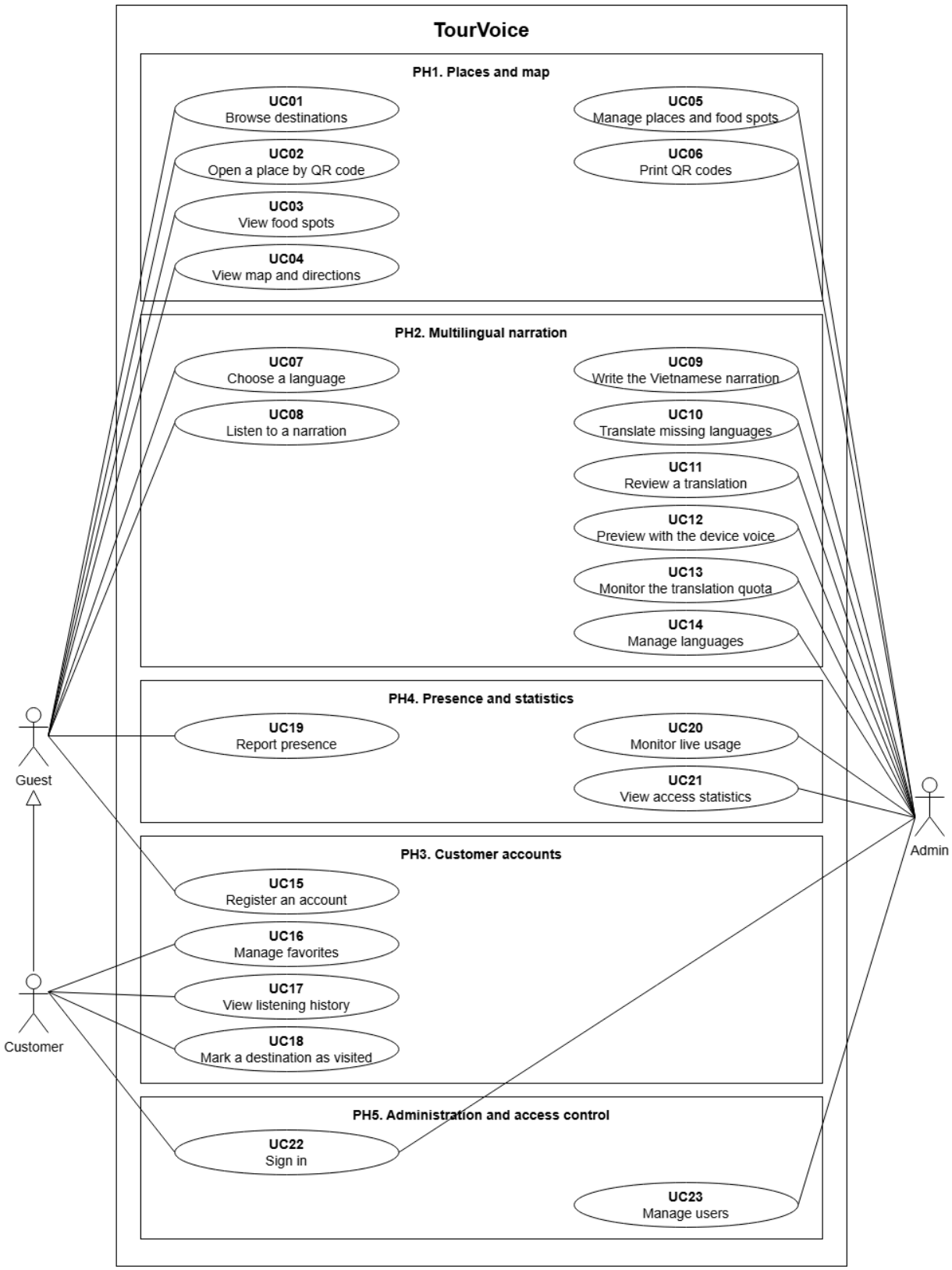
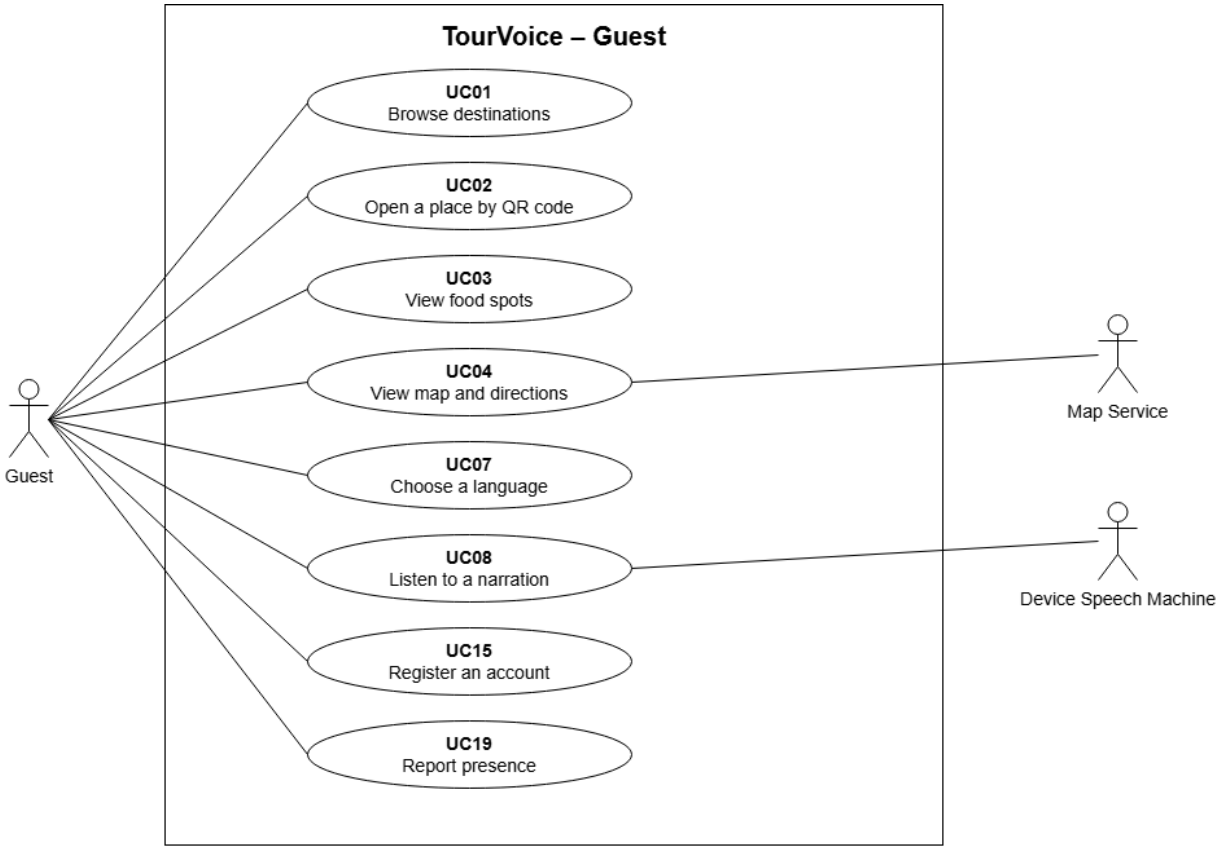



*Figure 3.3. Overall use case diagram of TourVoice*

### *3.5.2. Guest overview use case*

Figure 3.4 shows the eight use cases that a Guest reaches without an account. The point to notice is the two secondary actors on the right. Each is linked to exactly one use case: the map services only draw the map and the route of UC04, and the device speech engine only speaks the text that UC08 has already selected. Neither starts a use case and neither receives anything that identifies the visitor. The specifications in Tables 3.6 to 3.13 follow the order of the diagram.




#### *3.5.2.1. UC01 Browse destinations*

| **ID and Name** | **UC01: Browse destinations** |
|---|---|
| **Primary Actor** | Guest |

| | |
|---|---|
| **Secondary Actors** | N/A |
| **Description** | This feature allows the Guest to see the published destinations of the district and to choose one to visit. |
| **Trigger** | The Guest opens the TourVoice address or the Explore menu. |
| **Pre-conditions** | **PRE-1**. At least one destination is Published. |
| **Post-conditions** | **POST-1**. The Guest sees the cards of the published destinations in the chosen language.<br><br>**POST-2.** If the list cannot be loaded, an error message is shown and nothing is changed. |
| **Normal Flow** | 1. The Guest opens the Home screen (SCR01) (see UC01.E1).<br><br>2. The system determines the display language (UC07).<br><br>3. The system shows a card for each published destination with its cover image, name and short introduction.<br><br>4. The Guest taps “Explore”.<br><br>5. The system shows the Explore screen (SCR07), where destinations and food spots can be filtered by type.<br><br>6. The Guest taps a destination card and the system opens the Destination detail screen (SCR02); the visit continues in UC08. |

| | |
|---|---|
| **Alternative Flows** | 3a. A destination has no Reviewed text in the chosen language.<br><br>3a1. The card shows the Vietnamese name and introduction (BR-01). Return to step 4.<br><br>4a. The Guest taps a destination card directly on the Home screen.<br><br>4a1. Continue at step 6.<br><br>6a. The Guest taps a food spot card on the Explore screen.<br><br>6a1. The system opens the Food spot detail screen (SCR03); the visit continues in UC03. |
| **Exceptions** | UC01.E1. The Internet connection is lost.<br><br>1a. The system shows “Connection lost. Please try again.”<br><br>1a1. The Guest taps “Retry” – the use case continues at UC01.1.<br><br>1a2. The Guest closes the page – the use case stops. |
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | BR-01, BR-07, BR-08 |
| **Other Information** | Draft and Hidden places never appear in the list. No account is needed. |

| | |
|---|---|
| **Assumptions** | The Guest uses a current mobile browser. |

*Table 3.6. Specification of UC01 Browse destinations*

#### *3.5.2.2. UC02 Open a place by QR code*

| | |
|---|---|
| **ID and Name** | **UC02: Open a place by QR code** |
| **Primary Actor** | Guest |
| **Secondary Actors** | N/A |
| **Description** | This feature allows the Guest standing at a place to reach its page directly by scanning the QR code printed there. |
| **Trigger** | The Guest scans the QR code of a place with the phone camera. |
| **Pre-conditions** | **PRE-1.** The place is Published and its QR code has been printed (UC06). |
| **Post-conditions** | **POST-1.** The page of the place is open in the Guest's language and the visit is counted with the entry type QR (UC19).<br><br>**POST-2**. If the place cannot be shown, a message with a link to the Home screen is displayed. |

| | |
|---|---|
| **Normal Flow** | 1. The Guest scans the code; the camera opens the address /q/{slug} of the place (SCR09).<br><br>2. The system looks up the place by its slug (see UC02.E1, UC02.E2).<br><br>3. The system marks the entry of this visit as QR.<br><br>4. The system selects the language in this order: the choice saved on the device, the browser language, Vietnamese.<br><br>5. The system redirects the Guest to the Destination detail screen (SCR02) in that language.<br><br>6. The Guest taps “Listen”; the use case continues in UC08. |
| **Alternative Flows** | 5a. The code belongs to a food spot.<br><br>5a1. The system redirects to the Food spot detail screen (SCR03). Return to step 6. |
| **Exceptions** | UC02.E1. The slug is unknown, or the place is Draft or Hidden.<br><br>1a. The system shows “This place is not available.” with a link to the Home screen.<br><br>1a1. The Guest taps the link – the use case continues in UC01.<br><br>1a2. The Guest closes the page – the use case stops.<br><br>UC02.E2. The Internet connection is lost.<br><br>1a. The browser cannot open the address – the use case stops. |

| | |
|---|---|
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | BR-08, BR-18 |
| **Other Information** | The code carries no language (BR-18), so one printed code serves every visitor. The slug is locked after the first publication (BR-08), so a printed code stays valid. From the scan, one tap reaches the narration (O3). |
| **Assumptions** | The phone camera can read a QR code and open its link. |

*Table 3.7. Specification of UC02 Open a place by QR code*

#### *3.5.2.3. UC03 View food spots*

| | |
|---|---|
| **ID and Name** | **UC03: View food spots** |
| **Primary Actor** | Guest |
| **Secondary Actors** | N/A |
| **Description** | This feature allows the Guest to see the food guide of a destination, grouped by the five dish types, and to open one food spot. |

| | |
|---|---|
| **Trigger** | The Guest opens the Food and Culture screen or the food section of a destination. |
| **Pre-conditions** | **PRE-1**. The destination has a food guide with at least one published food spot (BR-02). |
| **Post-conditions** | **POST-1.** The Food spot detail screen (SCR03) is open, with its narration and directions available.<br><br>P**OST-2.** If no spot matches, the Guest stays on the list with a message. |
| **Normal Flow** | 1. The Guest opens the Food and Culture screen (SCR08) or the food section of the Destination detail screen (SCR02).<br><br>2. The system shows the food spots of the destination grouped by the five dish types.<br><br>3. The Guest chooses a dish type (see UC03.E1).<br><br>4. The system shows the spots of that type with image, name and short description.<br><br>5. The Guest taps a spot.<br><br>6. The system shows the Food spot detail screen (SCR03) with the description, the address, the “Listen” button (UC08) and the “Directions” button (UC04). |

| | |
|---|---|
| **Alternative Flows** | 4a. The Guest allows the browser to use the location (see UC03.E2).<br><br>4a1. The phone computes the distance to each spot and the list is sorted by distance. The position is never sent to the server (NFR-05). Return to step 5. |
| **Exceptions** | UC03.E1. No spot of the chosen type is published.<br><br>1a. The system shows “No food spot of this type yet.” – the use case continues at UC03.3.<br><br>UC03.E2. The Guest refuses the location permission.<br><br>1a. The list keeps its default order without distances – the use case continues at UC03.5. |
| **Priority** | Medium |
| **Frequency of Use** | High |
| **Business Rules** | BR-01, BR-02, BR-07 |
| **Other Information** | A food spot is a Place whose parent is a destination (Figure 3.2), so it has the same narration as a destination. |
| **Assumptions** | N/A |

*Table 3.8. Specification of UC03 View food spots*

#### *3.5.2.4. UC04 View map and directions*

| ID and Name | **UC04: View map and directions** |
|---|---|
| **Primary Actor** | Guest |
| **Secondary Actors** | Map services |
| **Description** | This feature allows the Guest to see the published places on a map and to open walking directions to one of them. |
| **Trigger** | The Guest opens the Map screen or taps “Directions” on a place. |
| **Pre-conditions** | **PRE-1.** Every published place has coordinates inside the service area (BR-03). |
| **Post-conditions** | **POST-1**. The map shows the places, or the Guest's map application shows the walking route.<br><br>**POST-2.** If the map cannot be drawn, the places are shown as a list with their addresses. |

| | |
|---|---|
| **Normal Flow** | 1. The Guest opens the Map screen (SCR04).<br><br>2. The system shows a marker for each published destination and food spot on map tiles from the map services (see UC04.E1).<br><br>3. The Guest taps a marker.<br><br>4. The system shows a short card with the name and a link to the page of the place.<br><br>5. The Guest taps “Directions”.<br><br>6. The system opens the walking route to the place in the Guest's map application. |
| **Alternative Flows** | 1a. The Guest taps “Directions” on SCR02 or SCR03.<br><br>1a1. Continue at step 6.<br><br>2a. The Guest allows the browser to use the location.<br><br>2a1. The Guest's own position is drawn on the map; it stays on the phone (NFR-05). Return to step 3.<br><br>4a. The Guest taps the link on the card.<br><br>4a1. The system opens the page of the place; the visit continues in UC08. |
| **Exceptions** | UC04.E1. The map tiles cannot be loaded. |

| | |
|---|---|
| | 1a. The system shows “The map is not available. Places are listed below.” with the list of places and addresses – the use case continues at UC04.5. |
| **Priority** | Medium |
| **Frequency of Use** | Medium |
| **Business Rules** | BR-03 |
| **Other Information** | Routing stays outside TourVoice; the system only passes the coordinates of the place to the map application. |
| **Assumptions** | The phone has a map application or can open directions in the browser. |

*Table 3.9. Specification of UC04 View map and directions*

#### *3.5.2.5. UC07 Choose a language*

| **ID and Name** | **UC07: Choose a language** |
|---|---|
| **Primary Actor** | Guest |
| **Secondary Actors** | N/A |

| | |
|---|---|
| **Description** | This feature allows the Guest to read every page and hear every narration in one of the five languages. |
| **Trigger** | The Guest opens a visitor page for the first time, or opens the language menu. |
| **Pre-conditions** | **PRE-1.** The language is enabled by the Admin (UC14). |
| **Post-conditions** | **POST-1.** The content is shown in the chosen language and the choice is remembered on the device.<br>**POST-2.** If the choice cannot be saved, it lasts only for the open page. |
| **Normal Flow** | 1. The Guest opens a visitor page for the first time.<br>2. The system proposes the browser language if it is enabled, otherwise Vietnamese.<br>3. The Guest opens the language menu in the header.<br>4. The system lists the enabled languages, at most five.<br>5. The Guest selects a language.<br>6. The system saves the choice on the device and shows the content again in that language (see UC07.E1). |

| | |
|---|---|
| **Alternative Flows** | 1a. The Guest has chosen a language on an earlier visit.<br><br>1a1. The saved choice is used and step 2 is skipped.<br><br>3a. The Guest keeps the proposed language.<br><br>3a1. The use case ends.<br><br>6a. A narration is playing.<br><br>6a1. The narration stops, because its text is replaced. Return to step 6. |
| **Exceptions** | UC07.E1. The content of a place is not Reviewed in the chosen language.<br><br>1a. The system shows the Vietnamese text with the notice “Not yet available in this language. Showing Vietnamese.” (BR-01) – the use case continues. |
| **Priority** | High |
| **Frequency of Use** | Medium |
| **Business Rules** | BR-01, BR-07, BR-18 |
| **Other Information** | The language is a setting of the device, not of the place or of the QR code (BR-18). |

| | |
|---|---|
| **Assumptions** | The browser allows the page to store a small setting. |

*Table 3.10. Specification of UC07 Choose a language*

#### *3.5.2.6. UC08 Listen to a narration*

| | |
|---|---|
| **ID and Name** | **UC08: Listen to a narration** |
| **Primary Actor** | Guest (and Customer by inheritance) |
| **Secondary Actors** | Device speech engine |
| **Description** | This feature allows the Guest to hear and read the reviewed narration of one place in the chosen language. |
| **Trigger** | The Guest opens a place page from a card (UC01), a QR code (UC02) or the map (UC04). |
| **Pre-conditions** | **PRE-1.** The place is Published, so its Vietnamese text is Reviewed (BR-08). |
| **Post-conditions** | **POST-1.** The Guest has heard the narration; at most one listen event is stored for the playback, without IP address or location (BR-09).<br>**POST-2.** If no voice is available, the Guest can still read the full text on the screen. |

| | |
|---|---|
| **Normal Flow** | 1. The Guest opens the Destination detail screen (SCR02) (see UC08.E2).<br><br>2. The system loads the place in the chosen language (UC07).<br><br>3. The system takes the Reviewed translation of that language.<br><br>4. The system shows the text and the player.<br><br>5. The Guest taps “Listen”.<br><br>6. The system asks the device speech engine for a voice of the language (see UC08.E1).<br><br>7. The system reads the text sentence by sentence and highlights the current sentence.<br><br>8. After 10 seconds of speech, the system records an anonymous listen event (BR-09).<br><br>9. The narration ends and the player returns to the start. |

| | |
|---|---|
| **Alternative Flows** | 1a. The Guest opens the Food spot detail screen (SCR03).<br><br>1a1. Return to step 2; the flow is the same.<br><br>3a. The language is not Reviewed for this place.<br><br>3a1. The system takes the Vietnamese text and shows a notice (BR-01). Return to step 4.<br><br>7a. The Guest taps “Pause”, skips a sentence or chooses one of the three speeds.<br><br>7a1. The system applies the action and continues reading. Return to step 7.<br><br>7b. The Guest starts another narration.<br><br>7b1. The current narration stops, because only one narration plays at a time. The use case starts again at step 6 for the new place.<br><br>8a. The Guest stops before 10 seconds of speech.<br><br>8a1. No listen event is recorded. The use case ends. |
| **Exceptions** | UC08.E1. The device has no voice for the language.<br><br>1a. The system shows “Your device has no voice for this language. You can read the text below or install a voice.”<br><br>1a1. The text stays on the screen – the use case stops.<br><br>UC08.E2. The Internet connection is lost before the page is loaded.<br><br>1a. The system shows “Connection lost. Please try again.” |

| | |
|---|---|
| | 1a1. The Guest taps “Retry” – the use case continues at UC08.1. |
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | BR-01, BR-07, BR-08, BR-09 |
| **Other Information** | Objective O3: at most 2 s from step 5 to speech and at most 3 taps from the Home screen to step 5. The flows are drawn in Figure 3.5. |
| **Assumptions** | The browser supports speech synthesis; the voices available depend on the device. |

*Table 3.11. Specification of UC08 Listen to a narration*

Figure 3.5 draws UC08. The point to notice is that the text is shown at step 4, before any speech is attempted. Both departures that take something away from the Guest, a missing translation (flow 3a) and a missing voice (UC08.E1), therefore still leave readable content on the screen. This ordering is what turns the two weaknesses accepted in Section 2.2 into planned fallbacks rather than failures.

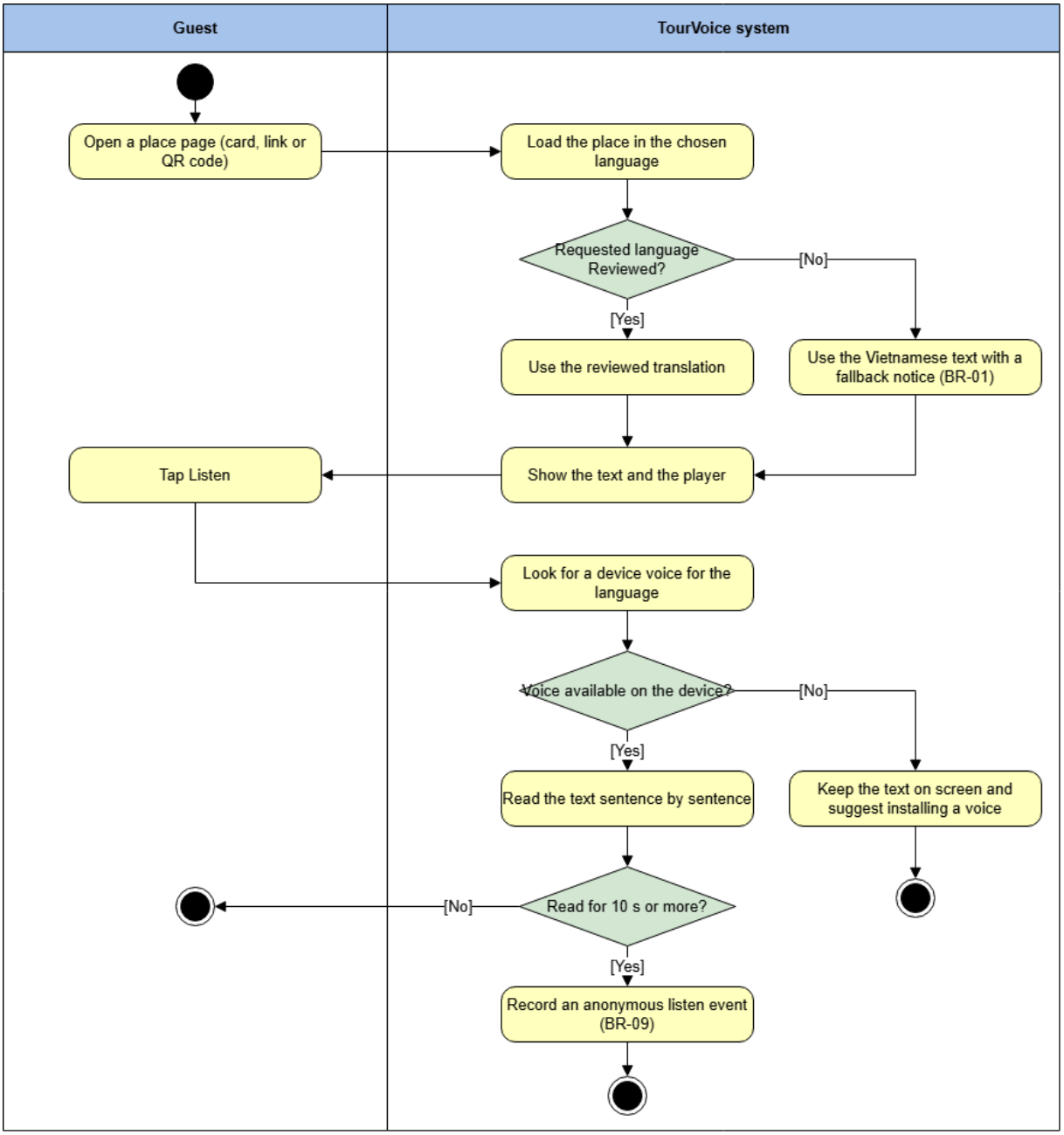



*Figure 3.5. Activity of UC08 Listen to a narration*

#### *3.5.2.7. UC15 Register an account*

| **ID and Name** | **UC15: Register an account** |
|---|---|
| **Primary Actor** | Guest |

| | |
|---|---|
| **Secondary Actors** | N/A |
| **Description** | This feature allows a Guest to create an account and become a Customer. |
| **Trigger** | The Guest taps “Sign in” in the header, or uses an account function without being signed in. |
| **Pre-conditions** | **PRE-1**. The Guest is not signed in and has no account with the email. |
| **Post-conditions** | **POST-1.** An account with the role Customer exists, the Customer is signed in and returns to the page where the use case started.<br><br>**POST-2.** If registration fails, no account is created. |
| **Normal Flow** | 1. The Guest taps “Sign in” in the header.<br><br>2. The system shows the Sign in and register screen (SCR05).<br><br>3. The Guest chooses “Register” and enters an email and a password.<br><br>4. The Guest taps “Register” (see UC15.E1 to UC15.E4).<br><br>5. The system stores the account with the role Customer and only a hash of the password (BR-12). |

| | |
|---|---|
| | 6. The system signs the Customer in and returns to the previous page. |
| **Alternative Flows** | 1a. The Guest taps the heart of a place (UC16) or “Mark as visited” (UC18) without being signed in.<br><br>1a1. Return to step 2; after step 6 the original action is completed.<br><br>3a. The Guest already has an account.<br><br>3a1. The Guest chooses “Sign in”; the use case continues in UC22. |
| **Exceptions** | UC15.E1. The email is not valid.<br><br>1a. The system shows “Email is not valid.” – the use case continues at UC15.3.<br><br>UC15.E2. The email is already registered.<br><br>1a. The system shows “This email is already registered.” – the use case continues at UC15.3.<br><br>UC15.E3. The password has fewer than 8 characters.<br><br>1a. The system shows “Password must have at least 8 characters.” (BR-12) – the use case continues at UC15.3.<br><br>UC15.E4. More than 10 attempts come from the same IP address within one minute.<br><br>1a. The system shows “Too many attempts. Please try again later.” – the use case stops. |

| | |
|---|---|
| **Priority** | Medium |
| **Frequency of Use** | Low |
| **Business Rules** | BR-12 |
| **Other Information** | Registration is never required for listening (O3). An Admin account cannot be created here; it is created in UC23. |
| **Assumptions** | The email is not verified in this version. |

*Table 3.12. Specification of UC15 Register an account*

#### *3.5.2.8. UC19 Report presence*

| | |
|---|---|
| **ID and Name** | **UC19: Report presence** |
| **Primary Actor** | Guest |
| **Secondary Actors** | N/A |
| **Description** | This feature keeps the Guest's device counted while a visitor page is open, without any action from the Guest. |
| **Trigger** | A visitor page is opened on the Guest's device. |

| | |
|---|---|
| **Pre-conditions** | **PRE-1.** The page has loaded and the browser can store a random device identifier. |
| **Post-conditions** | **POST-1.** The device is counted once in the live count while the page is open, and its access session is recorded.<br><br>**POST-2.** When the page is closed or the connection is lost, the device leaves the count within the delays of BR-15 and BR-17. |
| **Normal Flow** | 1. The Guest opens any visitor page.<br><br>2. The page reads the random device identifier stored on the phone, or creates one.<br><br>3. The page sends a join signal with the place and the entry type, QR or web.<br><br>4. The system stores the device as present under a hash of its identifier (BR-14) and opens or resumes its access session (BR-16).<br><br>5. While the page is open, it sends a heartbeat every 15 s (BR-15) (see UC19.E1).<br><br>6. The Guest closes the page; it sends a leave signal and the system removes the device. |

| | |
|---|---|
| **Alternative Flows** | 1a. The Guest opens a second tab on the same phone.<br><br>1a1. The same identifier is used, so the device is still counted once (BR-14).<br><br>5a. The page is hidden, for example the phone is locked.<br><br>5a1. Heartbeats pause and the device is removed 45 s after the last one. When the page is shown again, return to step 3.<br><br>6a. The connection is lost and no leave signal is sent.<br><br>6a1. The device is removed 45 s after its last heartbeat, found by a sweep every 5 s. The use case ends. |
| **Exceptions** | UC19.E1. The device sends more than 120 requests in one minute.<br><br>1a. The extra requests are refused without a message to the Guest; the device stays counted until its time-to-live expires – the use case continues at UC19.5. |
| **Priority** | High |
| **Frequency of Use** | High |
| **Business Rules** | BR-14, BR-15, BR-16 |
| **Other Information** | The use case has no screen of its own. It is drawn together with UC20 in Figure 3.9. |

| | |
|---|---|
| **Assumptions** | The identifier says nothing about the person; clearing the browser data creates a new device. |

*Table 3.13. Specification of UC19 Report presence*

### *3.5.3. Customer overview use case*

Figure 3.6 shows what a Customer adds to a Guest. The point to notice is the generalisation arrow from Customer to Guest: every use case of Figure 3.4 is inherited, so only the account functions are drawn. Each of them includes Sign in, because favorites, history and visits belong to an identity. Register an account stays with the Guest, since it is the step that turns a Guest into a Customer. The specifications follow in Tables 3.14 to 3.17.

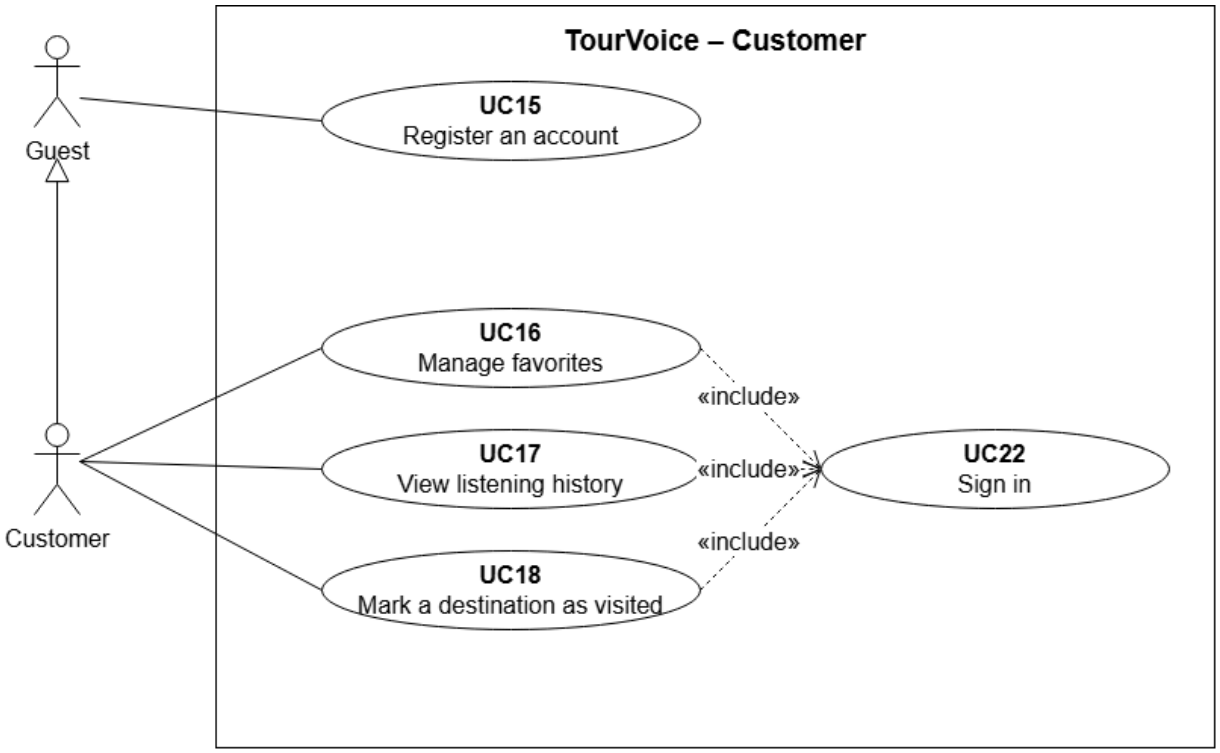



*Figure 3.6. Use case diagram of the Customer*

#### *3.5.3.1. UC16 Manage favorites*

| ID and Name | UC16: Manage favorites |
|---|---|
| **Primary Actor** | Customer |
| **Secondary Actors** | N/A |
| **Description** | This feature allows the Customer to keep a list of favorite places and to return to them later. |
| **Trigger** | The Customer taps the heart on a place, or opens the Favorites tab of the Account screen. |
| **Pre-conditions** | **PRE-1.** The Customer is signed in (UC22). |
| **Post-conditions** | **POST-1.** The favorite list of the account is updated and shown on the Account screen.<br>**POST-2.** If the Customer is not signed in, the list is unchanged. |
| **Normal Flow** | 1. The Customer taps the heart on SCR02 or SCR03 (see UC16.E1).<br>2. The system adds the place to the favorites and fills the heart.<br>3. The Customer opens the Account screen (SCR06) and selects “Favorites”. |

| | |
|---|---|
| | 4. The system shows the favorite places with image and name, newest first.<br><br>5. The Customer taps a favorite and the system opens its page. |
| **Alternative Flows** | 1a. The heart is already filled.<br><br>1a1. The system removes the place from the favorites and empties the heart. The use case ends.<br><br>5a. The Customer taps “Remove” on a favorite.<br><br>5a1. The system removes it from the list. Return to step 4.<br><br>4a. A favorite place is no longer Published.<br><br>4a1. It is shown greyed with “No longer available” and cannot be opened. |
| **Exceptions** | UC16.E1. The Customer is not signed in, or the session has expired.<br><br>1a. The system shows “Sign in to save favorites.” and opens SCR05.<br><br>1a1. The Customer signs in (UC22) or registers (UC15) – the use case continues at UC16.2.<br><br>1a2. The Customer closes the screen – the use case stops. |
| **Priority** | Low |

| | |
|---|---|
| **Frequency of Use** | Medium |
| **Business Rules** | BR-07, BR-08 |
| **Other Information** | A place appears at most once in the list. |
| **Assumptions** | N/A |

*Table 3.14. Specification of UC16 Manage favorites*

#### *3.5.3.2. UC17 View listening history*

| | |
|---|---|
| **ID and Name** | **UC17: View listening history** |
| **Primary Actor** | Customer |
| **Secondary Actors** | N/A |
| **Description** | This feature allows the Customer to see the narrations listened to while signed in and to open them again. |
| **Trigger** | The Customer opens the History tab of the Account screen. |
| **Pre-conditions** | **PRE-1**. The Customer is signed in (UC22). |

| | |
|---|---|
| **Post-conditions** | **POST-1**. The listens of the account are shown, newest first.<br><br>**POST-2.** Nothing is changed by this use case. |
| **Normal Flow** | 1. The Customer opens the Account screen (SCR06).<br><br>2. The Customer selects “History”.<br><br>3. The system shows the listen events of the account with place, language and time, newest first (see UC17.E1).<br><br>4. The Customer taps an entry and the system opens the page of that place. |
| **Alternative Flows** | 4a. The Customer leaves the history without opening an entry.<br><br>4a1. The use case ends. |
| **Exceptions** | UC17.E1. The account has no listen yet.<br><br>1a. The system shows “You have not listened to any narration yet.” with a link to the Explore screen – the use case stops.<br><br>UC17.E2. The session has expired.<br><br>1a. The system asks the Customer to sign in again (UC22) – the use case continues at UC17.3. |
| **Priority** | Low |

| | |
|---|---|
| **Frequency of Use** | Low |
| **Business Rules** | BR-09 |
| **Other Information** | Only listens of at least 10 seconds made while signed in appear here (BR-09); a Guest's listens stay anonymous. |
| **Assumptions** | N/A |

*Table 3.15. Specification of UC17 View listening history*

#### *3.5.3.3. UC18 Mark a destination as visited*

| | |
|---|---|
| **ID and Name** | **UC18: Mark a destination as visited** |
| **Primary Actor** | Customer |
| **Secondary Actors** | N/A |
| **Description** | This feature allows the Customer to record a visit to a destination and to follow the progress across all destinations. |
| **Trigger** | The Customer taps “Mark as visited” on a destination page. |
| **Pre-conditions** | **PRE-1.** The Customer is signed in (UC22). |

| | |
|---|---|
| | **PRE-2.** The place is a Published destination, not a food spot. |
| **Post-conditions** | **POST-1**. A visit is recorded and the progress is updated (BR-10).<br><br>**POST-2.** If the Customer is not signed in, nothing is recorded. |
| **Normal Flow** | 1. The Customer taps “Mark as visited” on the Destination detail screen (SCR02) (see UC18.E1).<br><br>2. The system records a visit of the destination with the date.<br><br>3. The system computes the progress as visited destinations divided by published destinations (BR-10).<br><br>4. The system shows the new progress, for example “2 of 4 destinations visited”.<br><br>5. The Customer opens the Account screen (SCR06) to see the visited destinations and the progress. |
| **Alternative Flows** | 1a. The destination is already marked as visited.<br><br>1a1. The Customer taps “Undo”; the system removes the visit. Return to step 3. |
| **Exceptions** | UC18.E1. The Customer is not signed in, or the session has expired.<br><br>1a. The system shows “Sign in to keep track of your visits.” and opens SCR05. |

| | |
|---|---|
| | 1a1. The Customer signs in (UC22) – the use case continues at UC18.2.<br><br>1a2. The Customer closes the screen – the use case stops. |
| **Priority** | Low |
| **Frequency of Use** | Low |
| **Business Rules** | BR-10 |
| **Other Information** | The visit is declared by the Customer; the system does not check the location (NFR-05). |
| **Assumptions** | When a destination is hidden, the progress is computed again over the published destinations only. |

*Table 3.16. Specification of UC18 Mark a destination as visited*

#### *3.5.3.4. UC22 Sign in*

| **ID and Name** | **UC22: Sign in** |
|---|---|
| **Primary Actor** | Customer, Admin |
| **Secondary Actors** | N/A |

| | |
|---|---|
| **Description** | This feature allows a Customer or an Admin to prove their identity and receive the functions of their role. |
| **Trigger** | The Customer taps “Sign in” on a visitor page, or the Admin opens the dashboard address. |
| **Pre-conditions** | **PRE-1.** The account exists. |
| **Post-conditions** | **POST-1.** The user is signed in with a session valid for 7 days for a Customer and 8 hours for an Admin. The Customer returns to the previous page; the Admin reaches the Overview screen (SCR11).<br><br>**POST-2.** If sign in fails, no session is created. |
| **Normal Flow** | 1. The user opens the Sign in and register screen (SCR05) or the Dashboard sign in screen (SCR10).<br><br>2. The user enters the email and the password.<br><br>3. The user taps “Sign in” (see UC22.E1, UC22.E2, UC22.E3).<br><br>4. The system compares the password with the stored hash (BR-12) and checks that the account is active (BR-11).<br><br>5. The system creates a session that carries the role of the account.<br><br>6. The system redirects the user according to the role. |

| | |
|---|---|
| **Alternative Flows** | 1a. The Admin opens a dashboard page without a valid session.<br><br>1a1. The system redirects to SCR10 and, after step 5, returns to the requested page.<br><br>1b. A Guest has no account.<br><br>1b1. The Guest chooses “Register”; the use case continues in UC15. |
| **Exceptions** | UC22.E1. The email or the password is wrong.<br><br>1a. The system shows “Email or password is incorrect.”, the same message in both cases – the use case continues at UC22.2.<br><br>UC22.E2. The account is locked.<br><br>1a. The system shows “This account is locked.” (BR-11) – the use case stops.<br><br>UC22.E3. A Customer account is used on the dashboard.<br><br>1a. The system shows “This account cannot access the dashboard.” – the use case stops. |
| **Priority** | High |
| **Frequency of Use** | Medium |
| **Business Rules** | BR-11, BR-12 |

| | |
|---|---|
| **Other Information** | UC22 is included by every Customer and Admin use case (Figures 3.6 and 3.7). The same message for a wrong email and a wrong password avoids revealing which emails are registered. |
| **Assumptions** | Password recovery is outside the scope of this version. |

*Table 3.17. Specification of UC22 Sign in*

### *3.5.4. Admin overview use case*

Figure 3.7 shows the eleven use cases of the Admin, all of which include Sign in. The point to notice is that the Admin's work is split into small goals instead of one "manage content" function. Write, Translate and Review are separate use cases because each has its own guard, BR-04, BR-05 and BR-06, and BR-07 respectively, and a Draft may wait days between translation and review. Preview extends Review rather than being included in it, because the Admin may approve a short text without listening to it. The specifications follow in Tables 3.18 to 3.28, with the activity of narration production after UC12 and the activity of the live count after UC20.

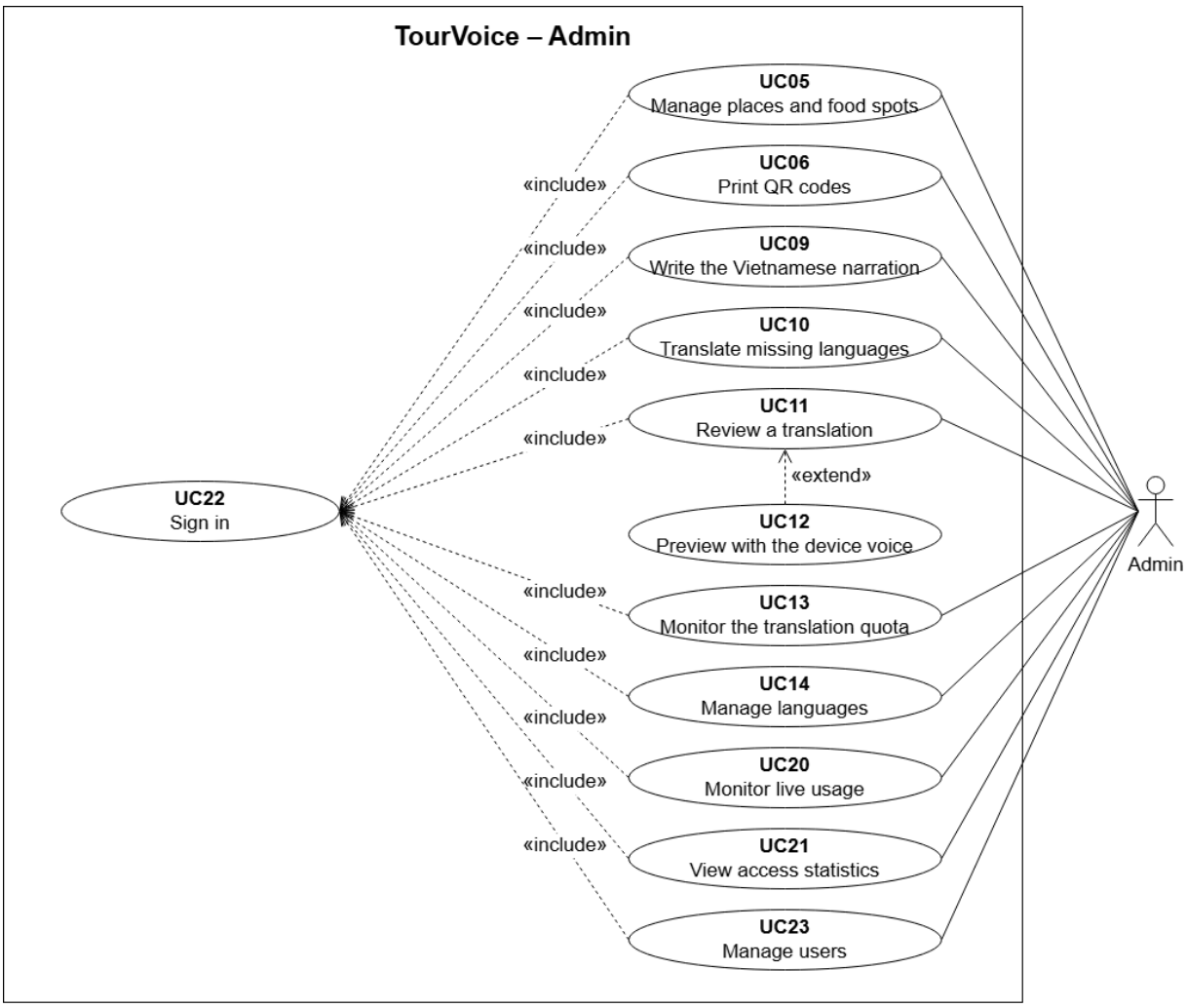



*Figure 3.7. Use case diagram of the Admin*

### *3.5.4.1. UC05 Manage places and food spots*

| **ID and Name** | **UC05: Manage places and food spots** |
|---|---|
| **Primary Actor** | Admin |
| **Secondary Actors** | N/A |

| | |
|---|---|
| **Description** | This feature allows the Admin to create, edit, publish, hide and delete destinations and food spots. |
| **Trigger** | The Admin opens the Places screen of the dashboard. |
| **Pre-conditions** | **PRE-1.** The Admin is signed in (UC22). |
| **Post-conditions** | **POST-1**. The place is saved as Draft, Published or Hidden, and the visitor pages show only Published places.<br><br>**POST-2**. If a check fails, the place keeps its previous state. |
| **Normal Flow** | 1. The Admin opens the Places screen (SCR12).<br><br>2. The system lists the places with type and status, and can filter them.<br><br>3. The Admin taps “New place” or an existing place.<br><br>4. The system opens the Place editor (SCR13) on the tab General and QR.<br><br>5. The Admin enters the name, the type (destination or food spot), the parent destination of a food spot, the address and the cover image (see UC05.E2, UC05.E3).<br><br>6. On the tab Location, the Admin picks the position on the map (see UC05.E1).<br><br>7. The Admin taps “Save”; the system saves the place as Draft. |

| | |
|---|---|
| | 8. The Admin taps “Publish” (see UC05.E4).<br><br>9. The system checks BR-08, publishes the place and locks its slug. |
| **Alternative Flows** | 3a. The Admin chooses “Delete” on a place (see UC05.E5).<br><br>3a1. The system asks for confirmation and deletes the place. The use case ends.<br><br>8a. The Admin chooses “Hide” on a Published place.<br><br>8a1. The place becomes Hidden and disappears from the visitor pages; the food spots of a hidden destination are hidden with it. The use case ends.<br><br>8b. The Admin leaves the place as Draft.<br><br>8b1. The use case ends; the place is invisible to visitors. |

| | |
|---|---|
| **Exceptions** | UC05.E1. The position lies outside the service area.<br><br>1a. The system shows “The location must be inside the service area.” (BR-03) – the use case continues at UC05.6.<br><br>UC05.E2. The cover image is larger than 2 MB or is not an image.<br><br>1a. The system shows “The image must be a picture file of at most 2 MB.” (NFR-04) – the use case continues at UC05.5.<br><br>UC05.E3. The parent of a food spot is not a destination with a food guide.<br><br>1a. The system shows “A food spot must belong to a destination with a food guide.” (BR-02) – the use case continues at UC05.5.<br><br>UC05.E4. A publication condition is missing.<br><br>1a. The system lists what is missing, for example “The Vietnamese narration is not reviewed.” (BR-08) – the use case stops; the Admin completes it in UC09.<br><br>UC05.E5. The place is not a Draft without food spots.<br><br>1a. The system shows “Only a Draft place without food spots can be deleted.” (BR-13) – the use case stops. |
| **Priority** | High |
| **Frequency of Use** | Medium |
| **Business Rules** | BR-02, BR-03, BR-08, BR-13 |

| | |
|---|---|
| **Other Information** | The narration of the place is managed on the Narration tab of the same editor (UC09 to UC12). |
| **Assumptions** | N/A |

*Table 3.18. Specification of UC05 Manage places and food spots*

#### *3.5.4.2. UC06 Print QR codes*

| **ID and Name** | **UC06: Print QR codes** |
|---|---|
| **Primary Actor** | Admin |
| **Secondary Actors** | N/A |
| **Description** | This feature allows the Admin to download the QR code of a published place for printing and placing on site. |
| **Trigger** | The Admin opens the tab General and QR of a place. |
| **Pre-conditions** | **PRE-1.** The Admin is signed in (UC22).<br>**PRE-2.** The place is Published, so its slug is locked (BR-08). |
| **Post-conditions** | **POST-1.** A file with the code of the address /q/{slug} is downloaded. |

| | |
|---|---|
| | POST-2. If the place is not Published, no code is produced. |
| **Normal Flow** | 1. The Admin opens the Place editor (SCR13) of a place (see UC06.E1).<br><br>2. The Admin selects the tab General and QR.<br><br>3. The system shows a preview of the code, which encodes only the address of the place (BR-18).<br><br>4. The Admin chooses the format: SVG, PNG or an A6 sheet ready to print.<br><br>5. The Admin taps “Download”.<br><br>6. The system produces the file and the browser saves it. |
| **Alternative Flows** | 4a. The Admin needs several formats.<br><br>4a1. Return to step 4 after each download. |
| **Exceptions** | UC06.E1. The place is not Published.<br><br>1a. The system shows “Publish the place before printing its QR code.” – the use case stops. |
| **Priority** | Medium |
| **Frequency of Use** | Low |

| | |
|---|---|
| **Business Rules** | BR-08, BR-18 |
| **Other Information** | Because the slug is locked and the code carries no language, a printed code never needs to be replaced when the content or the languages change. |
| **Assumptions** | Printing and placing the code on site are done outside the system. |

*Table 3.19. Specification of UC06 Print QR codes*

#### *3.5.4.3. UC09 Write the Vietnamese narration*

| | |
|---|---|
| **ID and Name** | **UC09: Write the Vietnamese narration** |
| **Primary Actor** | Admin |
| **Secondary Actors** | N/A |
| **Description** | This feature allows the Admin to write and approve the Vietnamese narration of a place, which is the source of every other language. |
| **Trigger** | The Admin opens the Narration tab of a place. |
| **Pre-conditions** | **PRE-1**. The Admin is signed in (UC22).<br>**PRE-2.** The place exists (UC05). |

| | |
|---|---|
| **Post-conditions** | **POST-1.** The Vietnamese text is Reviewed; foreign translations made from an older version are marked outdated (BR-05).<br><br>**POST-2.** If the text is only saved, it stays Draft and translation is not yet allowed (BR-04). |
| **Normal Flow** | 1. The Admin opens the Narration tab of the Place editor (SCR13).<br><br>2. The system shows the Vietnamese text and the status of the five languages.<br><br>3. The Admin writes or edits the Vietnamese text.<br><br>4. The Admin taps “Save” (see UC09.E1, UC09.E2); the system saves the text as Draft.<br><br>5. The Admin reads the text again and taps “Approve”.<br><br>6. The system marks the Vietnamese text Reviewed, records it as a new source version and marks older translations outdated (BR-05).<br><br>7. The system enables “Translate missing” (UC10). |
| **Alternative Flows** | 4a. The Admin taps “Approve” without saving first.<br><br>4a1. The text is saved and approved together. Continue at step 6.<br><br>5a. The Admin wants to finish later.<br><br>5a1. The use case ends with the text as Draft. |

| | |
|---|---|
| **Exceptions** | UC09.E1. The text is longer than 4,000 characters.<br><br>1a. The system shows “The narration must have at most 4,000 characters.” (BR-04) – the use case continues at UC09.3.<br><br>UC09.E2. The text is empty.<br><br>1a. The system shows “The narration cannot be empty.” – the use case continues at UC09.3.<br><br>UC09.E3. The session has expired.<br><br>1a. The system asks the Admin to sign in again (UC22); the unsaved text is kept in the editor – the use case continues at UC09.4. |
| **Priority** | High |
| **Frequency of Use** | Medium |
| **Business Rules** | BR-04, BR-05, BR-07 |
| **Other Information** | Editing a Reviewed Vietnamese text returns it to Draft (BR-07), and visitors keep the last Reviewed version until it is approved again. |
| **Assumptions** | The Admin writes the text; the system does not generate content. |

*Table 3.20. Specification of UC09 Write the Vietnamese narration*

#### *3.5.4.4. UC10 Translate missing languages*

| ID and Name | UC10: Translate missing languages |
|---|---|
| **Primary Actor** | Admin |
| **Secondary Actors** | MyMemory translation API |
| **Description** | This feature allows the Admin to obtain a machine draft in every foreign language that is missing or outdated, within the free daily quota. |
| **Trigger** | The Admin taps “Translate missing” on the Narration tab of a place. |
| **Pre-conditions** | **PRE-1.** The Admin is signed in (UC22).<br>**PRE-2.** The Vietnamese text is Reviewed (BR-04). |
| **Post-conditions** | **POST-1.** Each selected language has a Draft with the source Machine; no visitor sees it until it is Reviewed (BR-07).<br>**POST-2.** If the request is refused or fails, nothing is sent beyond the quota and the languages saved before the failure stay Draft. |

| | |
|---|---|
| **Normal Flow** | 1. The Admin taps “Translate missing” on the Place editor (SCR13) (see UC10.E1).<br><br>2. The system selects the languages that are missing or outdated (BR-05) (see UC10.E2).<br><br>3. The system adds the characters already sent today to the size of the request and checks that the total stays within 45,000 (BR-06) (see UC10.E3).<br><br>4. The system sends the Vietnamese text in segments to the translation service (see UC10.E4).<br><br>5. The system saves each language as Draft with the source Machine and the version of the Vietnamese text it came from.<br><br>6. The system logs the characters sent and the outcome in the translation usage log.<br><br>7. The system shows the drafts for review (UC11). |
| **Alternative Flows** | 1a. The Admin starts from a cell of the Narration matrix (SCR14).<br><br>1a1. Only the language of that cell is selected. Continue at step 3. |

| | |
|---|---|
| **Exceptions** | UC10.E1. The Vietnamese text is not Reviewed.<br><br>1a. The button is disabled with the hint “Approve the Vietnamese text before translating.” (BR-04) – the use case stops.<br><br>UC10.E2. No language is missing or outdated.<br><br>1a. The system shows “All languages are up to date.” and sends nothing – the use case stops.<br><br>UC10.E3. The request would exceed the daily limit.<br><br>1a. The system refuses the request with “Today's translation quota would be exceeded. Try again tomorrow.” (BR-06) – the use case stops.<br><br>UC10.E4. The translation service returns an error.<br><br>1a. The system shows “The translation service failed. Languages already translated are kept as Draft.” and logs the failure – the use case stops. |
| **Priority** | High |
| **Frequency of Use** | Medium |
| **Business Rules** | BR-04, BR-05, BR-06, BR-07 |
| **Other Information** | Objectives O1 (drafts for all five languages) and O2 (no paid service, quota never exceeded). |

| | |
|---|---|
| **Assumptions** | The free limit of the service stays at 50,000 characters per day. |

*Table 3.21. Specification of UC10 Translate missing languages*

#### *3.5.4.5. UC11 Review a translation*

| **ID and Name** | **UC11: Review a translation** |
|---|---|
| **Primary Actor** | Admin |
| **Secondary Actors** | N/A |
| **Description** | This feature allows the Admin to check, correct and approve the draft of one language so that visitors can see it. |
| **Trigger** | The Admin selects a Draft language on the Narration tab or in the Narration matrix. |
| **Pre-conditions** | **PRE-1.** The Admin is signed in (UC22).<br>**PRE-2.** A Draft exists for the language (UC10). |
| **Post-conditions** | **POST-1.** The translation is Reviewed and shown to visitors (BR-07).<br>POST-2. If the Admin does not approve, the translation stays Draft and visitors see the fallback of BR-01. |

| | |
|---|---|
| **Normal Flow** | 1. The Admin selects a Draft language on the Place editor (SCR13).<br><br>2. The system shows the Vietnamese source beside the draft (see UC11.E1).<br><br>3. The Admin reads the draft and may listen to it (UC12).<br><br>4. The Admin corrects the text where needed.<br><br>5. The Admin taps “Approve” (see UC11.E2).<br><br>6. The system saves the translation as Reviewed, with the source Edited if the text was changed, and shows it to visitors. |
| **Alternative Flows** | 1a. The Admin selects a cell of the Narration matrix (SCR14).<br><br>1a1. The system opens the same review panel. Return to step 2.<br><br>5a. The Admin taps “Save” instead of “Approve”.<br><br>5a1. The correction is kept as Draft. The use case ends.<br><br>6a. The Admin later edits a Reviewed translation.<br><br>6a1. It returns to Draft (BR-07). Return to step 3. |
| **Exceptions** | UC11.E1. The draft is outdated because the Vietnamese text has changed.<br><br>1a. The system shows “The Vietnamese text has changed. Translate again or correct before approving.” (BR-05) – the use case continues at UC11.4.<br><br>UC11.E2. The text is empty. |

| | |
|---|---|
| | 1a. The system shows “The translation cannot be empty.” – the use case continues at UC11.4. |
| **Priority** | High |
| **Frequency of Use** | Medium |
| **Business Rules** | BR-01, BR-05, BR-07 |
| **Other Information** | This step replaces the human translator of Figure 2.3: the service drafts, the Admin judges. |
| **Assumptions** | The Admin can read each of the five languages or has help to do so. |

*Table 3.22. Specification of UC11 Review a translation*

#### *3.5.4.6. UC12 Preview with the device voice*

| **ID and Name** | **UC12: Preview with the device voice** |
|---|---|
| **Primary Actor** | Admin |
| **Secondary Actors** | Device speech engine |
| **Description** | This feature allows the Admin to hear a translation as a visitor will hear it before approving it. It extends UC11 at step 3. |

| | |
|---|---|
| **Trigger** | The Admin taps “Preview” while reviewing a language. |
| **Pre-conditions** | **PRE-1.** The Admin is reviewing a translation (UC11). |
| **Post-conditions** | **POST-1.** The Admin has heard the text and returns to the review.<br>**POST-2.** If no voice exists, the review continues without listening. |
| **Normal Flow** | 1. The Admin taps “Preview” on the Narration tab (SCR13).<br>2. The system asks the device speech engine for a voice of the language (see UC12.E1).<br>3. The system reads the current text, including unsaved corrections, with the same player as the visitor page.<br>4. The reading ends and the use case returns to UC11 step 4. |
| **Alternative Flows** | 3a. The Admin taps “Stop”.<br>3a1. The reading stops. Continue at step 4. |
| **Exceptions** | UC12.E1. The Admin's device has no voice for the language.<br>1a. The system shows “This device has no voice for this language.” – the use case stops and the review continues. |
| **Priority** | Medium |

| | |
|---|---|
| **Frequency of Use** | Medium |
| **Business Rules** | N/A |
| **Other Information** | Preview is optional, which is why it extends UC11 instead of being included in it (Figure 3.7). |
| **Assumptions** | A preview is not a visitor's listen, so it records no listen event. |

*Table 3.23. Specification of UC12 Preview with the device voice*

Figure 3.8 places UC09 to UC12 inside the whole production of a narration. The point to notice is the loop on the left. A correction now returns only to Preview and Approve for one language, whereas in Figure 2.3 it returned to the writer and passed again through translation and recording for every language. This shortened loop is the concrete answer to problem P3.

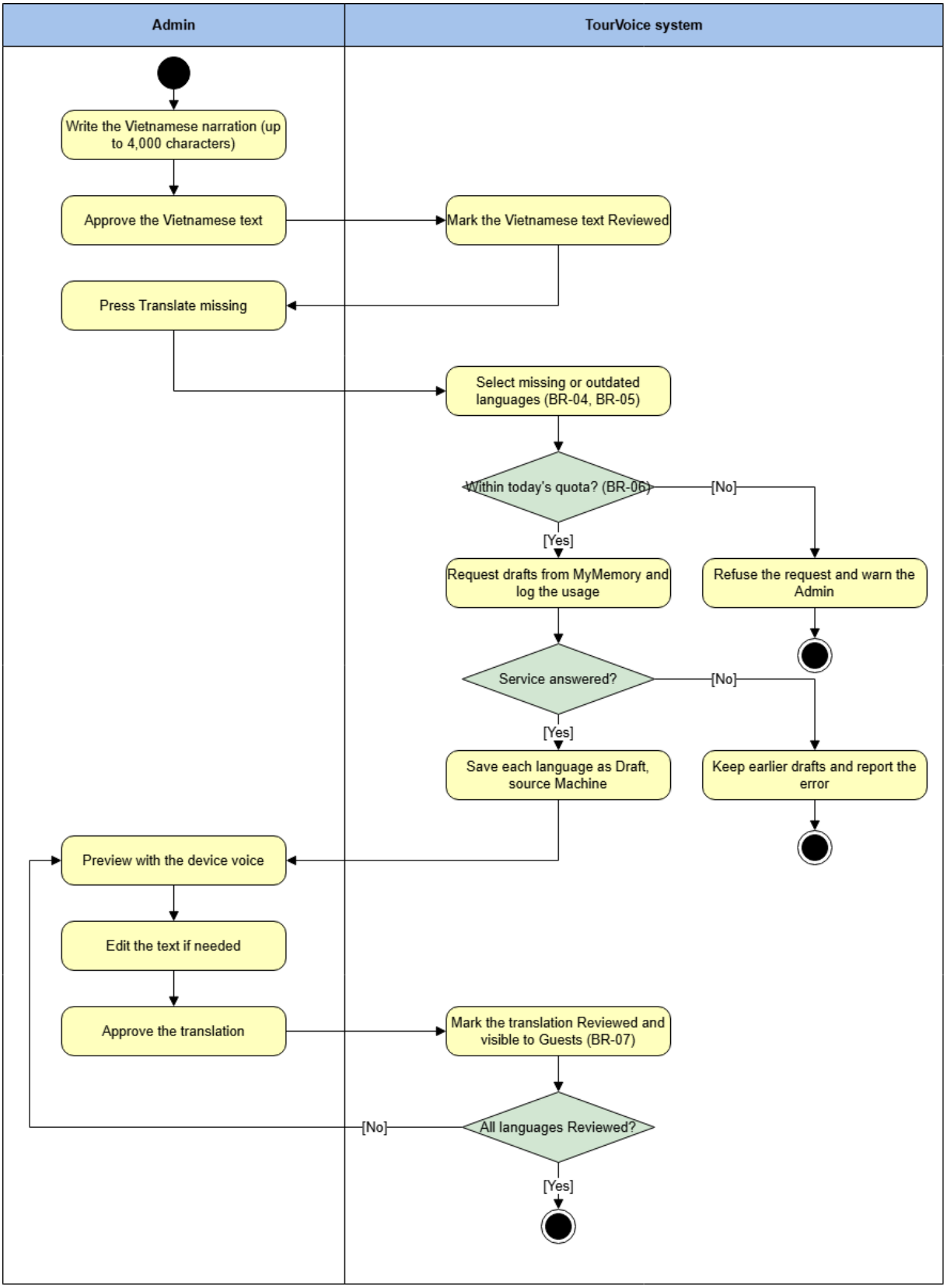



*Figure 3.8. Activity of PH2 narration production*

### *3.5.4.7. UC13 Monitor the translation quota*

| ID and Name | UC13: Monitor the translation quota |
|---|---|
| **Primary Actor** | Admin |
| **Secondary Actors** | N/A |
| **Description** | This feature allows the Admin to see how much of the daily translation quota is used and to plan translation work. |
| **Trigger** | The Admin opens the Overview or the Quotas screen. |
| **Pre-conditions** | **PRE-1.** The Admin is signed in (UC22). |
| **Post-conditions** | **POST-1.** The Admin sees the characters sent today against the 45,000 guard and the history of requests.<br><br>**POST-2.** Nothing is changed by this use case. |
| **Normal Flow** | 1. The Admin opens the Overview screen (SCR11).<br><br>2. The system shows a quota card with the characters sent today and the characters left before 45,000 (BR-06).<br><br>3. The Admin opens the Quotas screen (SCR17). |

| | |
|---|---|
| | 4. The system shows the translation usage log by day and by request, with the place, the languages, the characters and the outcome. |
| **Alternative Flows** | 1a. The Admin opens the Quotas screen directly.<br>1a1. Continue at step 4.<br>2a. The quota is nearly used.<br>2a1. The Admin postpones UC10 to the next day. The use case ends. |
| **Exceptions** | UC13.E1. No request has been sent today.<br>1a. The card shows 0 characters used – the use case continues at UC13.3. |
| **Priority** | Medium |
| **Frequency of Use** | Low |
| **Business Rules** | BR-06 |
| **Other Information** | The figures come from the same translation usage log that BR-06 checks, so what the Admin sees is what the guard applies (O2). |
| **Assumptions** | The day is counted by the server's local date. |

*Table 3.24. Specification of UC13 Monitor the translation quota*

#### *3.5.4.8. UC14 Manage languages*

| ID and Name | UC14: Manage languages |
|---|---|
| **Primary Actor** | Admin |
| **Secondary Actors** | N/A |
| **Description** | This feature allows the Admin to enable or disable a language for visitors and to see how complete each language is. |
| **Trigger** | The Admin opens the Languages screen. |
| **Pre-conditions** | **PRE-1.** The Admin is signed in (UC22). |
| **Post-conditions** | **POST-1.** The language is enabled or disabled, and the language menu of the visitor pages (UC07) follows.<br>**POST-2.** If the change is refused, the languages are unchanged. |
| **Normal Flow** | 1. The Admin opens the Languages screen (SCR15).<br>2. The system lists the five languages with code, name, state and the number of places with a Reviewed narration.<br>3. The Admin switches a language on or off (see UC14.E1). |

| | |
|---|---|
| | 4. The system asks for confirmation.<br><br>5. The Admin confirms and the system saves the change. |
| **Alternative Flows** | 5a. The Admin cancels.<br><br>5a1. Nothing is changed. The use case ends. |
| **Exceptions** | UC14.E1. The Admin tries to disable Vietnamese.<br><br>1a. The system shows "Vietnamese is the source and fallback language and cannot be disabled." (BR-01) – the use case stops. |
| **Priority** | Low |
| **Frequency of Use** | Low |
| **Business Rules** | BR-01 |
| **Other Information** | Disabling a language hides it from visitors but keeps its translations, so enabling it again restores them. |
| **Assumptions** | The set of five languages is fixed in this version; adding a new language is out of scope. |

*Table 3.25. Specification of UC14 Manage languages*

#### *3.5.4.9. UC20 Monitor live usage*

| ID and Name | UC20: Monitor live usage |
|---|---|
| **Primary Actor** | Admin |
| **Secondary Actors** | N/A (the counts come from Guest devices through UC19) |
| **Description** | This feature allows the Admin to see how many devices have a TourVoice page open now, by place and by entry (QR code or web). |
| **Trigger** | The Admin opens the Overview screen of the dashboard. |
| **Pre-conditions** | **PRE-1.** The Admin is signed in (UC22). |
| **Post-conditions** | **POST-1**. The count equals the number of distinct devices with a page open, within the stated delays; no device identifier reaches the dashboard (BR-14).<br><br>**POST-2.** If the live connection is lost, the last count stays visible with a warning. |

| | |
|---|---|
| **Normal Flow** | 1. The Admin opens the Overview screen (SCR11).<br><br>2. The system shows the current number of open devices, in total, by place and by entry (see UC20.E1).<br><br>3. When a device opens a page (UC19), the system adds it and pushes the new count within 5 s (BR-17).<br><br>4. While the page stays open, its heartbeat every 15 s keeps the device counted (BR-15).<br><br>5. When the page closes, the leave signal removes the device and the new count is pushed. |
| **Alternative Flows** | 3a. The device returns within 30 minutes.<br><br>3a1. Its earlier access session is resumed instead of a new one (BR-16). Return to step 4.<br><br>3b. The same phone opens several tabs.<br><br>3b1. It is counted once, because the count is by device (BR-14). Return to step 4.<br><br>5a. The device loses its connection and sends no leave signal.<br><br>5a1. It is removed when its 45 s time-to-live expires, found by a sweep every 5 s, and the new count is pushed. |
| **Exceptions** | UC20.E1. The live connection between the dashboard and the server is lost. |

| | |
|---|---|
| | 1a. The system shows “Live updates paused. Reconnecting…”, reconnects and reloads the current count – the use case continues at UC20.2. |
| **Priority** | High |
| **Frequency of Use** | Medium |
| **Business Rules** | BR-14, BR-15, BR-16, BR-17 |
| **Other Information** | Objective O5: the count changes within 5 s on open and close, and within 60 s on loss of connection. The flows are drawn in Figure 3.9. |
| **Assumptions** | The server and the dashboard keep a live connection while the Overview is open. |

*Table 3.26. Specification of UC20 Monitor live usage*

Figure 3.9 draws UC19 and UC20 together, because neither makes sense alone. The point to notice is the fork after the first push: from then on, the Guest's heartbeats and the Admin's view run in parallel, and they meet again only when the device leaves. The slowest path, a device that disappears without a leave signal, takes at most 45 + 5 = 50 seconds, inside the 60-second target of O5.

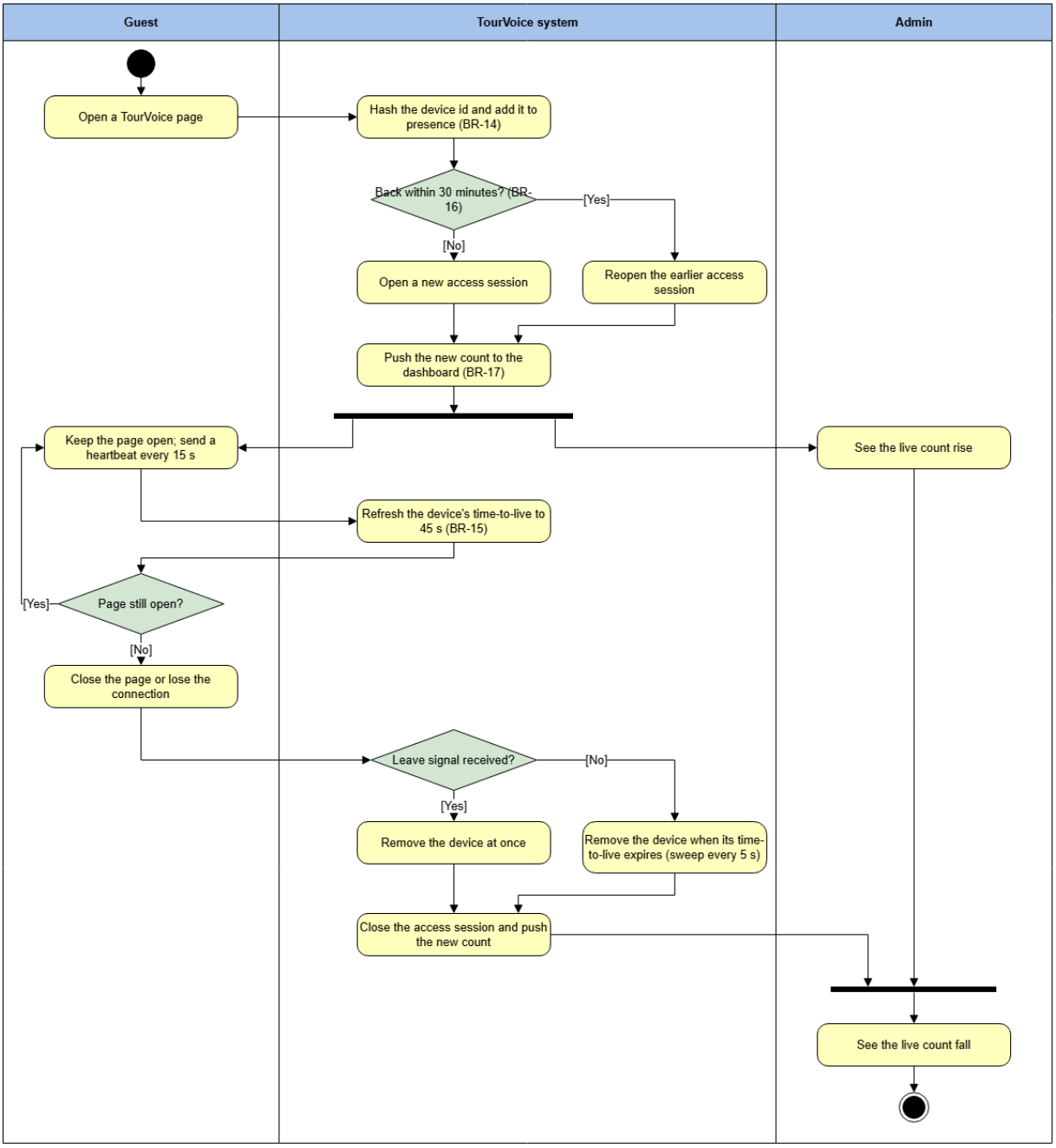



*Figure 3.9. Activity of PH4 live usage count*

### *3.5.4.10. UC21 View access statistics*

| **ID and Name** | **UC21: View access statistics** |
|---|---|
| **Primary Actor** | Admin |

| | |
|---|---|
| **Secondary Actors** | N/A |
| **Description** | This feature allows the Admin to see how TourVoice has been used over a period: visits, entries by QR code and listens by place and language. |
| **Trigger** | The Admin opens the statistics section of the Overview screen. |
| **Pre-conditions** | **PRE-1**. The Admin is signed in (UC22). |
| **Post-conditions** | **POST-1.** The Admin sees aggregated figures for the chosen period.<br>**POST-2.** Nothing is changed by this use case. |
| **Normal Flow** | 1. The Admin opens the Overview screen (SCR11) and scrolls to the statistics.<br>2. The Admin chooses a period.<br>3. The system shows the number of access sessions, the share of QR and web entries, and the listens by place and by language (see UC21.E1).<br>4. The Admin selects a place to see its own figures. |
| **Alternative Flows** | 2a. The Admin keeps the default period.<br>2a1. Continue at step 3. |

| | |
|---|---|
| **Exceptions** | UC21.E1. There is no data for the period.<br>1a. The system shows “No data for this period.” – the use case continues at UC21.2. |
| **Priority** | Medium |
| **Frequency of Use** | Low |
| **Business Rules** | BR-09, BR-14 |
| **Other Information** | Only aggregated counts are shown; no device identifier, IP address or position is displayed or stored (BR-14, NFR-05). |
| **Assumptions** | N/A |

*Table 3.27. Specification of UC21 View access statistics*

#### *3.5.4.11. UC23 Manage users*

| **ID and Name** | **UC23: Manage users** |
|---|---|
| **Primary Actor** | Admin |
| **Secondary Actors** | N/A |

| | |
|---|---|
| **Description** | This feature allows the Admin to see the accounts, create another Admin account, and lock or unlock an account. |
| **Trigger** | The Admin opens the Users screen. |
| **Pre-conditions** | **PRE-1.** The Admin is signed in (UC22). |
| **Post-conditions** | **POST-1.** The account is created, locked or unlocked; a locked account can no longer sign in (BR-11).<br><br>**POST-2.** If a check fails, the accounts are unchanged. |
| **Normal Flow** | 1. The Admin opens the Users screen (SCR16).<br><br>2. The system lists the accounts with email, role, state and creation date, and can search by email.<br><br>3. The Admin selects an account and taps “Lock” or “Unlock” (see UC23.E1).<br><br>4. The system asks for confirmation.<br><br>5. The Admin confirms and the system saves the new state. |

| | |
|---|---|
| **Alternative Flows** | 3a. The Admin taps “New Admin”.<br><br>3a1. The Admin enters an email and a first password (see UC23.E2, UC23.E3).<br><br>3a2. The system stores the account with the role Admin and a hash of the password (BR-12). The use case ends.<br><br>5a. The Admin cancels.<br><br>5a1. Nothing is changed. The use case ends. |
| **Exceptions** | UC23.E1. The account is the last active Admin.<br><br>1a. The system shows “At least one active Admin must remain.” (BR-11) – the use case stops.<br><br>UC23.E2. The email is already registered.<br><br>1a. The system shows “This email is already registered.” – the use case continues at UC23.3a1.<br><br>UC23.E3. The password has fewer than 8 characters.<br><br>1a. The system shows “Password must have at least 8 characters.” (BR-12) – the use case continues at UC23.3a1. |
| **Priority** | Medium |
| **Frequency of Use** | Low |

| | |
|---|---|
| **Business Rules** | BR-11, BR-12 |
| **Other Information** | Accounts are locked rather than deleted, so the history and statistics that refer to them stay consistent. |
| **Assumptions** | The first Admin account is created when the system is installed. |

*Table 3.28. Specification of UC23 Manage users*

### 3.6. Interface and Layout Analysis

The screens are derived from the use cases, not designed first. A screen is needed where an actor must see or enter something to reach a goal, and two use cases share a screen when they act on the same object at the same moment. Table 3.29 lists the resulting screens, numbered 01 to 09 for visitor pages and 10 to 17 for the dashboard; these are the codes used in the normal flows of Section 3.5. UC19 Report presence has no screen of its own, because every visitor page performs it.

| Code | Screen | Use cases served |
|---|---|---|
| SCR01 | Home | UC01, UC07 |
| SCR02 | Destination detail | UC03, UC04, UC08, UC16, UC18 |
| SCR03 | Food spot detail | UC04, UC08, UC16 |
| SCR04 | Map | UC04 |

| | | |
|---|---|---|
| SCR05 | Sign in and register | UC15, UC22 |
| SCR06 | Account | UC16, UC17, UC18 |
| SCR07 | Explore | UC01, UC03 |
| SCR08 | Food and Culture | UC03 |
| SCR09 | QR entry (redirects to the place) | UC02 |
| SCR10 | Dashboard sign in | UC22 |
| SCR11 | Overview | UC13, UC20, UC21 |
| SCR12 | Places | UC05 |
| SCR13 | Place editor (General and QR, Location, Narration, Food spots) | UC05, UC06, UC09 to UC12 |
| SCR14 | Narration matrix | UC10, UC11 |
| SCR15 | Languages | UC14 |
| SCR16 | Users | UC23 |
| SCR17 | Quotas | UC13 |

*Table 3.29. Screens and the use cases they serve*

The tap count of objective O3 can already be checked on this list. From the home page a Guest taps a destination card to open SCR02 and then taps Listen, which is two taps. From a QR code the Guest arrives on SCR02 through SCR09 and needs one tap. Both paths are within the limit of three.

Two screens carry the key use cases and are analysed at the level of layout. Figure 3.10 shows the destination page on a phone. The point to notice is the vertical order of regions 3, 4 and 5: the Listen button is visible without scrolling, the player sits directly under it, and the narration text follows the player. This is the order of steps 4 to 7 of UC08 in Table 3.11, and it keeps the fallback notice of BR-01 next to the content it qualifies.

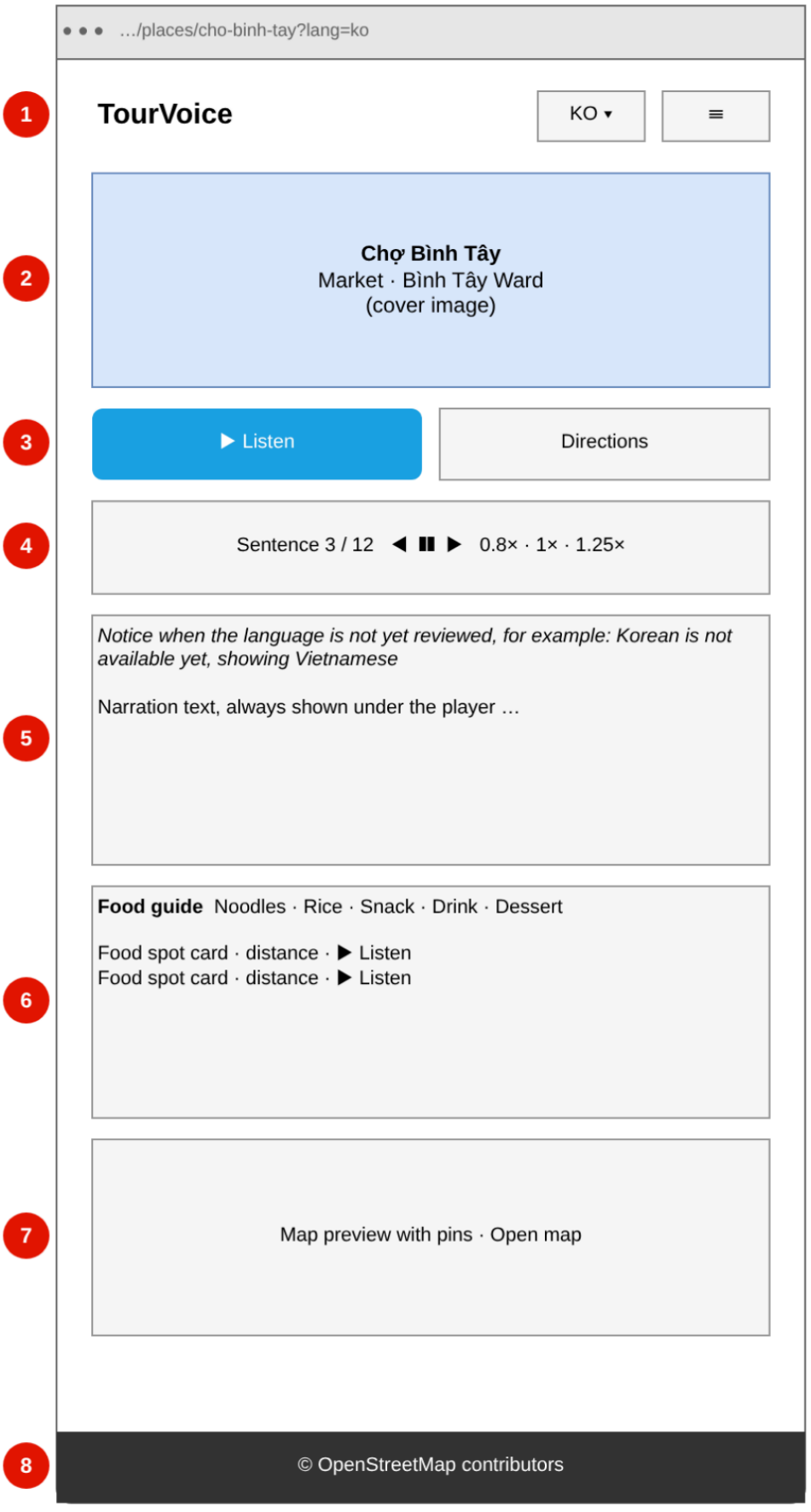

.../places/cho-binh-tay?lang=ko
1
TourVoice
KO ▾
≡
2
Chợ Bình Tây
Market · Bình Tây Ward
(cover image)
3
▶ Listen
Directions
4
Sentence 3 / 12 ◀ ▮▮ ▶ 0.8× · 1× · 1.25×
5
Notice when the language is not yet reviewed, for example: Korean is not available yet, showing Vietnamese
Narration text, always shown under the player …
6
Food guide Noodles · Rice · Snack · Drink · Dessert
Food spot card · distance · ▶ Listen
Food spot card · distance · ▶ Listen
7
Map preview with pins · Open map
8
© OpenStreetMap contributors

*Figure 3.10. Layout of the destination page on a phone (SCR02)*

Figure 3.11 shows the narration tab of the place editor. The point to notice is region 4, where all five languages are listed with their status and source. The Admin sees at once which languages are missing, which are Draft and which have an outdated source, so the rules BR-04 to BR-07 become visible states rather than error messages. The actions in region 6 appear i

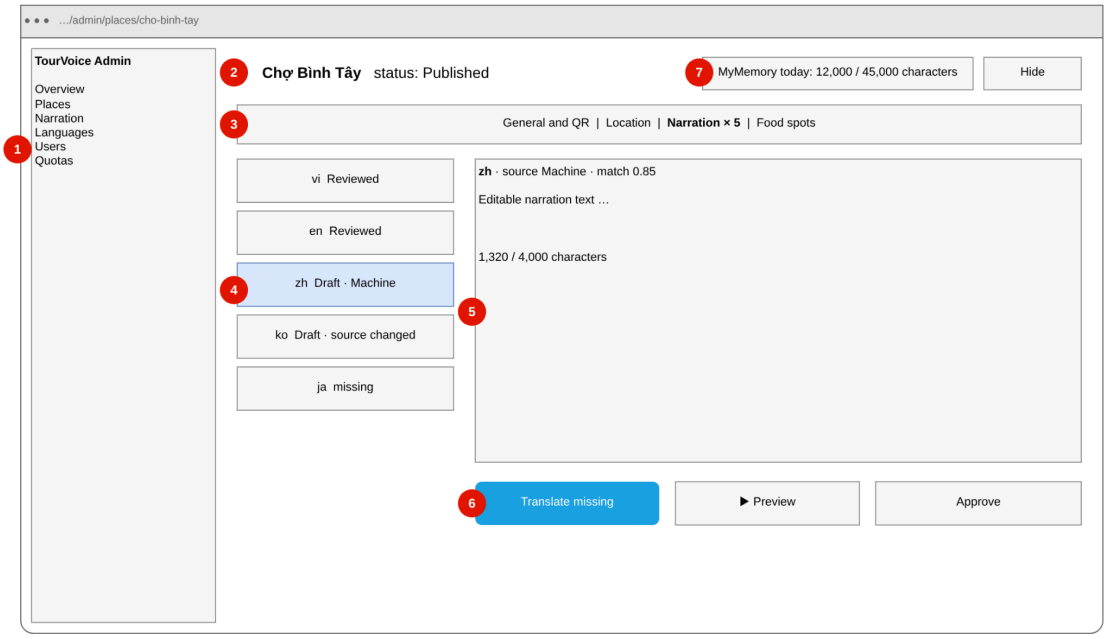



*Figure 3.11. Layout of the narration tab of the place editor (SCR13)*

The layouts fix the position and purpose of each region but not its detailed controls, colours or validation, which belong to the interface design in Section 4.4.

# CHAPTER 4: SYSTEM DESIGN

Chapter 3 described what TourVoice must do: three actors, ten business concepts, eighteen business rules and twenty-three use cases, without deciding how any of them is built. This chapter takes those results as its input and turns each one into a design decision, in the order announced in Section 1.6. Section 4.1 turns the business concepts into a database. Section 4.2 places the data, the logic and the three external services in an architecture and explains why each technology was chosen over its alternatives. Section 4.3 divides the logic into classes and checks the division against the use cases with sequence and state diagrams, grouped by actor as in Section 3.5. Section 4.4 turns the screen list of Section 3.6 into detailed screens. Section 4.5 closes the chapter by tracing every objective of Chapter 1 to the decisions that serve it, so that Chapter 5 knows what to measure.

Two principles guide every decision below. First, a decision must answer a requirement that can be named by its code (FR, BR, NFR, UCnn or On); a structure that answers none is left out. Second, where a requirement depends on a service outside TourVoice, the design states what happens when that service is missing, because Chapter 2 showed that a visitor standing at a destination cannot wait for a fix.

## 4.1. Database Design

The database is designed in three steps, each more concrete than the one before. The conceptual model decides which entities exist and how they relate, without columns. The physical model gives every entity its columns, types and keys for MySQL 8.4. The table structures then state, column by column, what each value means and which rule it serves. Working in this order keeps the early decisions, which are the hardest to change later, free of storage details.

Figure 4.1 is derived from the conceptual model of Figure 3.2: each business concept becomes an entity, and each line between concepts becomes a relationship with its cardinality. The point to notice is DEVICE_PRESENCE, drawn as a dashed white box. It is the only concept of Chapter 3 that is not stored. A presence lives for at most 45 seconds after its last heartbeat (BR-15) and changes every 15 seconds, so

writing it to the database would turn every open page into a stream of writes, while no question asked by the Admin needs its history. What the Admin does need, how many visits there were and where they started, is kept by ACCESS_SESSION, which is written only when a visit starts, resumes or ends. This split answers the question left open in Chapter 3: presence is fast and temporary, access sessions are slow and permanent, and each is stored where its lifetime fits.

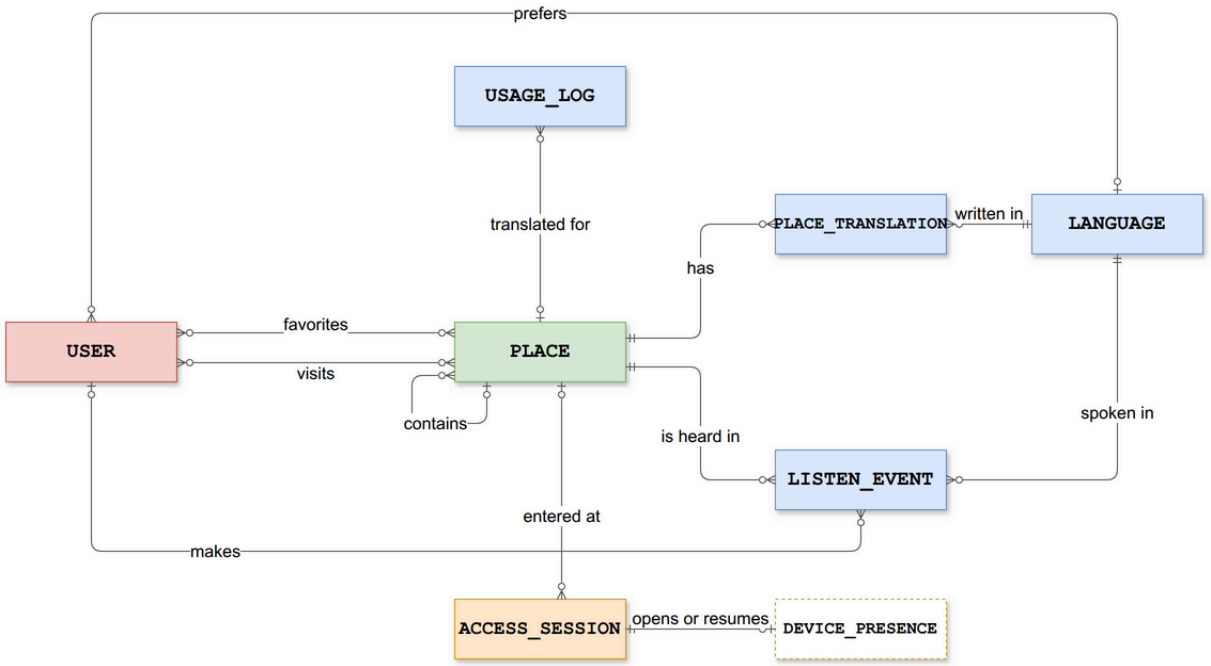



*Figure 4.1. Conceptual entity-relationship diagram of TourVoice*

Five further decisions shape the model.

- **One entity for destinations and food spots.** A food spot is a PLACE whose parent is a destination (the relationship *contains*). Both kinds share coordinates, a cover image, a status, translations and a QR code, so one entity avoids duplicating the whole narration pipeline. BR-02 is then a check on the parent, not a second table.
- **Translations as a separate entity with a composite key.** Each (place, language) pair has exactly one PLACE_TRANSLATION, with its own status and source. Adding a sixth language adds rows, not columns, and a missing

language is simply a missing row, which the narration matrix of SCR14 can show directly.

- **Languages as data.** LANGUAGE holds the code sent to MyMemory and the code used to pick a device voice. The codes are not the same as the site code (Chinese is zh on the site but zh-CN for both services, and a voice code carries a region such as vi-VN), so keeping them in a table instead of in code lets UC14 switch a language off without a release.
- **Usage logs as the memory of the quota guard.** USAGE_LOG records every call to MyMemory with its character count. BR-06 is then a sum over today's rows, so the guard survives a restart of the API, which a counter in memory would not.
- **Favorites and visits as relationships, not entities.** Both are many-to-many relations between USER and PLACE with a single date. They become two small tables in the physical model; their composite keys make a double favorite impossible without any code.

### *4.1.1. List of tables*

Table 4.1 lists the nine tables that result, grouped by subsystem with the colours of Section 3.2. The concept column links each table back to Figure 3.2.

| No | Table | Subsystem | Purpose | Concept in Figure 3.2 |
|---|---|---|---|---|
| 1 | Places | PH1 | Destinations and food spots, their location, status and order | Place |
| 2 | PlaceTranslations | PH2 | One narration per place and language, with review status and source | Place translation |
| 3 | Languages | PH2 | The five languages and their translation and speech codes | Language |

| | | | | |
|---|---|---|---|---|
| 4 | ListenEvents | PH2 | Listens longer than 10 s, anonymous or linked to a Customer | Listen event |
| 5 | UsageLogs | PH2 | Every call to MyMemory, the basis of the quota guard | Translation usage log |
| 6 | Favorites | PH3 | Places saved by a Customer | Favorite |
| 7 | Visits | PH3 | Destinations marked as visited by a Customer | Visit |
| 8 | AccessSessions | PH4 | Visits of a device, with entry and duration | Access session |
| 9 | Users | PH5 | Customer and Admin accounts | User |

*Table 4.1. List of tables*

Figures 4.2 and 4.3 give the physical model, split in two so that each fits a page. Figure 4.2 holds the content that visitors read; Figure 4.3 holds what is created by people using the system. The point to notice in Figure 4.2 is that UsageLogs refers to a place and a language only optionally: a failed or rejected call must still be logged, even when it cannot be tied to one translation.

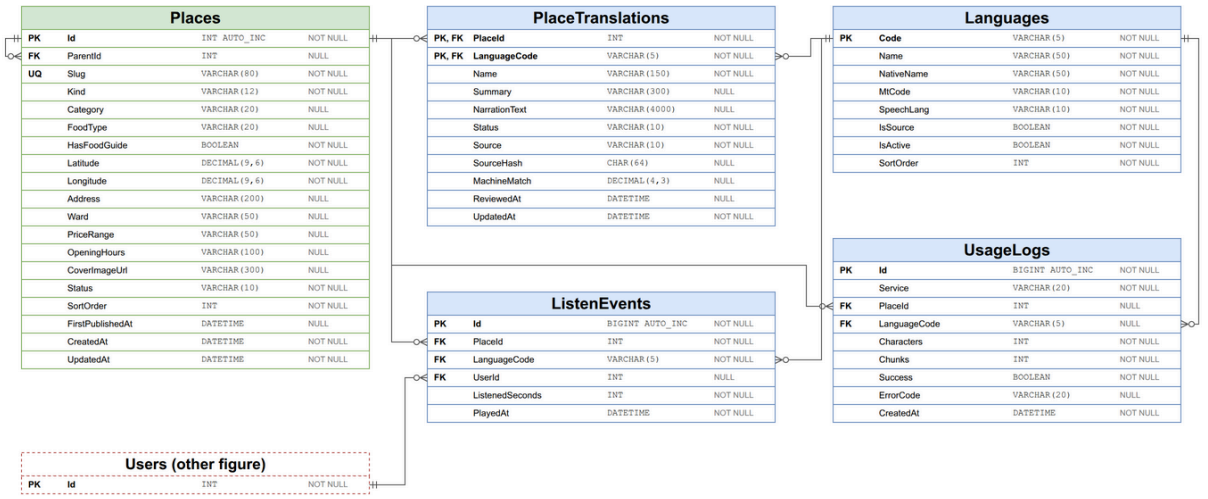



*Figure 4.2. Physical data model of places and narration (PH1, PH2)*

In Figure 4.3 the point to notice is the dashed box on the right. The live presence is shown next to AccessSessions because it opens, resumes and ends sessions, but it is not a table: the arrow is a runtime dependency, not a foreign key.

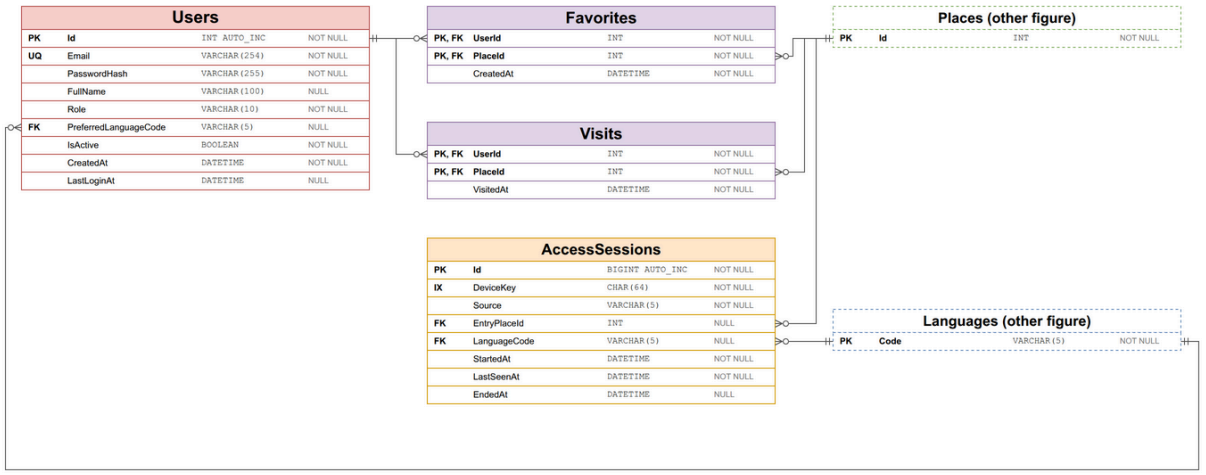



*Figure 4.3. Physical data model of accounts, presence and access (PH3, PH4, PH5)*

Three conventions apply to every table. All times are stored in UTC and converted to Asia/Ho_Chi_Minh only on screen, so the daily quota and the daily statistics use one clock. Every status is stored as text equal to the state name of Section 4.3.4 and limited by a CHECK constraint, so a state diagram and a column can be compared word for word. All text columns use utf8mb4, because the narrations contain Vietnamese, Chinese, Korean and Japanese characters.

### *4.1.2. Structure of table [ Table name ]*

Tables 4.2 to 4.10 follow the same format: the column, its type, its meaning, its key role, whether it may be empty and whether it must be unique.

| Field name | Data Type | Description | PK/FK | Not Null | Unique |
|---|---|---|---|---|---|
| Id | INT AUTO_INCREMENT | Identifier of the place | PK | Yes | Yes |

| | | | | | |
|---|---|---|---|---|---|
| ParentId | INT | Destination that contains this food spot; empty for a destination (BR-02) | FK → Places.Id | No | No |
| Slug | VARCHAR(80) | Address used in /places/{slug} and in the QR code; locked after the first publication (BR-08, BR-18) | | Yes | Yes |
| Kind | VARCHAR(12) | Destination or FoodSpot | | Yes | No |
| Category | VARCHAR(20) | Heritage, Market, FoodStreet or Park, for destinations | | No | No |
| FoodType | VARCHAR(20) | Noodles, Rice, Snack, Drink or Dessert, for food spots | | No | No |
| HasFoodGuide | BOOLEAN | Whether the destination shows a food guide (BR-02) | | Yes | No |
| Latitude, Longitude | DECIMAL(9,6) | Position, checked against the service area (BR-03) | | Yes | No |
| Address, Ward | VARCHAR(200), VARCHAR(50) | Postal address and new ward name | | No | No |
| PriceRange, OpeningHours | VARCHAR(50), VARCHAR(100) | Practical information shown to visitors | | No | No |
| CoverImageUrl | VARCHAR(300) | Path of the cover image in the uploads volume; required to publish (BR-08) | | No | No |
| Status | VARCHAR(10) | Draft, Published or Hidden (Figure 4.15) | | Yes | No |
| SortOrder | INT | Order of the place in lists | | Yes | No |

| | | | | | |
|---|---|---|---|---|---|
| FirstPublishedAt | DATETIME | Time of the first publication; once set, the slug cannot change | | No | No |
| CreatedAt, UpdatedAt | DATETIME | Audit times in UTC | | Yes | No |

*Table 4.2. Structure of table Places*

| **Field name** | **Data Type** | **Description** | **PK/FK** | **Not Null** | **Unique** |
|---|---|---|---|---|---|
| PlaceId | INT | Place being narrated | PK, FK → Places.Id | Yes | Yes (with LanguageCode) |
| LanguageCode | VARCHAR(5) | Language of this narration | PK, FK → Languages.Code | Yes | Yes (with PlaceId) |
| Name | VARCHAR(150) | Name of the place in this language | | Yes | No |
| Summary | VARCHAR(300) | One-line description shown on cards | | No | No |
| NarrationText | VARCHAR(4000) | Text read aloud by the device voice; at most 4,000 characters (BR-04) | | No | No |
| Status | VARCHAR(10) | Draft or Reviewed (Figure 4.16); only Reviewed is shown to visitors (BR-07) | | Yes | No |
| Source | VARCHAR(10) | Machine (from MyMemory) or Manual (written or edited by the Admin) | | Yes | No |

| | | | | | |
|---|---|---|---|---|---|
| SourceHash | CHAR(64) | SHA-256 of the Reviewed Vietnamese text this translation was made from; a different hash marks it outdated (BR-05) | | No | No |
| MachineMatch | DECIMAL(4,3) | Match score returned by MyMemory, from 0 to 1 | | No | No |
| ReviewedAt | DATETIME | Time of the last approval | | No | No |
| UpdatedAt | DATETIME | Time of the last change | | Yes | No |

*Table 4.3. Structure of table PlaceTranslations*

| Field name | Data Type | Description | PK/FK | Not Null | Unique |
|---|---|---|---|---|---|
| Code | VARCHAR(5) | Code used by the site: vi, en, zh, ko, ja | PK | Yes | Yes |
| Name, NativeName | VARCHAR(50) | Name in English and in the language itself, for the selector | | Yes | No |
| MtCode | VARCHAR(10) | Code sent to MyMemory (vi, en, zh-CN, ko, ja) | | Yes | No |
| SpeechLang | VARCHAR(10) | BCP-47 code used to choose a device voice (vi-VN, en-US, zh-CN, ko-KR, ja-JP) | | Yes | No |
| IsSource | BOOLEAN | True only for Vietnamese, the language every translation starts from | | Yes | No |
| IsActive | BOOLEAN | Whether the language is offered to visitors (UC14) | | Yes | No |
| SortOrder | INT | Order in the language selector | | Yes | No |

*Table 4.4. Structure of table Languages*

| Field name | Data Type | Description | PK/FK | Not Null | Unique |
|---|---|---|---|---|---|
| Id | BIGINT AUTO_INCREMENT | Identifier of the listen | PK | Yes | Yes |
| PlaceId | INT | Place listened to | FK → Places.Id | Yes | No |
| LanguageCode | VARCHAR(5) | Language of the narration actually read | FK → Languages.Code | Yes | No |
| UserId | INT | Customer who listened; empty for a Guest | FK → Users.Id | No | No |
| ListenedSeconds | INT | Seconds of speech, at least 10 (BR-09) | | Yes | No |
| PlayedAt | DATETIME | Time of the listen in UTC | | Yes | No |

*Table 4.5. Structure of table ListenEvents*

| Field name | Data Type | Description | PK/FK | Not Null | Unique |
|---|---|---|---|---|---|
| Id | BIGINT AUTO_INCREMENT | Identifier of the call | PK | Yes | Yes |
| Service | VARCHAR(20) | External service called; MyMemory today | | Yes | No |
| PlaceId | INT | Place being translated, when known | FK → Places.Id | No | No |
| LanguageCode | VARCHAR(5) | Target language, when known | FK → Langua | No | No |

| | | | | | |
|---|---|---|---|---|---|
| | | | ges.Code | | |
| Characters | INT | Characters sent; summed per UTC day for the quota guard (BR-06) | | Yes | No |
| Chunks | INT | Number of requests of at most 500 bytes | | Yes | No |
| Success | BOOLEAN | Whether the service returned a translation | | Yes | No |
| ErrorCode | VARCHAR(20) | Status returned on failure, or QuotaGuard when the call was refused locally | | No | No |
| CreatedAt | DATETIME | Time of the call in UTC | | Yes | No |

*Table 4.6. Structure of table UsageLogs*

| Field name | Data Type | Description | PK/FK | Not Null | Unique |
|---|---|---|---|---|---|
| UserId | INT | Customer who saved the place | PK, FK → Users.Id | Yes | Yes (with PlaceId) |
| PlaceId | INT | Saved place | PK, FK → Places.Id | Yes | Yes (with UserId) |
| CreatedAt | DATETIME | Time it was saved | | Yes | No |

*Table 4.7. Structure of table Favorites*

| Field name | Data Type | Description | PK/FK | Not Null | Unique |
|---|---|---|---|---|---|
| UserId | INT | Customer who visited | PK, FK → Users.Id | Yes | Yes (with PlaceId) |

| | | | | | |
|---|---|---|---|---|---|
| PlaceId | INT | Visited destination; progress counts these rows against published destinations (BR-10) | PK, FK → Places.Id | Yes | Yes (with UserId) |
| VisitedAt | DATETIME | Time it was marked | | Yes | No |

*Table 4.8. Structure of table Visits*

| Field name | Data Type | Description | PK/FK | Not Null | Unique |
|---|---|---|---|---|---|
| Id | BIGINT AUTO_INCREMENT | Identifier of the session | PK | Yes | Yes |
| DeviceKey | CHAR(64) | SHA-256 of the device identifier (BR-14); indexed to find the last session of a device | | Yes | No |
| Source | VARCHAR(5) | Entry: QR (through /q/{slug}) or Web | | Yes | No |
| EntryPlaceId | INT | First place opened, when the entry was a place page | FK → Places.Id | No | No |
| LanguageCode | VARCHAR(5) | Language in use at the start | FK → Languages.Code | No | No |
| StartedAt | DATETIME | Start of the session | | Yes | No |
| LastSeenAt | DATETIME | Last heartbeat written at the end or on resume; used to apply the 30-minute rule (BR-16) | | Yes | No |

| | | | | | |
|---|---|---|---|---|---|
| EndedAt | DATETIME | End of the session; empty while the device is present | | No | No |

*Table 4.9. Structure of table AccessSessions*

| Field name | Data Type | Description | PK/FK | Not Null | Unique |
|---|---|---|---|---|---|
| Id | INT AUTO_INCREMENT | Identifier of the user | PK | Yes | Yes |
| Email | VARCHAR(254) | Sign-in name, stored in lower case | | Yes | Yes |
| PasswordHash | VARCHAR(255) | Salted hash produced by PasswordHasher; the password itself is never stored (BR-12) | | Yes | No |
| FullName | VARCHAR(100) | Name shown in the account and the dashboard; optional at registration | | No | No |
| Role | VARCHAR(10) | Customer or Admin | | Yes | No |
| PreferredLanguageCode | VARCHAR(5) | Language chosen by the Customer | FK → Languages.Code | No | No |
| IsActive | BOOLEAN | False for a locked account, which cannot sign in (BR-11) | | Yes | No |

| | | | | | |
|---|---|---|---|---|---|
| CreatedAt, LastLoginAt | DATETIME | Registration and last sign-in times | | Yes, No | No |

*Table 4.10. Structure of table Users*

Two rules could not be expressed as constraints and are enforced by the business layer instead: BR-11, which requires at least one active Admin, and BR-13, which allows only a Draft place without food spots to be deleted. Both depend on other rows, which a CHECK constraint cannot see. Section 4.3 shows where they are applied.

## 4.2. System Architecture Design

The architecture answers two questions that the database cannot: where each part of TourVoice runs, and which part may depend on which. This section first justifies the technologies against their alternatives, then places them on machines, then divides the server into layers, and finally designs the two boundaries that Chapter 3 marked as risky: the external services and the programming interface between the browser and the server.

### *4.2.1. Technology decisions*

Each choice in Table 4.11 was made against at least one realistic alternative, and the reason is stated as the requirement it serves. Most rows are decided by objective O2, zero external cost, which rules out every paid service and pushes work to the visitor's own device wherever the device can do it.

| Concern | Choice | Alternatives considered | Reason for the choice | Objective |
|---|---|---|---|---|
| Narration voice | Device voice through the Web Speech API | Recorded audio; cloud text-to-speech | No audio is produced, stored or downloaded, so speech costs nothing and starts without a transfer; a missing voice falls back to text (Table 4.12) | O2, O3 |

| | | | | |
|---|---|---|---|---|
| Translation | MyMemory free API, then review by the Admin | Manual translation only; paid machine translation | The free 50,000 characters per day cover the 47,200 characters of the content in two days [4]; the review step of Section 2.1.3 keeps the quality | O1, O2 |
| Server | ASP.NET Core on .NET 10, three layers | Node.js with Express; one project without layers | Strong typing and separate projects make the layer rule checkable by a test on every pull request | O6 |
| Database | MySQL 8.4 in Docker, EF Core migrations | MongoDB; SQL Server | The data is relational, with composite keys and constraints (Section 4.1); MySQL is free and runs in the same compose file as the API | O2, O6 |
| Live count | SignalR push and an in-memory presence store | Polling the server every few seconds; Redis | A push reaches the dashboard within 5 s without extra load; one API container keeps memory consistent, and the store sits behind an interface so Redis can replace it later | O4, O5 |
| Visitor and dashboard pages | React, Vite and TypeScript, served by the Cloudflare Tunnel | Server-rendered pages; a native app | A page opened from a QR code needs no installation, and the static bundle is served from a free CDN close to the visitor | O3, O4 |
| Map and directions | Leaflet with OpenStreetMap tiles; Google Maps URLs for directions | Google Maps JavaScript API | No key and no billing; walking directions are delegated to an app the visitor already has | O2 |
| Sign-in | JWT signed with HS256, passwords hashed by PasswordHasher | Server sessions with cookies; an external identity provider | One stateless endpoint serves both roles, and the API keeps no session state that would need sharing | O4 |

| | | | | |
|---|---|---|---|---|
| Delivery | GitHub Actions and the GitHub container registry | Manual build and copy | Every change is tested and deployed the same way, and a failed health check returns to the previous image | O6 |

*Table 4.11. Technology decisions and rejected alternatives*

### *4.2.2. Deployment architecture*

Figure 4.4 places these technologies on four kinds of node: the visitor's phone, the Admin's laptop, a static host and a single Docker host, with three external systems around them. The point to notice is what does not cross the network. Speech is produced inside the visitor's browser, so the server never sends audio, and the only large transfer, the page bundle, comes from the CDN rather than from the Docker host. The second point is that only the data access layer has arrows to MySQL, the uploads volume and MyMemory: every external dependency of the server leaves through one layer.

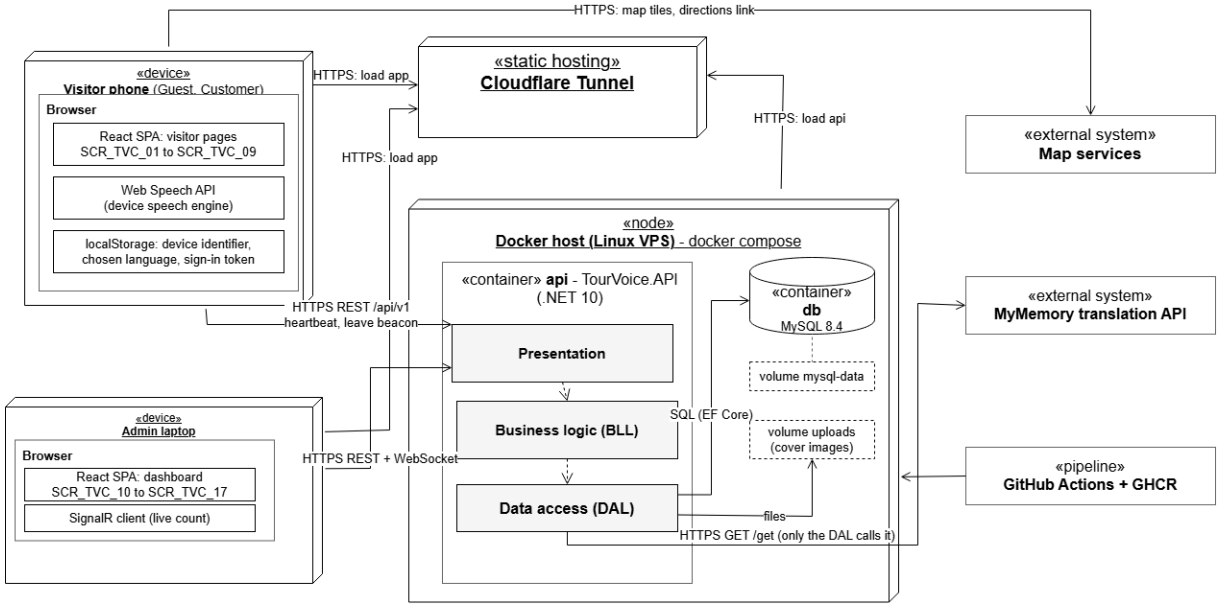



*Figure 4.4. Deployment architecture of TourVoice*

The API runs as exactly one container. This is a deliberate limit, not an oversight: the live count is held in the memory of that container, and two containers would each count only their own devices. The limit is acceptable because the target

load of O4, about fifty devices, is far below what one container serves. The way out is already prepared: the presence store is reached through the interface IPresenceStore, so moving it to Redis and adding a SignalR backplane changes the data access layer only.

### *4.2.3. Three-layer architecture*

Inside the API, the code is divided into three projects, shown in Figure 4.5. Each layer may use only the layer directly below it: Presentation calls the business layer through service interfaces and data transfer objects, and the business layer calls the data access layer through repository interfaces and entities. The point to notice is the red dashed arrow on the left. A direct reference from Presentation to the data access layer would let a controller read the database without passing through the business rules, so the rule "BR-01 to BR-18 live only in the BLL" would hold only by habit. To make it hold by construction, the start-up code of the API registers only the business layer, which registers the data access layer itself, and an architecture test with NetArchTest fails the pipeline on every pull request that adds a forbidden reference (O6).

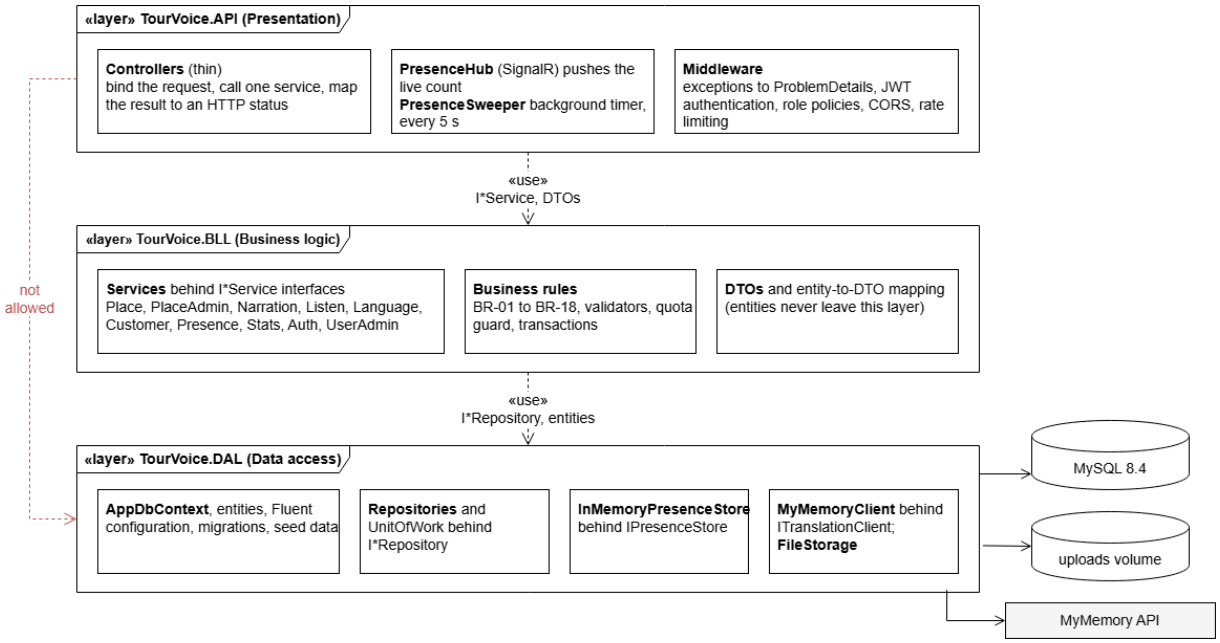



Figure 4.5. Three-layer architecture of the TourVoice API

One dependency needs to point upward: the business layer decides when the live count changes, but the SignalR hub that pushes it lives in Presentation. The design inverts

this dependency. The business layer declares an interface IPresenceNotifier, and Presentation implements it with SignalRPresenceNotifier. The business layer therefore knows that a count must be announced, but not how, and the architecture test still sees no reference from a lower layer to a higher one.

#### *4.2.4. Fallback for each external service*

Section 3.1 named three indirect actors and promised a fallback for each. Table 4.12 designs them. The common principle is that a failure of a service must never leave a visitor with an empty screen; at worst the visitor reads instead of listening, or sees an address instead of a map.

| External actor | Used in | Failure | How it is detected | Designed response | Rules |
|---|---|---|---|---|---|
| MyMemory translation API | UC10 | Service down, error status, or daily limit reached | Response status other than 200; local sum of today's characters before each call | The call is refused with 429 before 45,000 characters; on a failed call the API returns 502 and languages already saved stay Draft. Visitors are not affected, because they see only Reviewed text, and the Admin can still write a Manual translation | BR-06, BR-07 |
| Map services (OpenStreet Map tiles, Google Maps URLs) | UC04 | Tiles do not load; directions cannot open | Tile load error in the browser | Place lists, addresses and distances do not depend on tiles and stay visible; Directions is a plain link with the stored coordinates, so it works with any maps application | BR-03 |

| | | | | | |
|---|---|---|---|---|---|
| Device speech engine (Web Speech API) | UC08, UC12 | No speech support, or no voice for the language | speechSynthesis missing, or no voice matching SpeechLang | The narration text is always shown under the player, so the page is complete without sound; the player shows a hint to install the voice. Speech starts inside the tap handler so that iOS allows it | BR-01, NFR-06 |

*Table 4.12. Fallback for each external service*

The same principle covers the language itself. When the requested language has no Reviewed translation, the API returns the Vietnamese text with a flag, and the page shows the notice of BR-01 instead of an empty narration (Figure 4.20).

### *4.2.5. Programming interface*

The browser and the API exchange JSON over REST under the prefix /api/v1, plus one SignalR hub for the live count. Endpoints are grouped by the actor that may call them, following the grouping of the use cases in Section 3.5, so that one authorisation policy covers a whole group. Table 4.13 shows representative endpoints of each group.

| Group | Actor | Authorisation | Representative endpoints | Use cases |
|---|---|---|---|---|
| Public | Guest (and Customer) | None; at most 120 requests per minute per device and 1,200 per IP address | GET /languages, GET /places?lang=, GET /places/{slug}?lang=, GET /places/{slug}/foods, GET /explore, GET /map/markers, POST /listen-events, POST /presence/heartbeat, POST /presence/leave | UC01 to UC04, UC07, UC08, UC19 |
| Account | Guest | None, with a stricter limit of 10 attempts per minute per IP address | POST /auth/register, POST /auth/login | UC15, UC22 |

| | | | | |
|---|---|---|---|---|
| Customer | Customer | JWT with role Customer | GET /me, GET /me/favorites, GET /me/history, GET /me/visits, PUT and DELETE /me/favorites/{placeId} and /me/visits/{placeId} | UC16 to UC18 |
| Admin | Admin | JWT with role Admin | /admin/dashboard, /admin/presence, /admin/stats/access, /admin/places (create, update, publish, hide, cover), /admin/places/{id}/translations (save, translate, approve), /admin/narrations/matrix, /admin/languages, /admin/users, /admin/usage; SignalR hub /hubs/presence | UC05, UC06, UC09 to UC14, UC20, UC21, UC23 |

*Table 4.13. Endpoint groups of the API by actor*

Two decisions keep this interface small. The QR code has no endpoint of its own: the page /q/{slug} of the frontend reads the place and redirects, and the image of the code is generated in the Admin's browser, which also satisfies BR-18 because the code can only contain the address. The detail of a place carries isFavorite and isVisited when the request comes from a signed-in Customer, so SCR02 needs one request instead of three.

Errors follow one format. A middleware in Presentation turns every exception into a ProblemDetails response, and Table 4.14 fixes which status each kind of failure receives, so that the screens of Section 4.4 can react to the status alone.

| Status | Meaning | Example |
|---|---|---|
| 400 Bad Request | Input is malformed or out of range | Coordinates outside the service area (BR-03); password shorter than 8 characters (BR-12) |
| 401 Unauthorized | No valid sign-in | Expired token on /me/favorites; wrong email or password |

| | | |
|---|---|---|
| 403 Forbidden | Signed in, but the role is not allowed | A Customer token on /admin/places; a locked account (BR-11) |
| 404 Not Found | The resource does not exist or is not visible | Slug of a Hidden place opened from an old QR code |
| 409 Conflict | The request clashes with existing data | Email already registered (UC15.E2); slug already used |
| 422 Unprocessable Entity | A business rule forbids the action now | Publish without a Reviewed Vietnamese text (BR-08); translate before the Vietnamese text is Reviewed (BR-04); approve an empty text or one longer than 4,000 characters; delete a Published place (BR-13) |
| 429 Too Many Requests | A limit has been reached | Quota guard (BR-06); request rate per device or IP address |
| 502 Bad Gateway | An external service failed | MyMemory returned an error status |

*Table 4.14. Error statuses of the API*

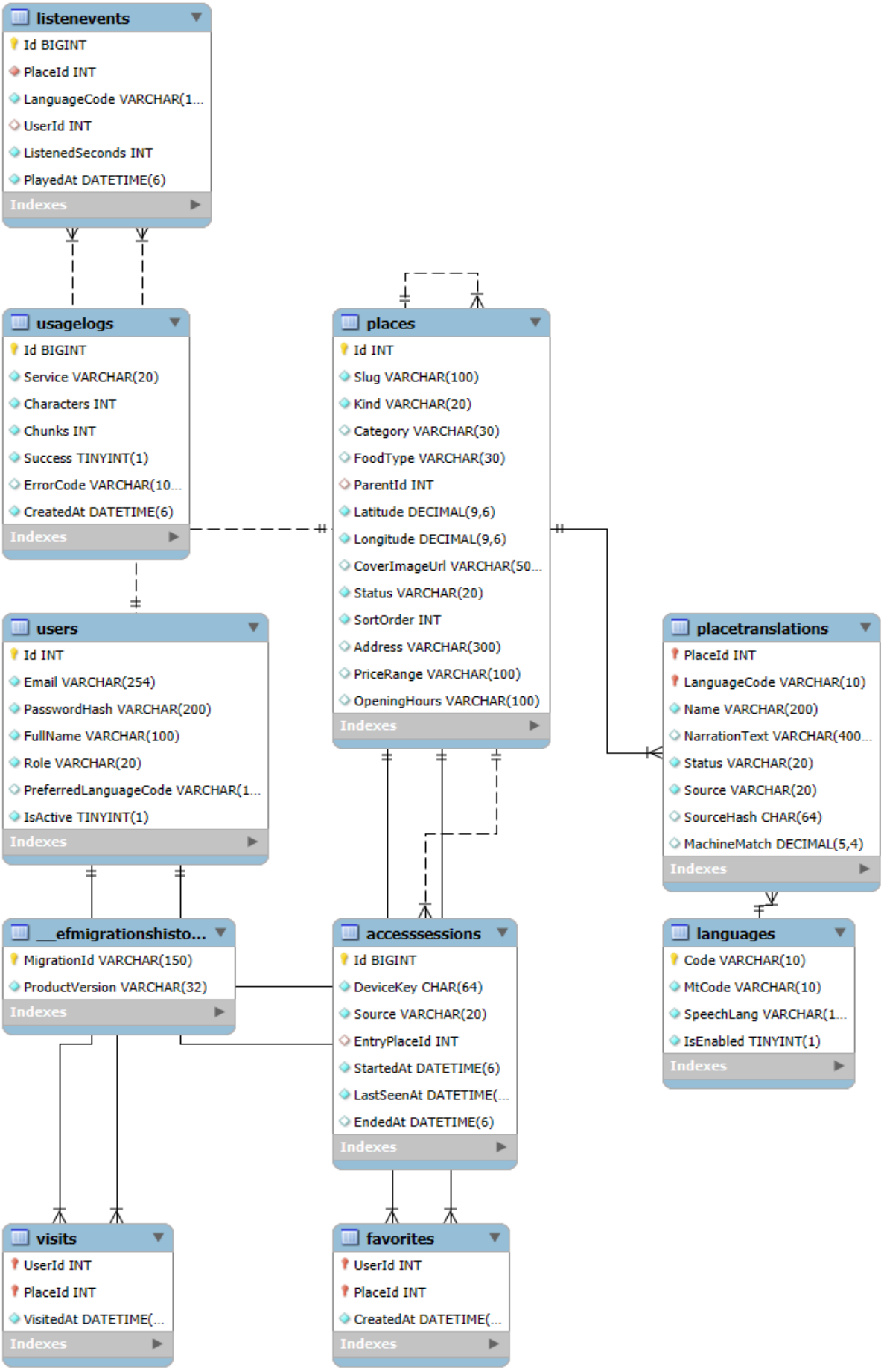
Internal details such as stack traces never appear in a response (NFR-04); they go to the server log only.

## 4.3. Class Design

The architecture fixes three layers; this section decides which classes live in them and checks the decision against the use cases. The method has three steps. The class diagram assigns every business rule to one service. The class specifications state the methods of each service. The sequence diagrams then replay the most important use cases through those methods, actor by actor, and the state machines verify that the status columns of Section 4.1 can only change in the ways the rules allow. A use case that could not be replayed with the methods of the specification would reveal a missing method; none was found for the eight flows below.

### *4.3.1. Class diagram*

Figure 4.6 shows the services of the business layer, the interfaces they depend on and the entities of the data access layer, in one band per subsystem. The point to notice is that the services are divided by subsystem and by actor at the same time. PH1 has two services, PlaceService for the Guest and PlaceAdminService for the Admin, because the two actors read the same places under different rules: a Guest sees only Published places and Reviewed text, while the Admin sees everything. Splitting them keeps each authorisation policy of Table 4.13 on one class. The second point is that every arrow is dashed and ends at an interface: a service never names a repository class, MyMemoryClient or the in-memory store, so each can be replaced by a test double, which is how the business-layer coverage of O6 is reached without a database.


listenevents
Id BIGINT
PlaceId INT
LanguageCode VARCHAR(1...
UserId INT
ListenedSeconds INT
PlayedAt DATETIME(6)
Indexes
usagelogs
Id BIGINT
Service VARCHAR(20)
Characters INT
Chunks INT
Success TINYINT(1)
ErrorCode VARCHAR(10...
CreatedAt DATETIME(6)
Indexes
places
Id INT
Slug VARCHAR(100)
Kind VARCHAR(20)
Category VARCHAR(30)
FoodType VARCHAR(30)
ParentId INT
Latitude DECIMAL(9,6)
Longitude DECIMAL(9,6)
CoverImageUrl VARCHAR(50...
Status VARCHAR(20)
SortOrder INT
Address VARCHAR(300)
PriceRange VARCHAR(100)
OpeningHours VARCHAR(100)
Indexes
users
Id INT
Email VARCHAR(254)
PasswordHash VARCHAR(200)
FullName VARCHAR(100)
Role VARCHAR(20)
PreferredLanguageCode VARCHAR(1...
IsActive TINYINT(1)
Indexes
placetranslations
PlaceId INT
LanguageCode VARCHAR(10)
Name VARCHAR(200)
NarrationText VARCHAR(400...
Status VARCHAR(20)
Source VARCHAR(20)
SourceHash CHAR(64)
MachineMatch DECIMAL(5,4)
Indexes
__efmigrationshisto...
MigrationId VARCHAR(150)
ProductVersion VARCHAR(32)
Indexes
accesssessions
Id BIGINT
DeviceKey CHAR(64)
Source VARCHAR(20)
EntryPlaceId INT
StartedAt DATETIME(6)
LastSeenAt DATETIME(...
EndedAt DATETIME(6)
Indexes
languages
Code VARCHAR(10)
MtCode VARCHAR(10)
SpeechLang VARCHAR(1...
IsEnabled TINYINT(1)
Indexes
visits
UserId INT
PlaceId INT
VisitedAt DATETIME(...
Indexes
favorites
UserId INT
PlaceId INT
CreatedAt DATETIME(...
Indexes

### *4.3.2. Class specification*

Tables 4.15 to 4.24 specify the ten services in the format of the class diagram: the method, its return type, its accessibility and what it does. Private methods do not appear in Figure 4.6; they are listed here because each one is the single place where a business rule is applied.

**PH1 Places and map**

| No | Method | Return Type | Accessibility | Description |
|---|---|---|---|---|
| 1 | GetList(lang) | PlaceCardDto[] | public | Published destinations in display order with name and summary in lang (UC01) |
| 2 | GetDetail(slug, lang, userId?) | PlaceDetailDto | public | One Published place with its narration, fallback flag and, for a Customer, isFavorite and isVisited (UC02, UC08) |
| 3 | GetFoods(slug, lang) | FoodDto[] | public | Published food spots of a destination, by food type (UC03) |
| 4 | GetMarkers() | MarkerDto[] | public | Positions and kinds of all Published places for the map (UC04) |
| 5 | PickTranslation(place, lang) | Translation | private | Returns the Reviewed translation of lang, otherwise the Vietnamese one with fallback = true (BR-01, BR-07) |

*Table 4.15. Class specification of PlaceService*

| No | Method | Return Type | Accessibility | Description |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| 1 | Create(dto) | PlaceDto | public | Creates a Draft destination or food spot after checking the parent and the area (BR-02, BR-03) (UC05) |
| 2 | Update(id, dto) | PlaceDto | public | Changes the fields of a place; refuses a new slug after the first publication (BR-08) |
| 3 | Publish(id) | PlaceDto | public | Moves the place to Published and sets FirstPublishedAt once (UC05, Figure 4.12) |
| 4 | Hide(id) | void | public | Moves the place and its food spots to Hidden |
| 5 | Delete(id) | void | public | Deletes a Draft place without food spots (BR-13) |
| 6 | SetCover(id, file) | string | public | Checks type and size (at most 2 MB) and stores the cover through IFileStorage |
| 7 | CheckPublishable(place) | string[] | private | Lists what BR-08 still misses: Reviewed Vietnamese text, valid position, cover image |

*Table 4.16. Class specification of PlaceAdminService*

**PH2 Multilingual narration**

| No | Method | Return Type | Accessibility | Description |
|---|---|---|---|---|
| 1 | SaveSource(placeId, text) | TranslationDto | public | Saves the Vietnamese text as Draft, at most 4,000 characters (UC09) |

| | | | | |
|---|---|---|---|---|
| 2 | TranslateMissing(placeId) | TranslationDto[] | public | Translates the missing and outdated languages through ITranslationClient and stores them as Draft, Machine (UC10, Figure 4.13) |
| 3 | Approve(placeId, lang, text) | TranslationDto | public | Moves a translation to Reviewed, records the source hash and whether the text was edited (UC11, Figure 4.14) |
| 4 | GetMatrix() | MatrixDto | public | Status of every place in every language for SCR14 |
| 5 | GetQuotaToday() | QuotaDto | public | Characters used today against 45,000 (UC13) |
| 6 | CheckQuota(characters) | void | private | Refuses the call when today's sum plus characters exceeds 45,000 (BR-06) |
| 7 | ComputeHash(text) | string | private | SHA-256 of the Reviewed Vietnamese text, compared with SourceHash to find outdated languages (BR-05) |

*Table 4.17. Class specification of NarrationService*

| No | Method | Return Type | Accessibility | Description |
|---|---|---|---|---|
| 1 | RecordListen(dto, userId?) | void | public | Stores a listen of at least 10 s, linked to the Customer when signed in, without IP address or location (BR-09) |

*Table 4.18. Class specification of ListenService*

| No | Method | Return Type | Accessibility | Description |
|---|---|---|---|---|
| 1 | GetActive() | LanguageDto[] | public | Active languages in selector order, with their speech codes (UC07) |
| 2 | Update(code, dto) | LanguageDto | public | Changes names, codes or activity of a language; Vietnamese cannot be switched off (UC14) |

*Table 4.19. Class specification of LanguageService*

**PH3 Customer accounts**

| No | Method | Return Type | Accessibility | Description |
|---|---|---|---|---|
| 1 | AddFavorite(userId, placeId) | void | public | Saves a Published place; a second call changes nothing (UC16) |
| 2 | RemoveFavorite(userId, placeId) | void | public | Removes a saved place (UC16) |
| 3 | MarkVisited(userId, placeId) | VisitDto | public | Marks a destination as visited and returns the new progress (UC18) |
| 4 | GetHistory(userId) | ListenDto[] | public | Listens of the Customer, newest first (UC17) |
| 5 | GetProgress(userId) | ProgressDto | public | Visited destinations divided by published destinations (BR-10) |

*Table 4.20. Class specification of CustomerService*

**PH4 Presence and statistics**

| No | Method | Return Type | Accessibility | Description |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| 1 | Beat(deviceId, slug, source) | void | public | Refreshes the presence for 45 s; for a new device, opens or resumes an access session and announces the count (UC19, Figure 4.9) |
| 2 | Leave(deviceId) | void | public | Removes the presence at once, ends the session and announces the count |
| 3 | Sweep() | void | public | Called every 5 s; removes expired presences, ends their sessions and announces the count (BR-15) |
| 4 | GetLiveCount() | int | public | Number of devices present now (UC20) |
| 5 | HashKey(deviceId) | string | private | SHA-256 of the device identifier; the identifier itself goes no further (BR-14) |

*Table 4.21. Class specification of PresenceService*

| No | Method | Return Type | Accessibility | Description |
|---|---|---|---|---|
| 1 | GetOverview() | OverviewDto | public | Figures of SCR11: listens today by language, sessions today, quota, drafts to review |
| 2 | GetAccessStats (from, to) | AccessStatsDto | public | Sessions by entry and by place, and the daily peak of devices computed from AccessSessions (UC21) |

Table 4.22. Class specification of StatsService

**PH5 Administration and access control**

| No | Method | Return Type | Accessibility | Description |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| 1 | Register(dto) | AuthResult | public | Creates a Customer with a hashed password and signs it in (UC15, Figure 4.8) |
| 2 | Login(email, password) | AuthResult | public | Verifies the password and the active flag and issues a token carrying the role (UC22, Figure 4.10) |
| 3 | IssueToken(user) | string | private | JWT signed with HS256, valid 7 days for a Customer and 8 hours for an Admin |

*Table 4.23. Class specification of AuthService*

| No | Method | Return Type | Accessibility | Description |
|---|---|---|---|---|
| 1 | List(page) | UserDto[] | public | Users, 20 per page (UC23) |
| 2 | SetActive(id, active) | void | public | Locks or unlocks an account (BR-11) |
| 3 | SetRole(id, role) | void | public | Changes the role between Customer and Admin |
| 4 | EnsureOneActiveAdmin(change) | void | private | Refuses a change that would leave no active Admin (BR-11) |

*Table 4.24. Class specification of UserAdminService*

### 4.3.3. Sequence diagrams

The sequence diagrams replay eight flows through the three layers. They were chosen because each carries at least one business rule or one external actor; a flow that only reads a list adds nothing that SD-01 does not already show. Table 4.25 maps them to the use cases, grouped by actor as in Section 3.5.

| Actor | Diagram | Use cases | What the diagram checks |
|---|---|---|---|
| Guest | SD-01 Listen to a narration | UC02, UC08 | Language fallback, device voice, listen recorded after 10 s |
| Guest | SD-07 Register an account | UC15 | Unique email, password hash |
| Guest | SD-04 Report presence and monitor live usage | UC19, UC20 | Heartbeat, 30-minute rule, push to the dashboard, sweep |
| Customer | SD-03 Sign in | UC22 | Password check, locked account, role in the token |
| Customer | SD-08 Save a favorite and mark a visit | UC16, UC18 | Sign-in required, idempotent writes, progress |
| Admin | SD-06 Publish a place and print its QR code | UC05, UC06 | Publication conditions, slug lock, QR in the browser |
| Admin | SD-02 Translate missing languages | UC10 | Review first, no repeated translation, quota guard |
| Admin | SD-05 Review and approve a translation | UC11, UC12 | Preview on the device, approval, outdated languages |

*Table 4.25. Sequence diagrams and the use cases they realise*

#### *4.3.3.1. Guest sequences*

Figure 4.7 follows a Guest from a QR scan to speech. The point to notice is the first alt fragment: the business layer, not the page, decides between the requested language and the Vietnamese fallback, so a Guest always receives text, and a page can never show a Draft by mistake. The second point is that speech involves no server at all; the API is called again only after 10 seconds of speech, to record the listen.

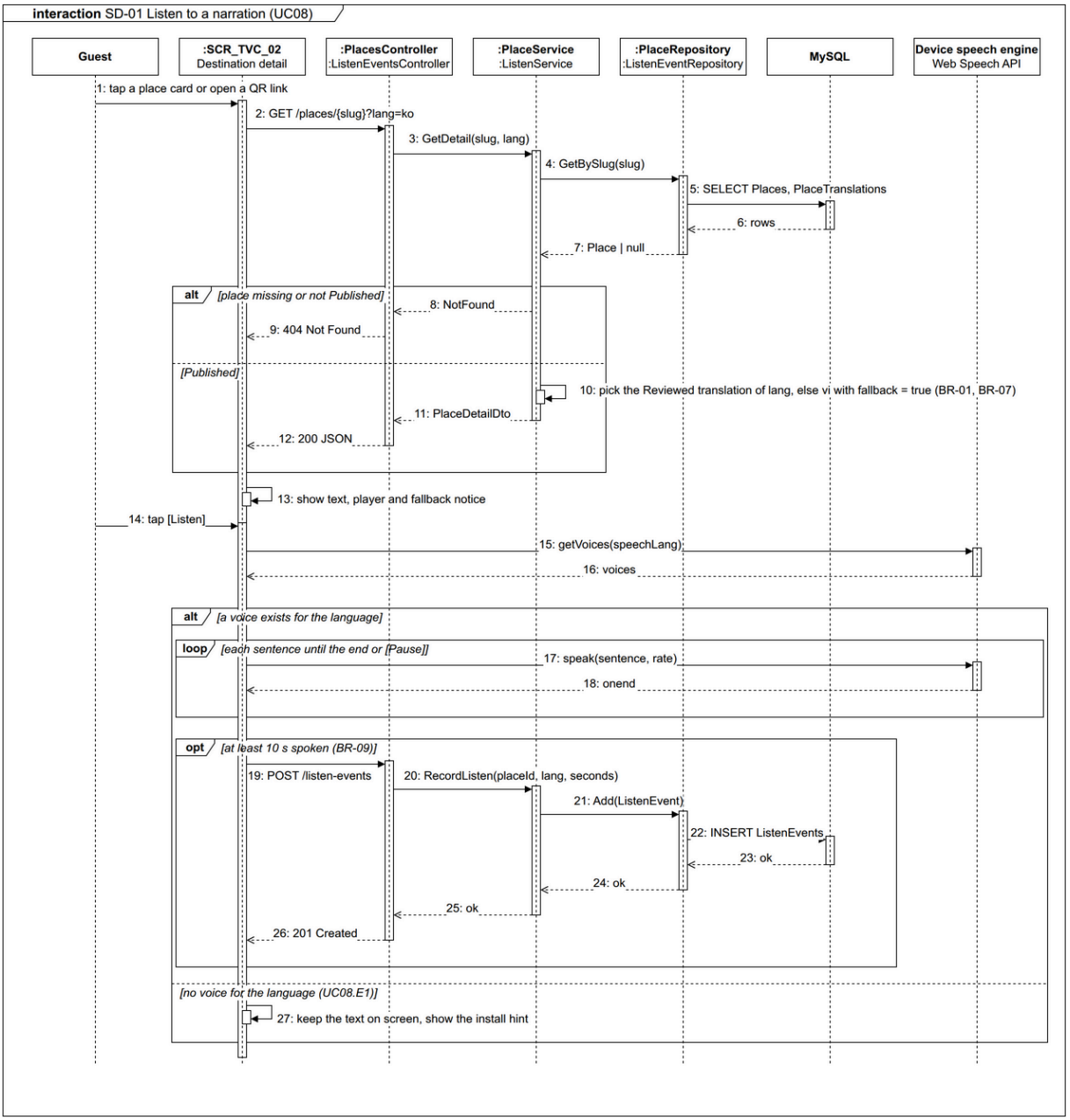



*Figure 4.7. SD-01 Listen to a narration*

Figure 4.8 shows registration. The point to notice is that the check for an existing email is made twice, once by the query and once by the unique index of Users.Email, so two registrations sent at the same moment cannot both succeed.

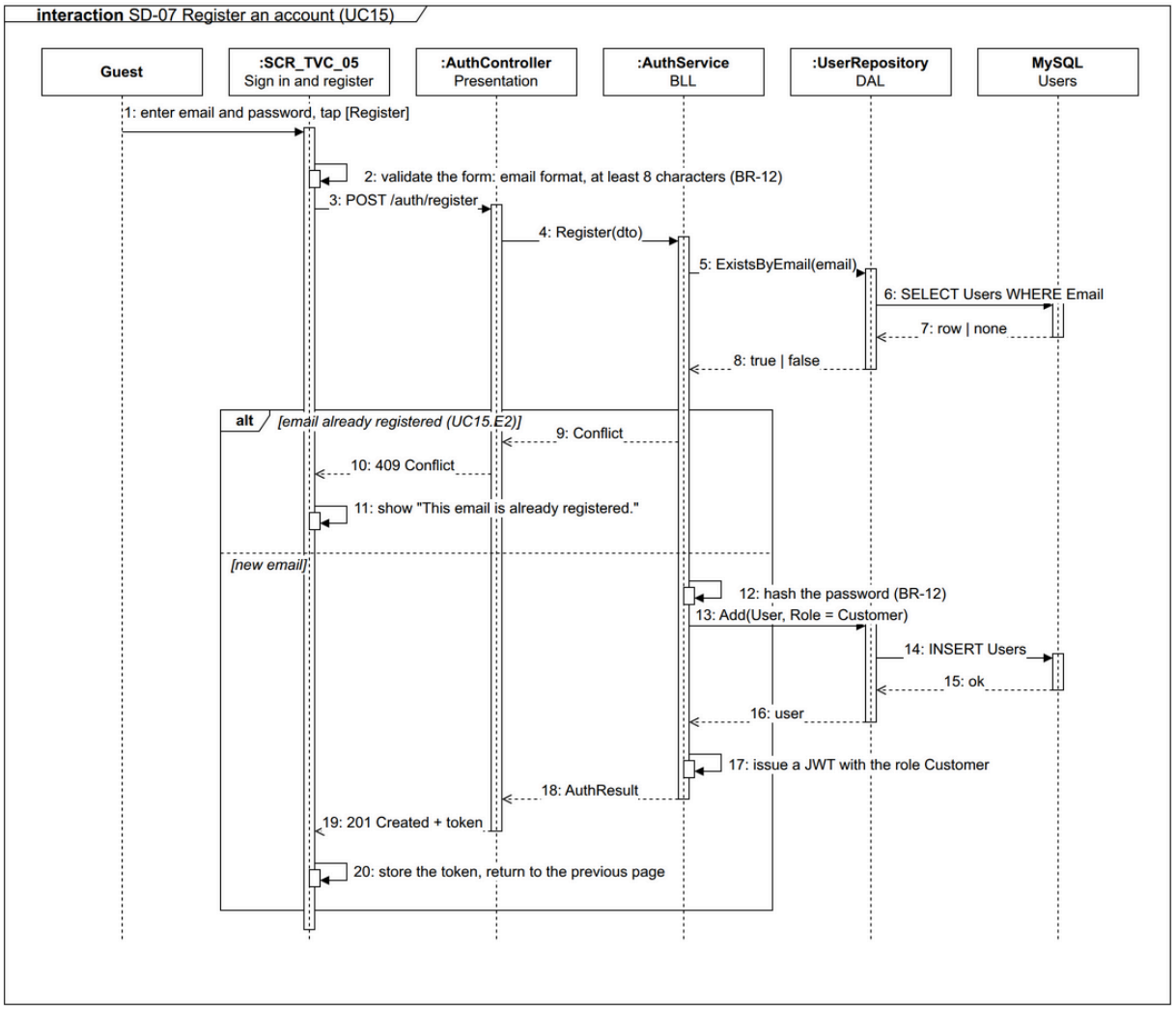



*Figure 4.8. SD-07 Register an account*

Figure 4.9 is the longest diagram, because it joins two actors: the Guest's phone reports presence and the Admin's dashboard receives the count. The point to notice is the three ways a device can leave, each ending in the same Notify call: a heartbeat that never comes back, a leave beacon, and the sweep. Notify is the inverted dependency of Section 4.2.3, which lets the business layer push without knowing SignalR.

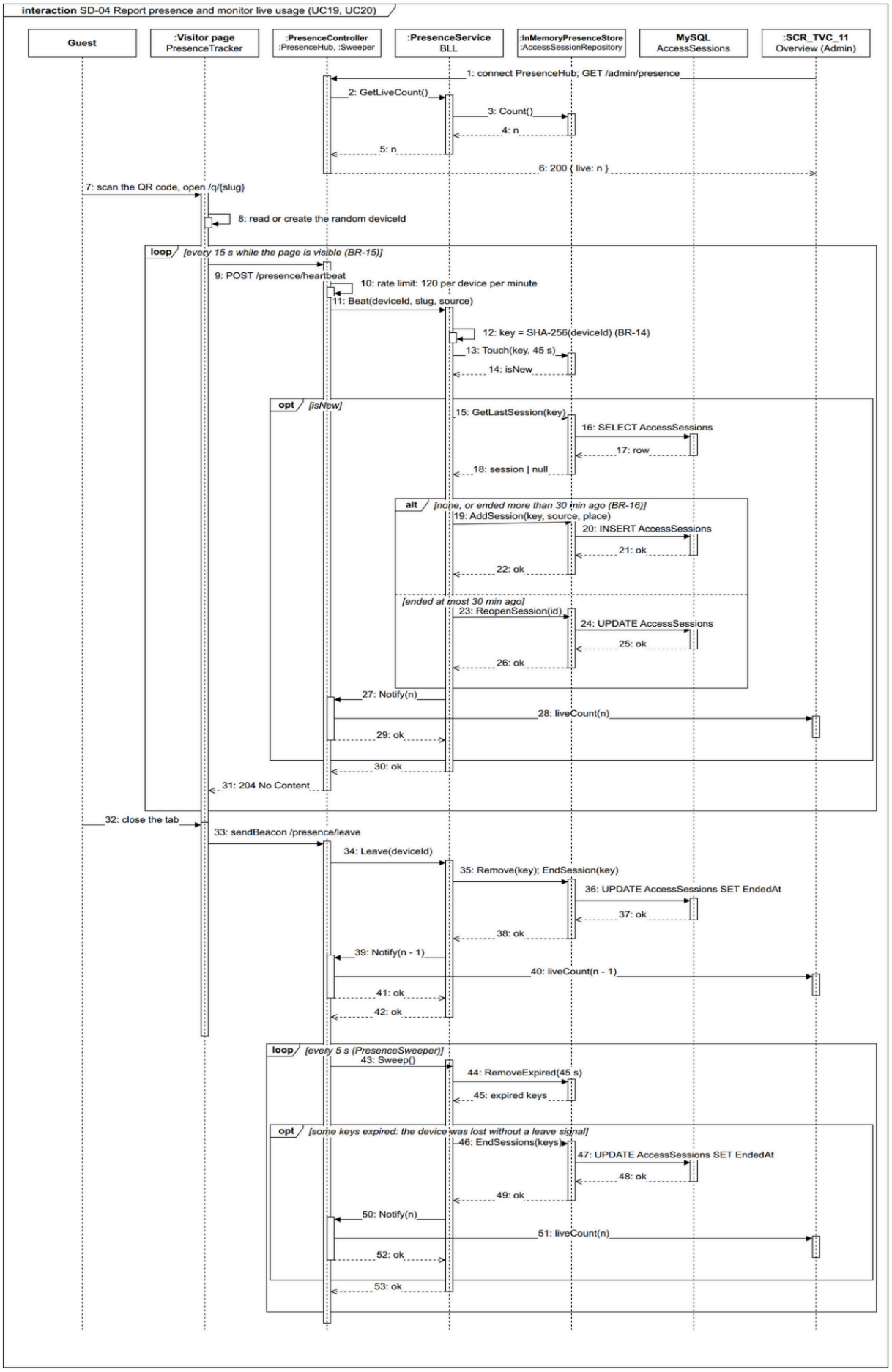

interaction SD-04 Report presence and monitor live usage (UC19, UC20)
Guest
:Visitor page
PresenceTracker
:PresenceController
:PresenceHub, :Sweeper
:PresenceService
BLL
:InMemoryPresenceStore
:AccessSessionRepository
MySQL
AccessSessions
:SCR_TVC_11
Overview (Admin)
1: connect PresenceHub; GET /admin/presence
2: GetLiveCount()
3: Count()
4: n
5: n
6: 200 { live: n }
7: scan the QR code, open /q/{slug}
8: read or create the random deviceId
loop
[every 15 s while the page is visible (BR-15)]
9: POST /presence/heartbeat
10: rate limit: 120 per device per minute
11: Beat(deviceId, slug, source)
12: key = SHA-256(deviceId) (BR-14)
13: Touch(key, 45 s)
14: isNew
opt
[isNew]
15: GetLastSession(key)
16: SELECT AccessSessions
17: row
18: session | null
alt
[none, or ended more than 30 min ago (BR-16)]
19: AddSession(key, source, place)
20: INSERT AccessSessions
21: ok
22: ok
[ended at most 30 min ago]
23: ReopenSession(id)
24: UPDATE AccessSessions
25: ok
26: ok
27: Notify(n)
28: liveCount(n)
29: ok
30: ok
31: 204 No Content
32: close the tab
33: sendBeacon /presence/leave
34: Leave(deviceId)
35: Remove(key); EndSession(key)
36: UPDATE AccessSessions SET EndedAt
37: ok
38: ok
39: Notify(n - 1)
40: liveCount(n - 1)
41: ok
42: ok
loop
[every 5 s (PresenceSweeper)]
43: Sweep()
44: RemoveExpired(45 s)
45: expired keys
opt
[some keys expired: the device was lost without a leave signal]
46: EndSessions(keys)
47: UPDATE AccessSessions SET EndedAt
48: ok
49: ok
50: Notify(n)
51: liveCount(n)
52: ok
53: ok

*Figure 4.9. SD-04 Report presence and monitor live usage*

### *4.3.3.2. Customer sequences*

Figure 4.10 shows that the Customer and the Admin share one sign-in flow. The point to notice is the last alt fragment: a Customer who signs in on the dashboard receives a valid token, but the dashboard refuses it on the screen, and every Admin endpoint refuses it again with 403. The role is checked where it is used, not only where it is issued.

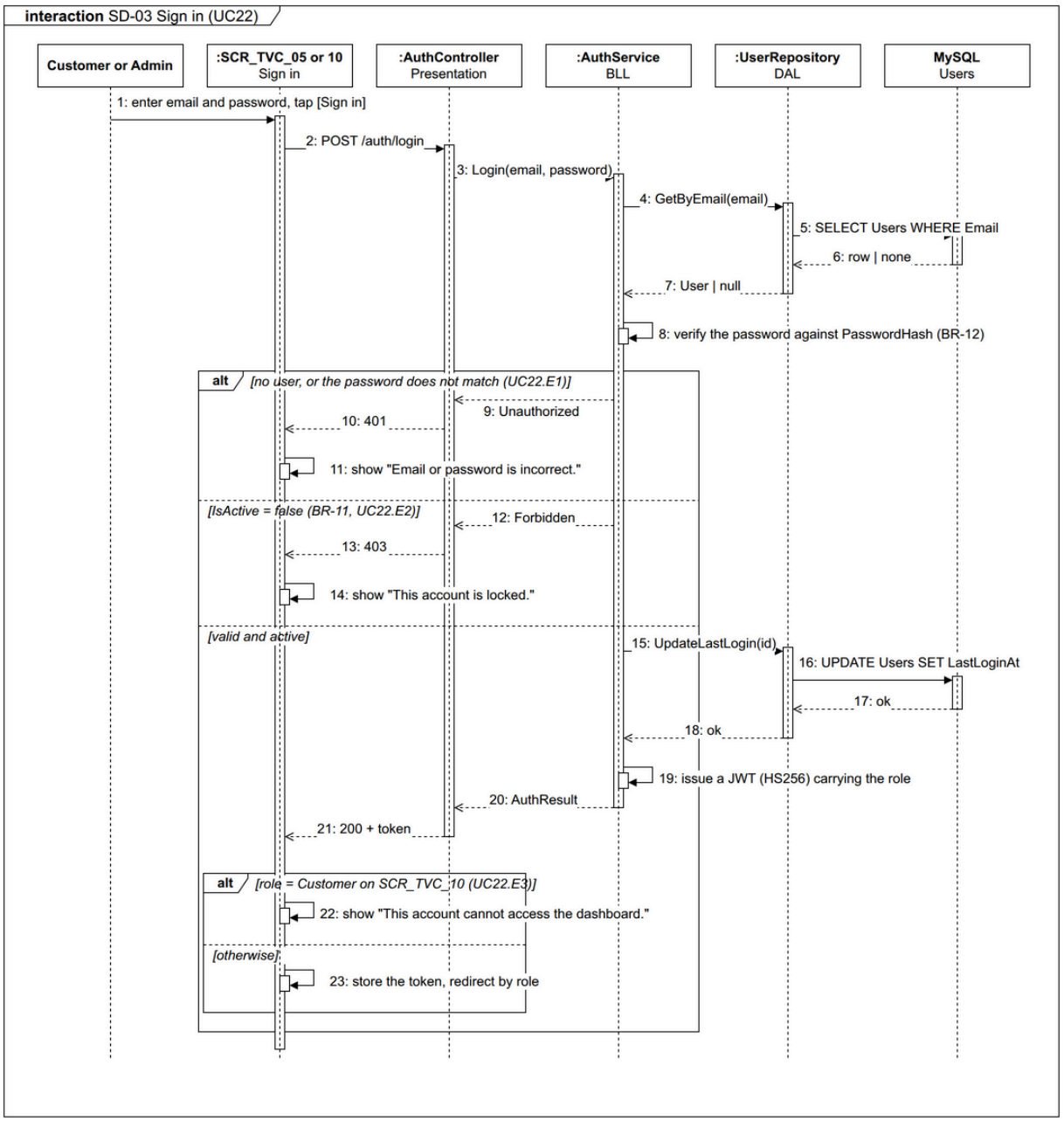


*Figure 4.10. SD-03 Sign in*

Figure 4.11 shows the two account actions of a visit. The point to notice is the first alt fragment: when the token is missing, the page opens the sign-in screen and then completes the original action, which is alternative flow 1a of UC16. Both writes are idempotent because of the composite keys of Tables 4.7 and 4.8.

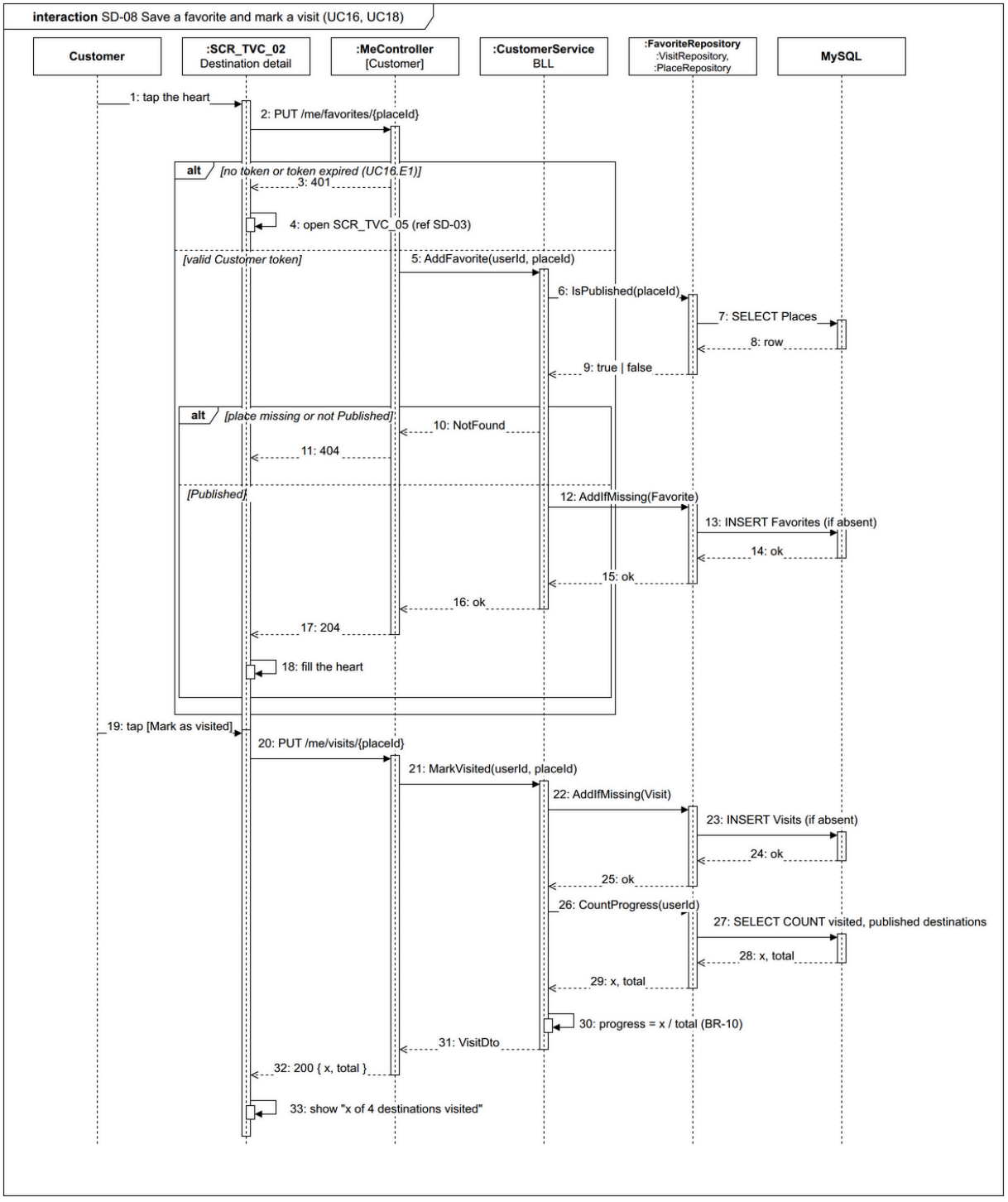


*Figure 4.11. SD-08 Save a favorite and mark a visit*

#### *4.3.3.3. Admin sequences*

Figure 4.12 shows publication. The point to notice is that the QR code is drawn after publication, in the browser, from the slug only: the server never stores an image, and the code cannot contain a language (BR-18).

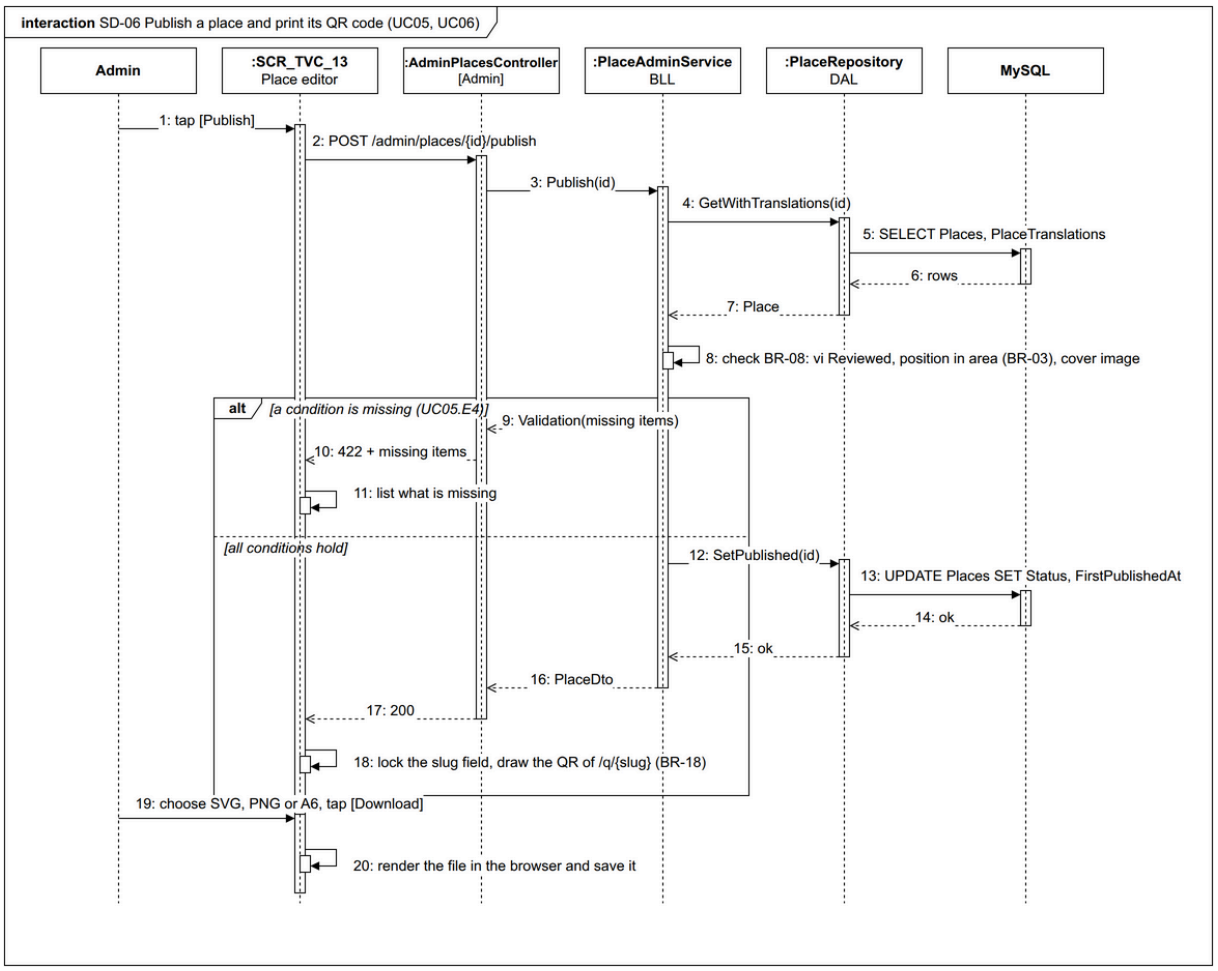



*Figure 4.12. SD-06 Publish a place and print its QR code*

Figure 4.13 is the core of the narration pipeline. The point to notice is the order of the three guards before MyMemory is called: the Vietnamese text must be Reviewed (BR-04), only missing or outdated languages are sent (BR-05), and the characters to send must fit under 45,000 for the day (BR-06). Each guard can only reduce the number of characters sent, which is how O2 is protected by design rather than by care.

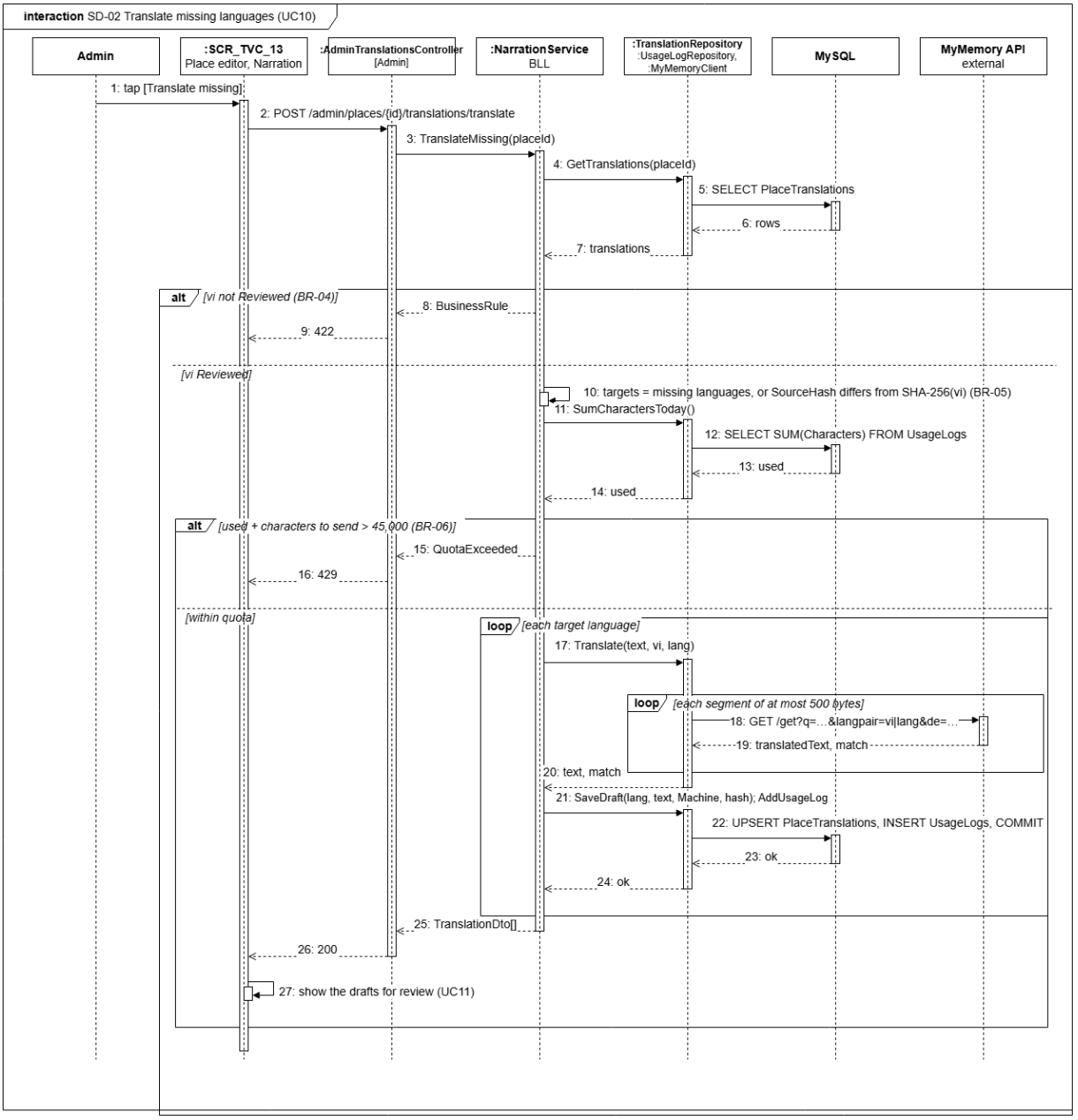



*Figure 4.13. SD-02 Translate missing languages*

Figure 4.14 shows the human half of the pipeline. The point to notice is the opt fragment at the end: approving a new Vietnamese text changes the hash that every other language was made from, so all of them become outdated at once, without any extra column.

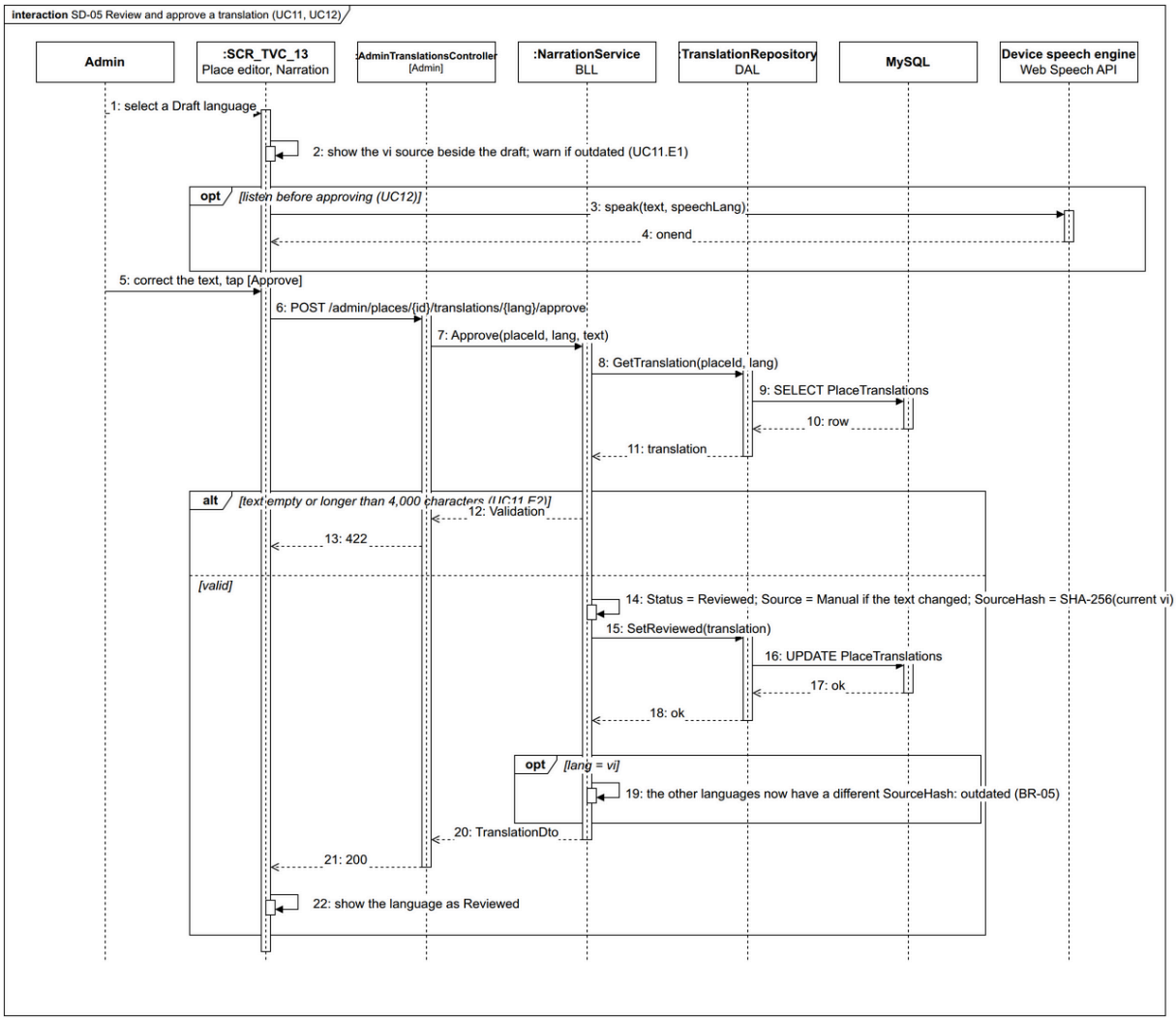



*Figure 4.14. SD-05 Review and approve a translation*

### *4.3.4. State machines*

Three objects of TourVoice have a life cycle that the rules constrain: a place, a translation and the presence of a device. Their state machines are drawn below, and their state names are the values stored in the database (Section 4.1).

Figure 4.15 shows a place. The point to notice is what is missing: there is no transition back to Draft, and only a Draft can be deleted. Together with the slug lock, this guarantees that a QR code, once printed, always reaches either a page or the notice of a Hidden place.

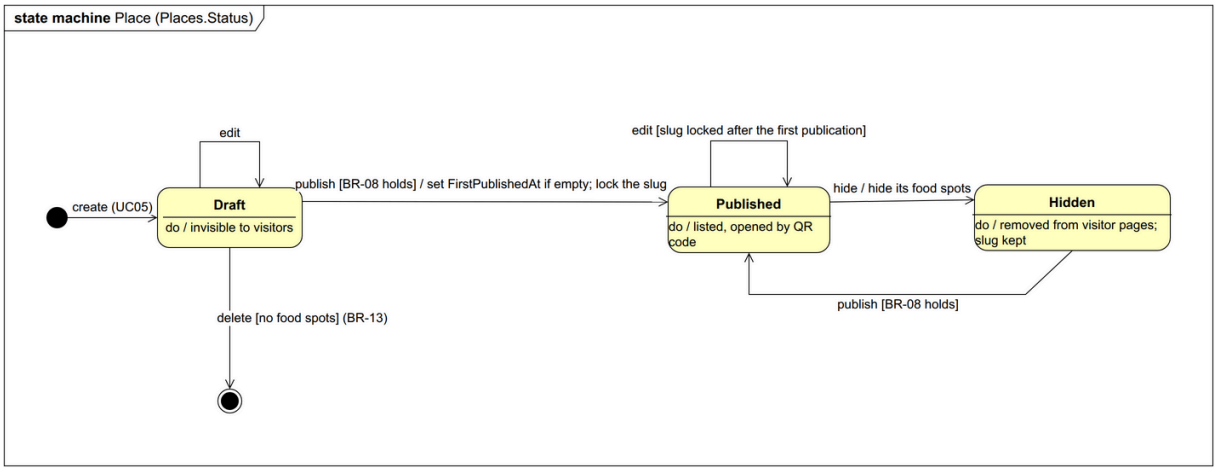



*Figure 4.15. State machine of a place*

Figure 4.16 shows a translation. The point to notice is that "outdated" is not a state. It is derived by comparing SourceHash with the hash of the current Vietnamese text, so it can never disagree with the text it describes, and it does not hide a Reviewed translation from visitors while a new one is prepared.

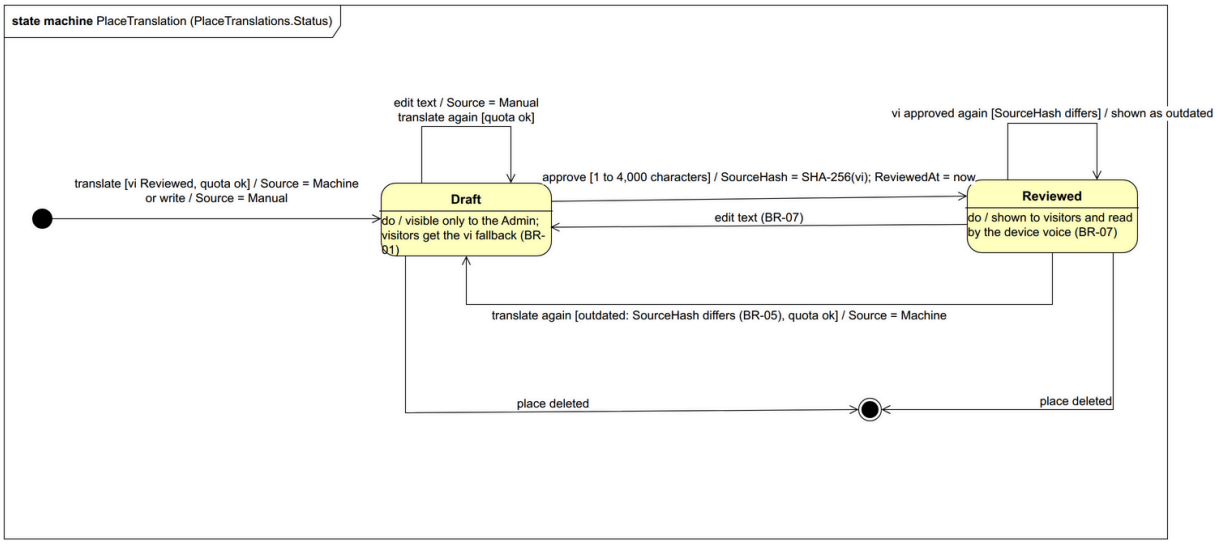



*Figure 4.16. State machine of a translation*

Figure 4.17 shows the presence of a device, which lives only in memory. The point to notice is the dashed region: a device in the background still counts until its time-to-live ends, because a visitor who locks the phone while listening is still at the destination. The worst case for a device lost without a signal is 45 + 5 = 50 seconds, inside the 60 seconds of O5.

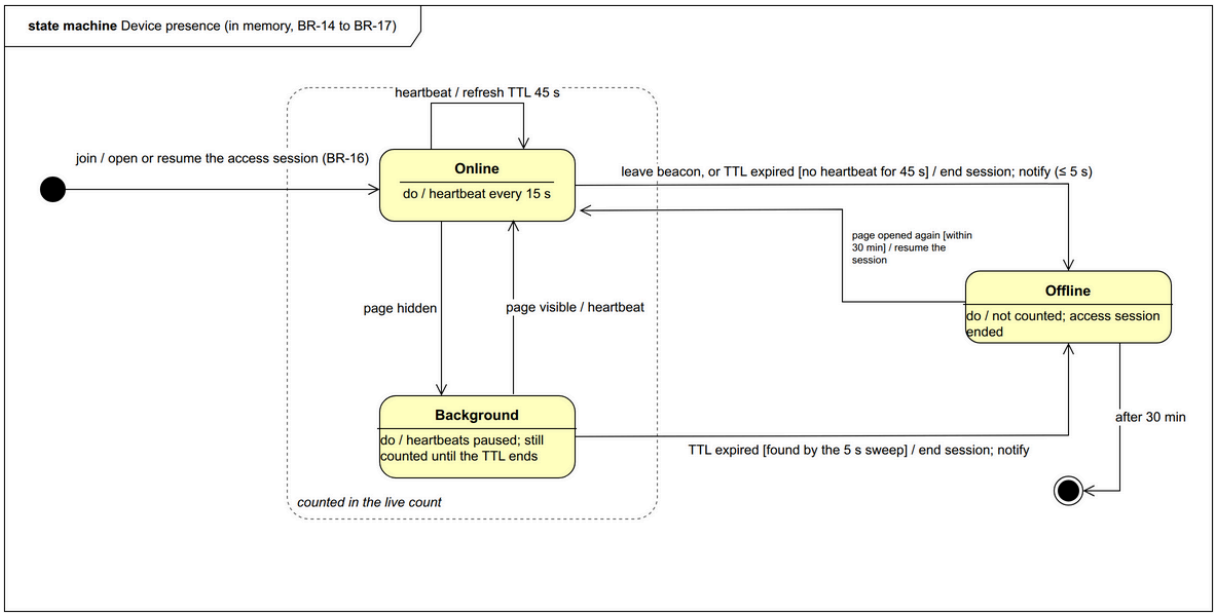



*Figure 4.17. State machine of the presence of a device*

## 4.4. Detailed User Interface Design

Section 3.6 listed seventeen screens and drew two of them as layouts, deciding what each region contains. This section completes the design in three steps: it fixes the visual rules that every screen shares, draws the routes between screens, and details the five screens that carry the main flows of the three actors, each with a table of its items. The other screens are built from the same components and are described more briefly at the end.

### *4.4.1. Design principles and tokens*

Five principles come directly from the requirements. The visitor pages are designed for a phone first, from 360 px wide, because a visitor arrives by scanning a QR code (NFR-06). A narration is reachable in two taps from the Home screen, a card and then Listen, which leaves one tap of margin under O3. The narration text is always on screen under the player, so a missing voice never leaves the page empty (Table 4.12). Every screen has one primary action in the primary colour, so the next step is never ambiguous. Status is never shown by colour alone: every badge also carries a word, because the Admin must tell Draft from Reviewed at a glance on any display.

The colours and type are defined once as design tokens, listed in Table 4.26, and every screen refers to the tokens rather than to raw values. Text on the primary colour and body text on the background both exceed the contrast of 4.5:1 required by NFR-06, and every control is at least 44 px high.

| Token | Value | Used for |
|---|---|---|
| Primary | #147D78 | Main buttons, links, selected state |
| Primary tint | #DDF3EF | Chip backgrounds, light highlight, current sentence |
| Background | #F7F9FA | Page background |
| Surface | #FFFFFF | Cards, tables, drawers |
| Text | #202B33 | Main text |
| Muted | #697780 | Secondary text |
| Border | #E3E9EC | Borders and dividers |
| Accent | #E9B86E | Highlights and mild warnings, such as the fallback notice and the quota |
| Font | Inter | All text; the browser's fonts take over for Chinese, Korean and Japanese characters |

*Table 4.26. Design tokens of the interface*

### *4.4.2. Navigation between screens*

Figure 4.18 draws the routes between the screens of Table 3.29. The point to notice is that the visitor web has two entrances, not one. A visitor who opens the site starts at SCR01, but a visitor who scans a code starts at SCR09, which redirects straight to the place, so the most frequent path of the PRD scenario never passes through Home. The dashboard, by contrast, has a single entrance through its sign-in screen, and every other dashboard screen is reachable from the sidebar.

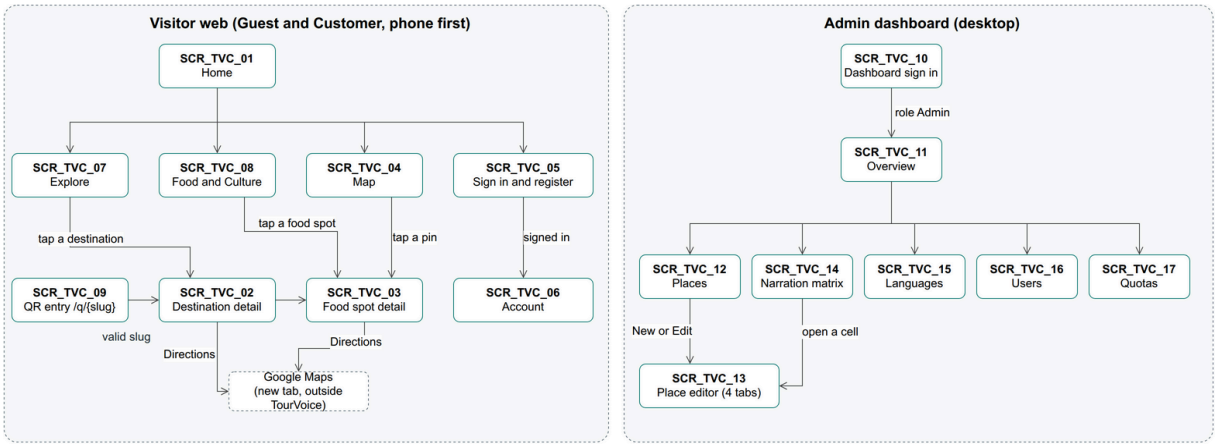



*Figure 4.18. Navigation between the screens*

### *4.4.3. Guest and Customer screens*

The figures below number each region with a red dot; the table that follows each figure describes the item with that number. The columns give the kind of control, the type of data, whether the item is input (I), output (O) or both, whether it is mandatory, its default and its allowed range.

**SCR01 Home.** Figure 4.19 shows the Home screen in English. The point to notice is the line under each destination name, which tells the visitor in advance whether a reviewed narration exists in the chosen language, so the fallback of BR-01 never comes as a surprise.

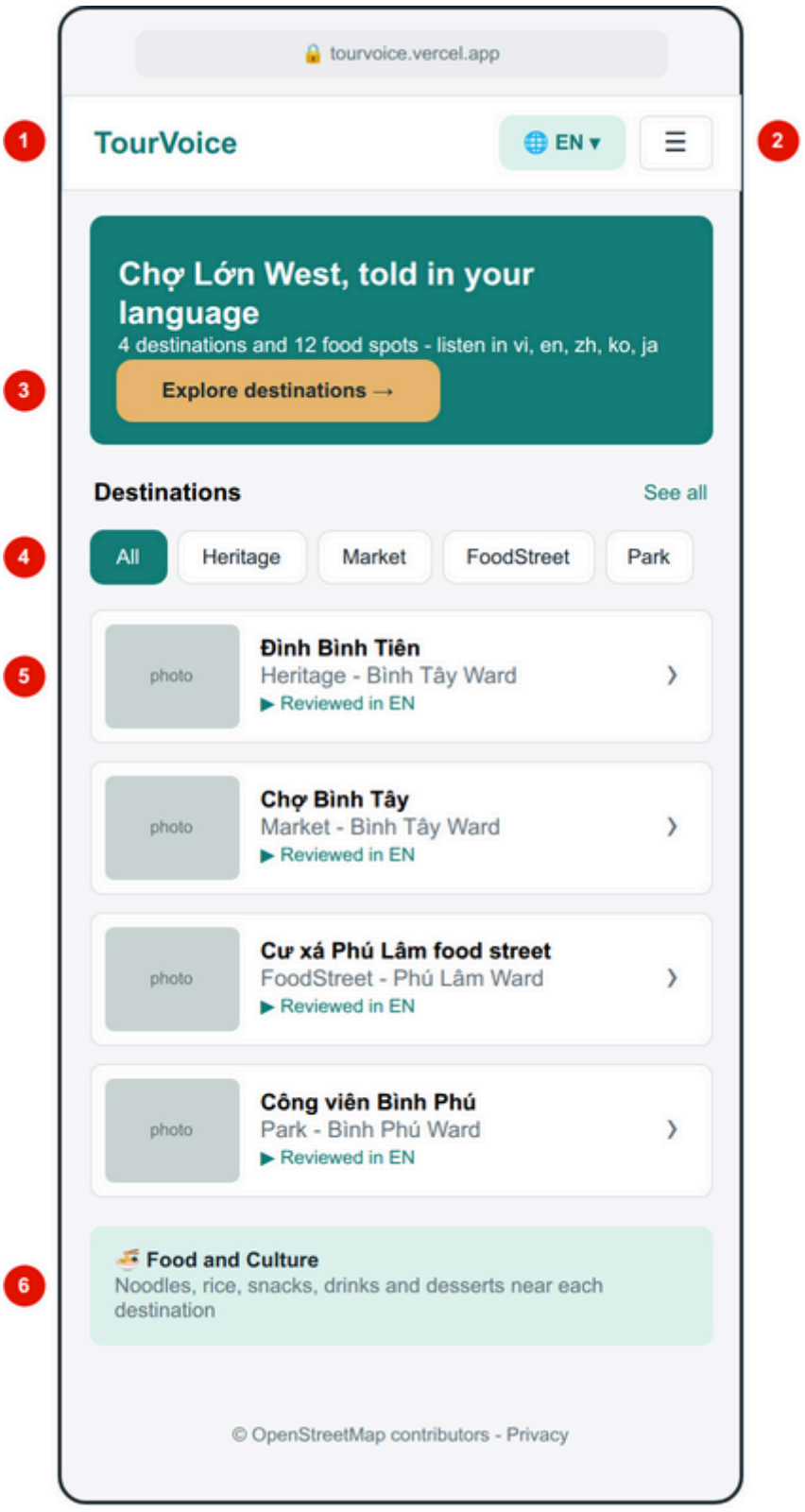



*Figure 4.19. SCR01 Home*

| No | Item | Control | Data type | I/O | Mandatory | Default | Range and notes |
|---|---|---|---|---|---|---|---|

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | Language selector | Drop-down | Text | I/O | Yes | Saved choice, else the browser language, else vi | vi, en, zh, ko, ja; changes text, labels and voice together (UC07) |
| 2 | Menu | Button opening a drawer | | I | No | Closed | Explore, Food and Culture, Map, Sign in or Account |
| 3 | Explore destinations | Button | | I | No | | Opens SCR07 |
| 4 | Category filter | Chip group | Text | I | No | All | All, Heritage, Market, FoodStreet, Park |
| 5 | Destination card | Card with link | Name, category, ward, review status | O | | | Published destinations in SortOrder; opens SCR02 (UC01) |
| 6 | Food and Culture | Card with link | | I | No | | Opens SCR08 (UC03) |

*Table 4.27. Items of SCR01 Home*

**SCR02 Destination detail.** Figure 4.20 details the layout of Figure 3.10 with a Korean visitor at Chợ Bình Tây before the Korean text has been reviewed. The point to notice is item 9: the notice of BR-01 sits between the player and the text, so the visitor reads why the voice speaks Vietnamese before the voice starts. Items 3 and 4 are the two largest controls and are visible without scrolling on a 360 px screen.

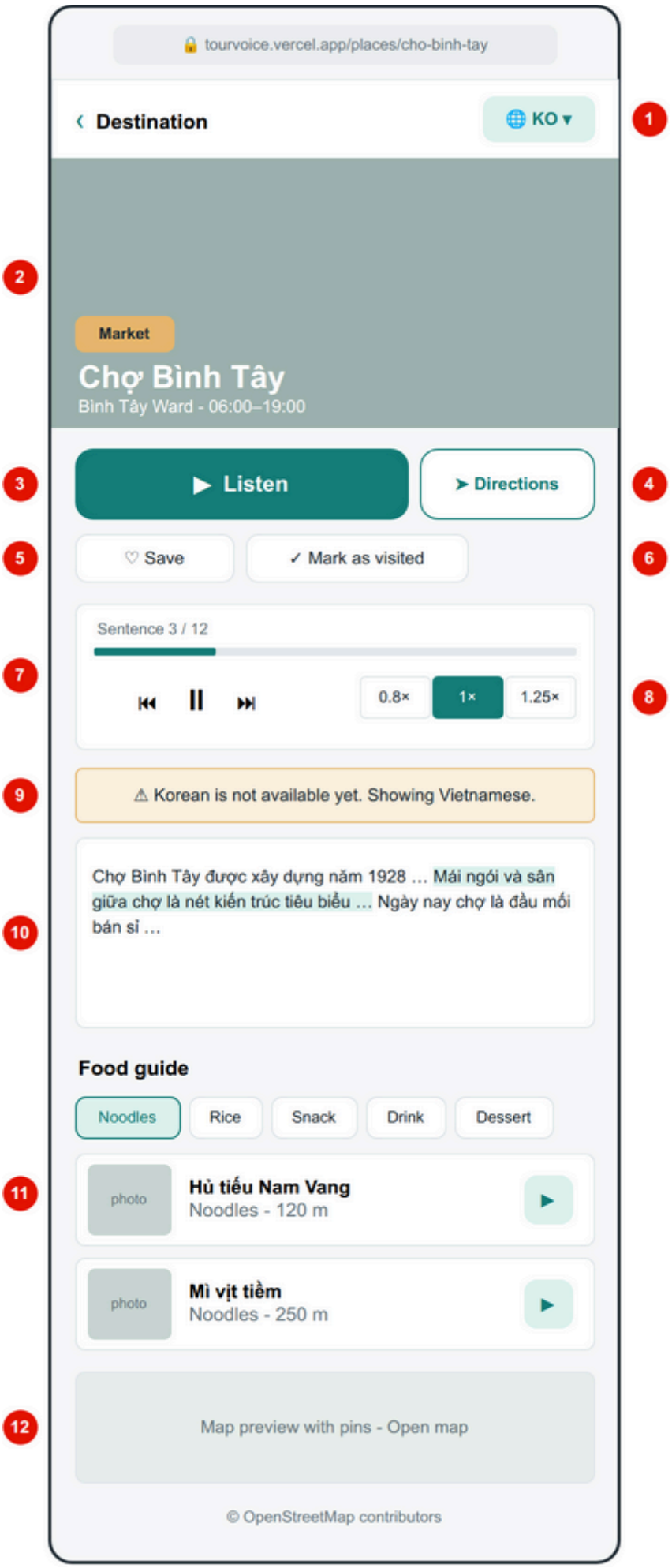

tourvoice.vercel.app/places/cho-binh-tay
Destination
KO
Market
Chợ Bình Tây
Bình Tây Ward - 06:00–19:00
Listen
Directions
Save
Mark as visited
Sentence 3 / 12
0.8×
1×
1.25×
Korean is not available yet. Showing Vietnamese.
Chợ Bình Tây được xây dựng năm 1928 … Mái ngói và sân giữa chợ là nét kiến trúc tiêu biểu … Ngày nay chợ là đầu mối bán sỉ …
Food guide
Noodles
Rice
Snack
Drink
Dessert
photo
Hủ tiếu Nam Vang
Noodles - 120 m
photo
Mì vịt tiềm
Noodles - 250 m
Map preview with pins - Open map
© OpenStreetMap contributors
1
2
3
4
5
6
7
8
9
10
11
12

*Figure 4.20. SCR02 Destination detail*

| No | Item | Control | Data type | I/O | Mandatory | Default | Range and notes |
|---|---|---|---|---|---|---|---|
| 1 | Language selector | Drop-down | Text | I/O | Yes | Current language | As in Table 4.27 |
| 2 | Cover and title | Image with labels | Image, name, category, ward, hours | O | | | Cover at most 2 MB |
| 3 | Listen | Button | | I | | | Starts speech inside the tap (UC08); becomes Pause while speaking |
| 4 | Directions | Link button | Coordinates | I | | | Opens Google Maps in a new tab (UC04) |
| 5 | Save | Toggle | Boolean | I/O | No | isFavorite | Signed out: opens SCR05 first (UC16) |
| 6 | Mark as visited | Toggle | Boolean | I/O | No | isVisited | Signed out: opens SCR05 first (UC18) |
| 7 | Player | Progress bar and buttons | Integer | I/O | | Sentence 1 | Previous, pause or resume, next sentence; one narration at a time |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 8 | Speed | Segmented control | Decimal | I | No | 1× | 0.8×, 1×, 1.25× |
| 9 | Notice | Banner | Text | O | | Hidden | Shown when fallback is true (BR-01) or no voice exists (UC08.E1) |
| 10 | Narration text | Read-only text | Text | O | | | At most 4,000 characters; the sentence being read is highlighted |
| 11 | Food guide | Chip group and cards | Food type, name, distance | I/O | | First type with spots | Noodles, Rice, Snack, Drink, Dessert; shown only when HasFoodGuide (BR-02) |
| 12 | Map preview | Map | Coordinates | O | | | Opens SCR04; shows the OpenStreetMap attribution |

*Table 4.28. Items of SCR02 Destination detail*

**SCR05 Sign in and register.** Figure 4.21 shows the Register tab after the exception UC15.E2. The point to notice is item 7: the screen offers a way back without an account, because the account is never needed to listen.

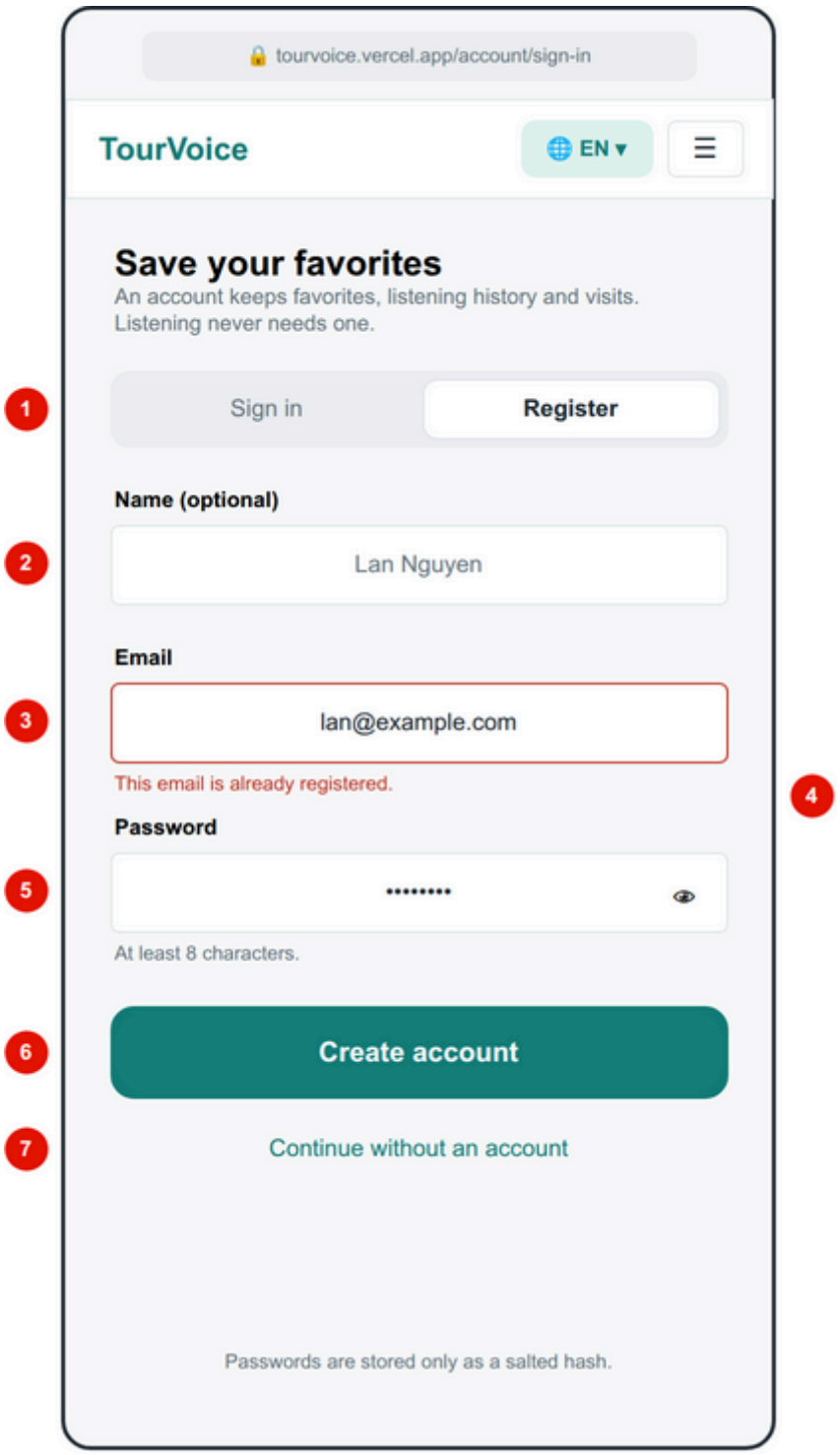



Figure 4.21. SCR05 Sign in and register

| No | Item | Control | Data type | I/O | Mandatory | Default | Range and notes |
|---|---|---|---|---|---|---|---|

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | Sign in or Register | Segmented control | | I | Yes | Sign in | Register is preselected when the Guest came from an account action |
| 2 | Name | Text box | Text | I | No | Empty | At most 100 characters; Register only |
| 3 | Email | Text box | Text | I | Yes | Empty | Valid format, at most 254 characters, not yet registered (UC15.E1, UC15.E2) |
| 4 | Error message | Label | Text | O | | Hidden | Messages of UC15.E1 to UC15.E4 and UC22.E1, UC22.E2 |
| 5 | Password | Password box with show toggle | Text | I | Yes | Empty | At least 8 characters (BR-12) |
| 6 | Create account or Sign in | Button | | I | | Disabled until the form is valid | Returns to the previous page after success |
| 7 | Continue without an account | Link | | I | No | | Returns to the previous page |

*Table 4.29. Items of SCR05 Sign in and register*

### *4.4.4. Admin screens*

The dashboard is designed for a laptop, with a fixed sidebar and the same tokens as the visitor web. Two screens carry most of the Admin's work.

**SCR11 Overview.** Figure 4.22 is the screen the Admin keeps open during a visit. The point to notice is the order of the cards: the live count comes first and the quota last in the top row, so the two figures that change during a day, one every few seconds and one with every translation, are at the two ends of the eye's path. The table at the bottom turns the overview into a work list: each Draft translation is one click from its editor.

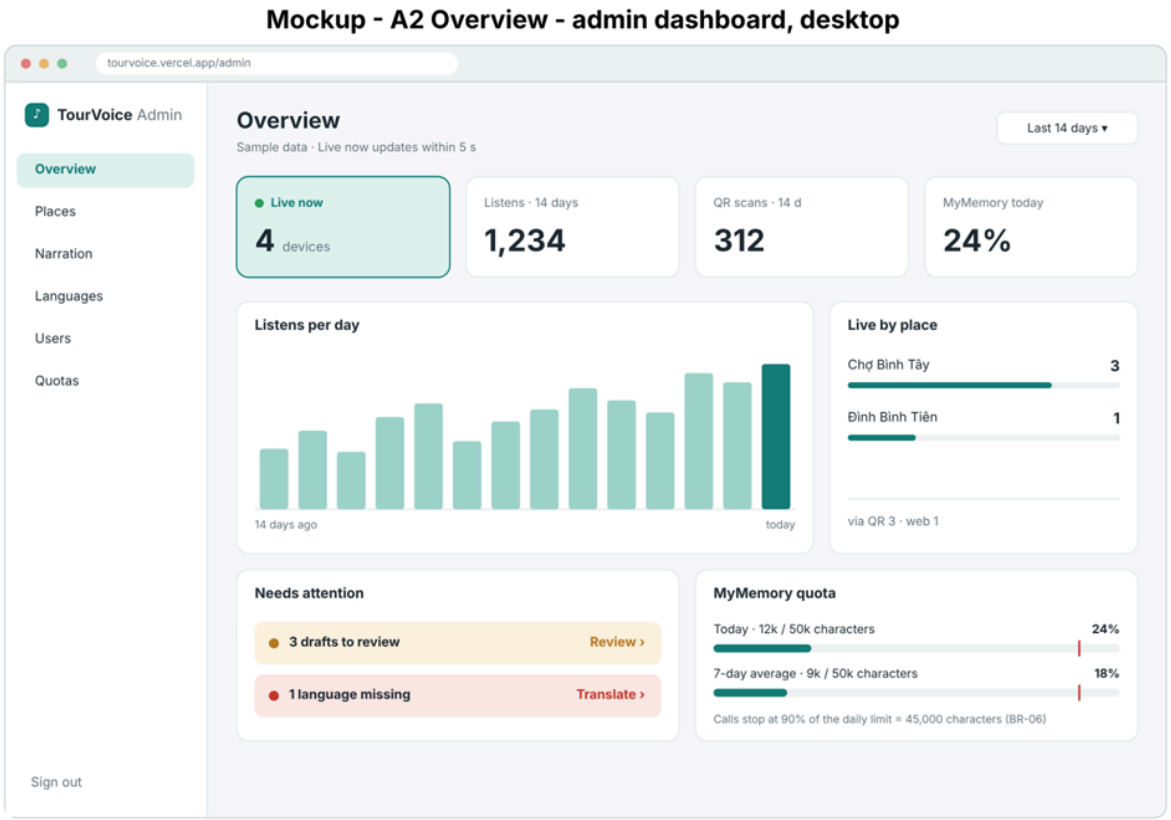



*Figure 4.22. SCR11 Overview*

| No | Item | Control | Data type | I/O | Mandatory | Default | Range and notes |
|---|---|---|---|---|---|---|---|

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | Sidebar | Navigation list | | I | | Current screen highlighted | Overview, Places, Narration, Languages, Users, Quotas |
| 2 | Live now | Live counter | Integer, by place and by entry | O | | Count at load | Updated by SignalR within 5 s (BR-17); shows "Live updates paused. Reconnecting …" when the connection is lost (UC20.E1) |
| 3 | Listens today | Counter | Integer, by language | O | | | From ListenEvents |
| 4 | MyMemory today | Counter and bar | Integer | O | | | 0 to 45,000 characters (BR-06); links to SCR17 |
| 5 | Period | Drop-down | Date range | I | No | Today | Today, 7 days, 30 days, custom (UC21) |
| 6 | Listens by language | Bar chart | Integer | O | | Last 7 days | Five languages |
| 7 | Access sessions by entry and place | Stacked bar chart | Integer | O | | Selected period | QR code and direct entry per destination (UC21) |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 8 | Waiting for review | Table with links | Place, language, source, time | O/I | | Oldest first | Opens SCR13 on the Narration tab (UC11) |

*Table 4.30. Items of SCR11 Overview*

**SCR13 Place editor, Narration tab.** Figure 4.23 details the layout of Figure 3.11. The point to notice is item 5: the five languages show four different situations at once, Reviewed, Draft, Outdated and Missing, each with a word and a colour, so the Admin sees the whole pipeline of Figure 4.16 for one place without opening any language. The actions in item 9 to 11 follow the order of the pipeline from left to right.

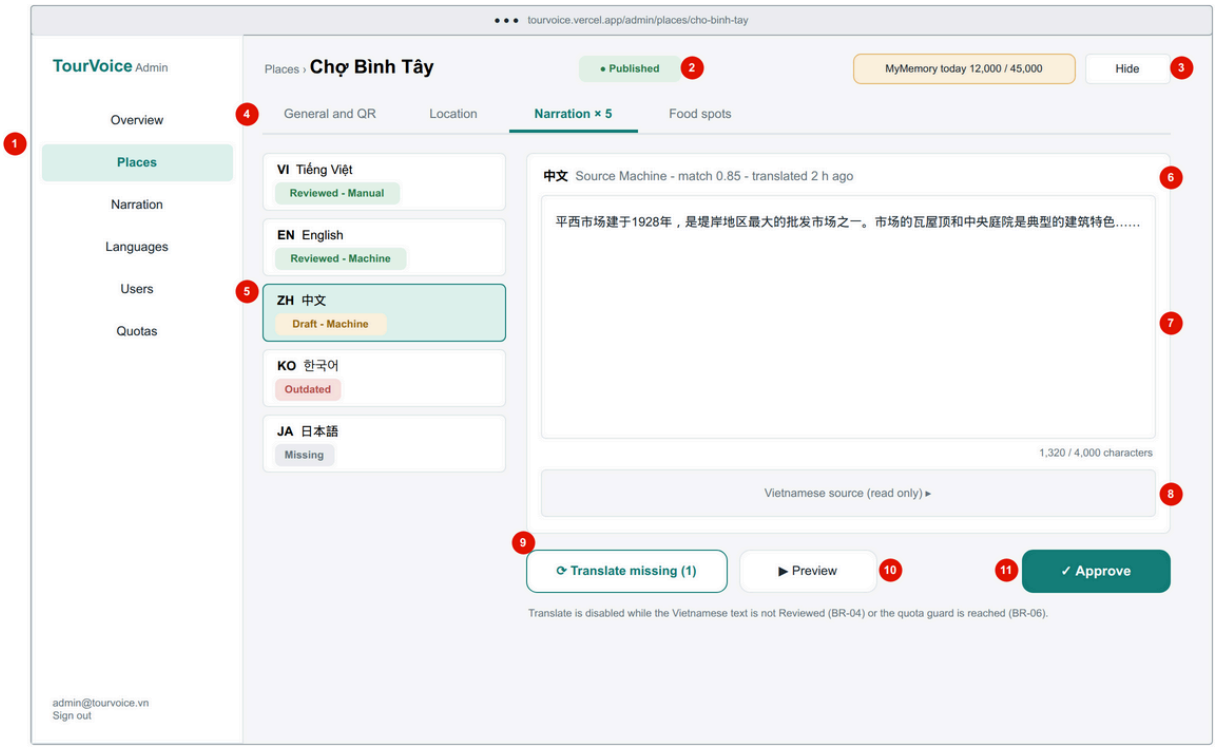



*Figure 4.23. SCR13 Place editor, Narration tab*

| No | Item | Control | Data type | I/O | Mandatory | Default | Range and notes |
|---|---|---|---|---|---|---|---|
| 1 | Sidebar | Navigation list | | I | | Places highlighted | As in Table 4.30 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | Place status | Badge | Text | O | | | Draft, Published or Hidden (Figure 4.15) |
| 3 | Status action | Button | | I | | | Publish for a Draft or Hidden place, listing what BR-08 still misses; Hide for a Published place |
| 4 | Tabs | Tab bar | | I | | General and QR | General and QR, Location, Narration × 5, Food spots |
| 5 | Languages | List with badges | Status and source | O/I | | vi selected | Reviewed with source, Draft with source, Outdated, Missing |
| 6 | Translation header | Label | Source, match score, time | O | | | Match score only for a Machine source |
| 7 | Narration editor | Multi-line text box | Text | I/O | Yes, to approve | Current text | 1 to 4,000 characters, with a counter (UC11.E2) |
| 8 | Vietnamese source | Collapsible read-only text | Text | O | | Collapsed | Shown beside a translation for comparison |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 9 | Translate missing | Button | | I | | | Enabled only when the Vietnamese text is Reviewed and the quota allows (BR-04, BR-06); shows how many languages it will translate |
| 10 | Preview | Button | | I | | | Reads the text with the device voice; nothing is sent (UC12) |
| 11 | Approve | Button | | I | | | Moves the language to Reviewed (UC11) |

*Table 4.31. Items of SCR13 Place editor, Narration tab*

#### *4.4.5. Other screens*

The remaining twelve screens reuse the components above, so Table 4.32 records only how each differs from the screen it is built from.

| Screen | Built from | Main difference |
|---|---|---|
| SCR03 Food spot detail | SCR02 | No food guide; adds price range and opening hours; Back returns to its destination |
| SCR04 Map | Map preview of SCR02 | Full-screen Leaflet map; pins by kind; a tapped pin opens a card with Listen and Directions |
| SCR06 Account | Cards of SCR01 | Tabs Favorites, History and Visited, with progress "x of 4 destinations" (BR-10) |
| SCR07 Explore | SCR01 | All Published places with category and food type filters |

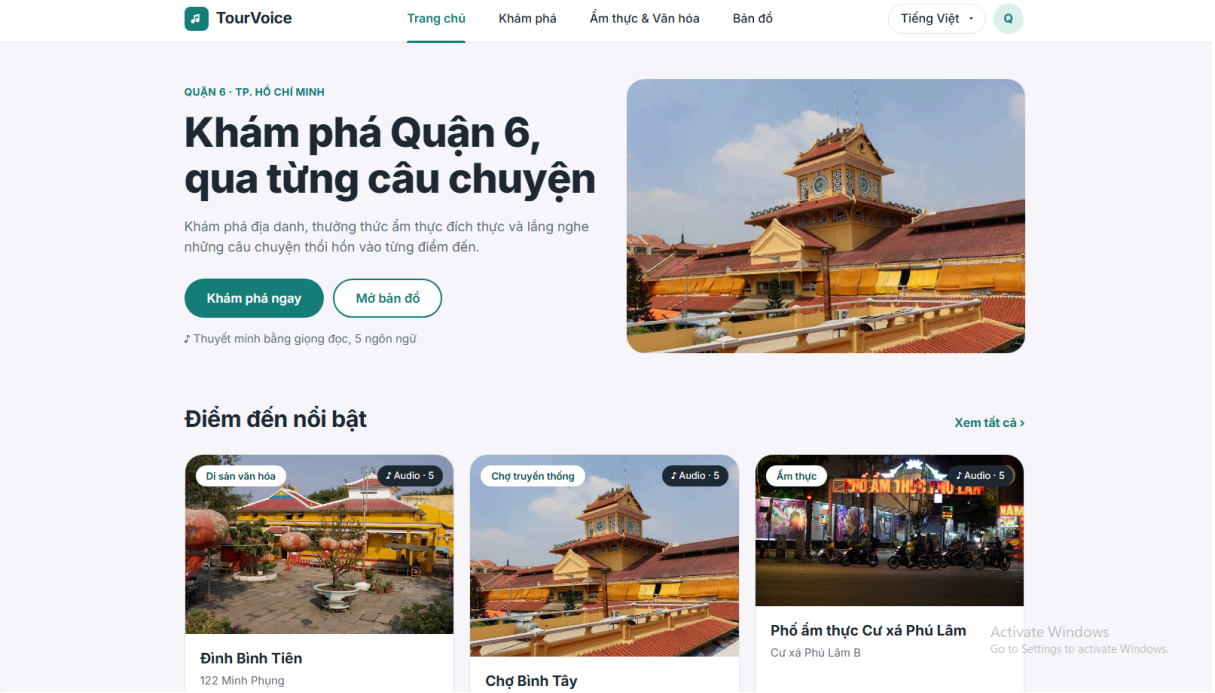
| | | |
|---|---|---|
| SCR08 Food and Culture | Food guide of SCR02 | Food spots of all destinations by type, with distance |
| SCR09 QR entry | None | Loading state only; redirects, or shows a notice for an unknown or Hidden slug |
| SCR10 Dashboard sign in | SCR05 | Sign in only, desktop width |
| SCR12 Places | Table of SCR11 | Paginated table, 20 rows, filter by status and kind, New place |
| SCR14 Narration matrix | Badges of SCR13 | Places by five languages; each cell opens SCR13 on that language |
| SCR15 Languages | Table of SCR11 | Five rows with codes and an active switch (UC14) |
| SCR16 Users | Table of SCR11 | Lock, unlock and change role, refused when no active Admin would remain (BR-11) |
| SCR17 Quotas | Cards of SCR11 | Usage today and over 30 days from UsageLogs, with failed calls (UC13) |

*Table 4.32. Other screens and their differences*

# CHAPTER 5: RESULTS

## 5.1. Implementation Results

TourVoice has been implemented across its whole functional scope: each of the twenty-three use cases and seventeen screens of Chapter 3 has working code, built in thirteen increments over three days. The structure follows the design of Chapter 4, with eight deviations recorded in Section 5.1.5. This section reports how the system was built, what its architecture became, which functions each actor can use, and where the result departs from the design.



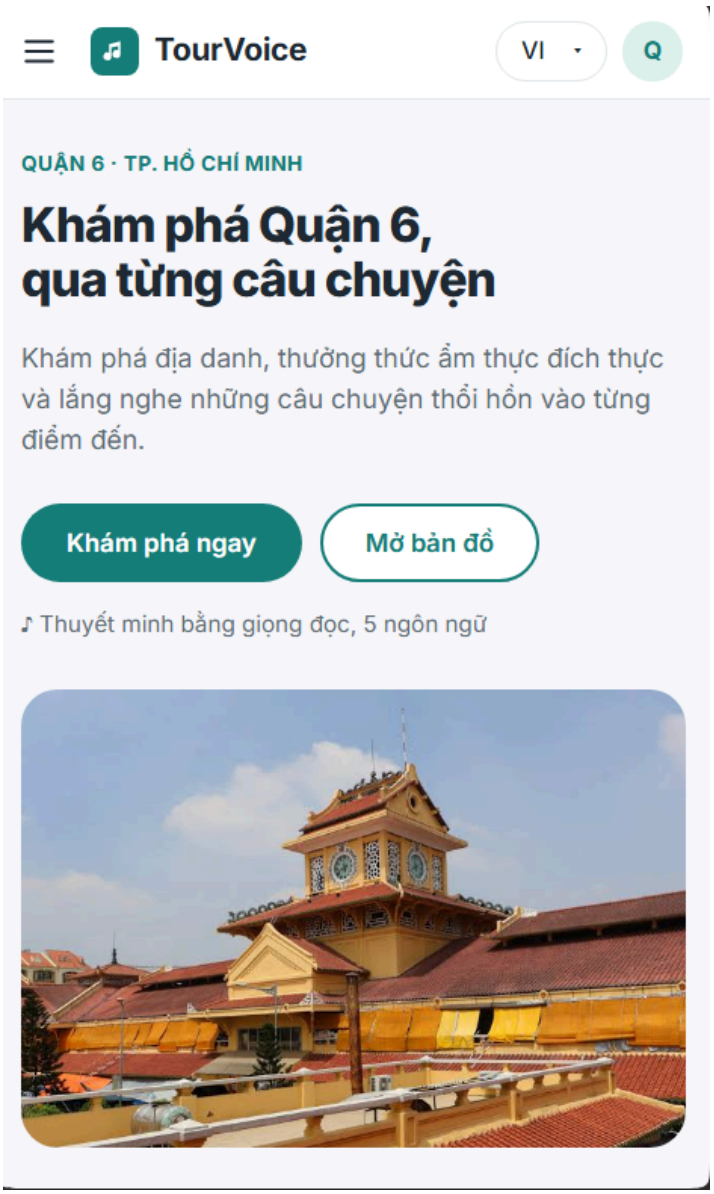
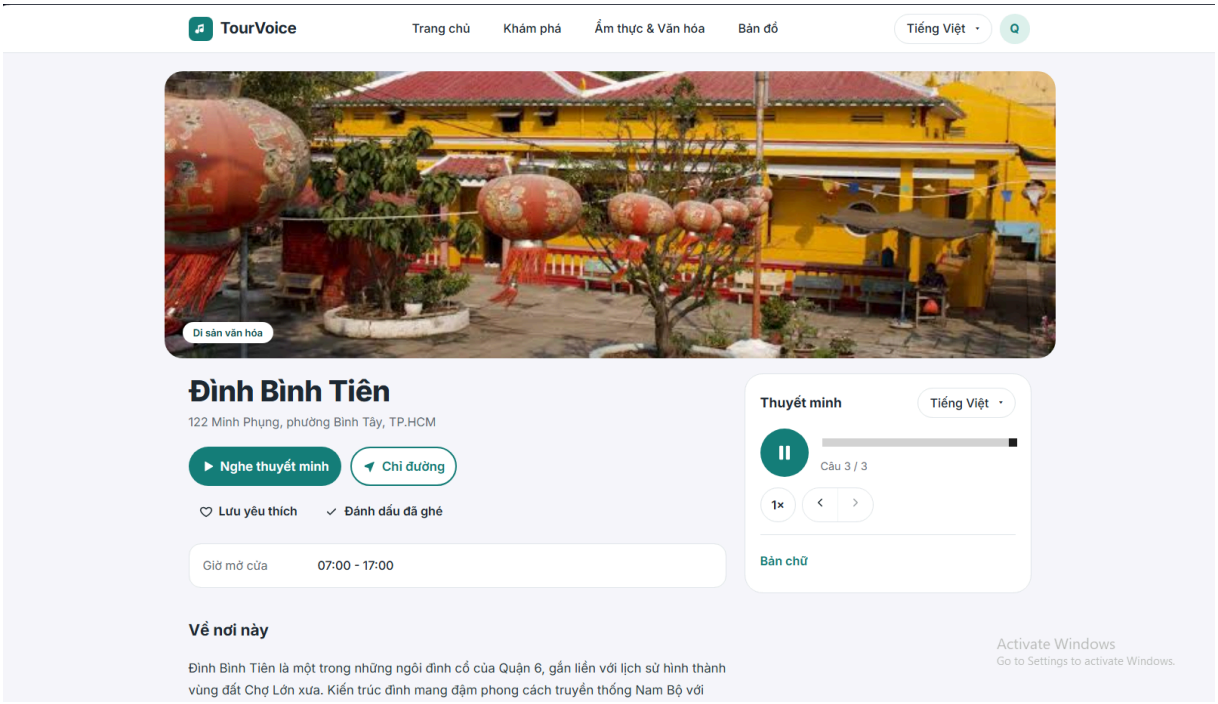



*Figure 5.1. Visitor website homepage*


TourVoice
Trang chủ
Khám phá
Ẩm thực & Văn hóa
Bản đồ
Tiếng Việt
Q
Di sản văn hóa
Đình Bình Tiên
122 Minh Phụng, phường Bình Tây, TP.HCM
Nghe thuyết minh
Chỉ đường
Lưu yêu thích
Đánh dấu đã ghé
Giờ mở cửa
07:00 - 17:00
Thuyết minh
Tiếng Việt
Câu 3 / 3
1×
Bản chữ
Về nơi này
Đình Bình Tiên là một trong những ngôi đình cổ của Quận 6, gắn liền với lịch sử hình thành vùng đất Chợ Lớn xưa. Kiến trúc đình mang đậm phong cách truyền thống Nam Bộ với
Activate Windows
Go to Settings to activate Windows.

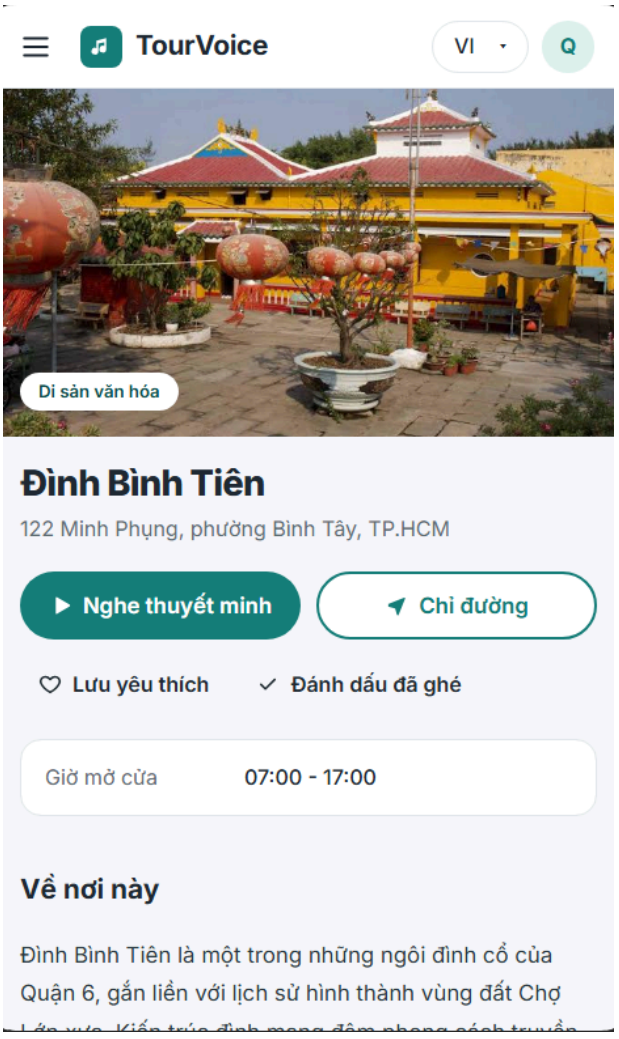

TourVoice
VI
Q
Di sản văn hóa
Đình Bình Tiên
122 Minh Phụng, phường Bình Tây, TP.HCM
Nghe thuyết minh
Chỉ đường
Lưu yêu thích
Đánh dấu đã ghé
Giờ mở cửa
07:00 - 17:00
Về nơi này
Đình Bình Tiên là một trong những ngôi đình cổ của Quận 6, gắn liền với lịch sử hình thành vùng đất Chợ

*Figure 5.2. Destination details and narration player*

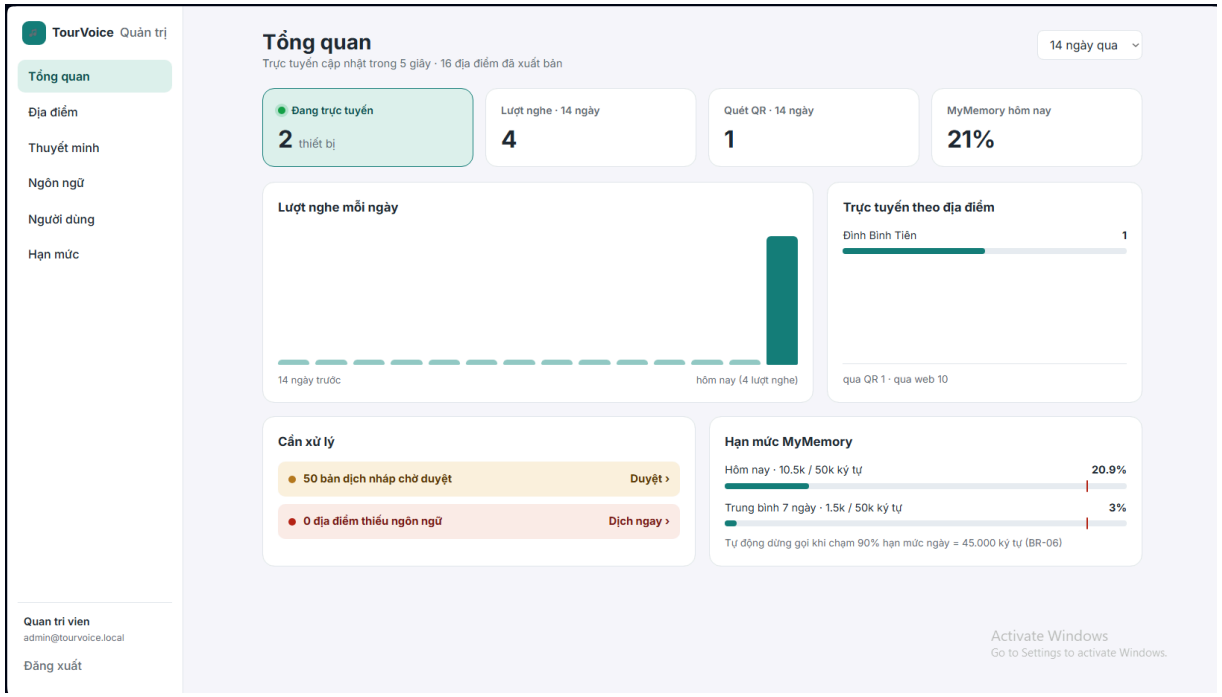



*Figure 5.3. Administrative API or management interface*

## 5.2. Evaluation

### *5.2.1. Strengths*

TourVoice provides a practical approach to multilingual tourism information without requiring visitors to install a native mobile application. Its browser-based narration reduces the need to generate and store separate audio files. The five-language interface, QR entry points, and map-based exploration are designed to make destination information easier to access.

The backend's three-layer architecture separates API handling, business logic, and data access. This structure supports maintainability and allows architecture rules to be checked automatically. The project also includes automated testing tools and a CI workflow, providing a foundation for consistent verification as the system evolves.

### *5.2.2. Limitations*

Some aspects of the system still require implementation or practical verification. In particular, the repository README identifies the administrative

dashboard interface as unfinished. The quality and availability of browser-generated voices may differ across devices, and the MyMemory translation output requires review before publication.

The actual quality of the user experience also depends on the availability and accuracy of destination content, translations, map coordinates, and browser speech support. Performance under concurrent access, narration response time, and real-time presence updates should be measured through practical tests rather than assumed from the design requirements.

The system should therefore be evaluated based on the features that can be demonstrated in the current implementation. Planned functions and performance targets must not be reported as verified results until evidence has been collected.

## 5.3. System Testing

### *5.3.1. Testing Approach*

Testing focuses on the main visitor and backend functions, including public place retrieval, authentication, authorization, translation management, narration-related behaviour, and automated build verification. Backend tests and architecture tests can be run from the server directory. Frontend linting, tests, and production build verification can be run from the client directory.

The repository provides the following verification commands:

- Backend tests and architecture checks: cd server followed by dotnet test.
- Frontend linting: cd client followed by npm run lint.
- Frontend tests: cd client followed by npm test.
- Frontend build: cd client followed by npm run build.

API behaviour can also be checked through Swagger in the Development environment. Browser-based functions, including narration, language switching, QR navigation, and map display, should be verified by running the application and testing the corresponding pages.

### *5.3.2. Test Cases and Results*

Table 5.1 presents representative test cases for the main system functions. The actual results and status must be updated after the tests have been executed. No test is marked as passed without execution evidence.

| No | Test case | Expected result | Actual result | Status |
|---|---|---|---|---|
| 1 | Retrieve the public place list | The API returns available published places. | Passed | Executed |
| 2 | Retrieve place details by slug | The API returns the matching place and applicable language content. | Passed | Executed |
| 3 | Log in with valid credentials | The API returns an authentication result and JWT when credentials are valid. | Passed | Executed |
| 4 | Access an admin endpoint without authorization | The request is rejected. | Passed | Executed |
| 5 | Request machine translation | The API processes the translation request or returns an appropriate error. | Passed | Executed |
| 6 | Play narration in a supported browser | The browser reads the available narration text using a supported voice. | Passed | Executed |
| 7 | Open a destination through its QR URL | The browser navigates to the corresponding destination page. | Passed | Executed |
| 8 | Run the automated backend test suite | The test command completes and reports the actual passed, failed, or skipped tests. | Test summary: total: 195, failed: 0, succeeded: 195, skipped: 0, duration: 13.4s | Executed |

| | | | | |
|---|---|---|---|---|
| | | | Build succeeded with 4 warning(s) in 33.6s | |
| 9 | Run frontend tests and build | The commands report their actual test and build results. | Test Files 7 passed (7) Tests 44 passed (44) Start at 15:37:24 Duration 5.44s (environment 75%, setup 14%, tests 4%, transform 3%, import 3%) | Executed |

*Table 5.1. Test results*

### *5.3.3. Testing Evidence*

The following evidence should be inserted after practical verification:

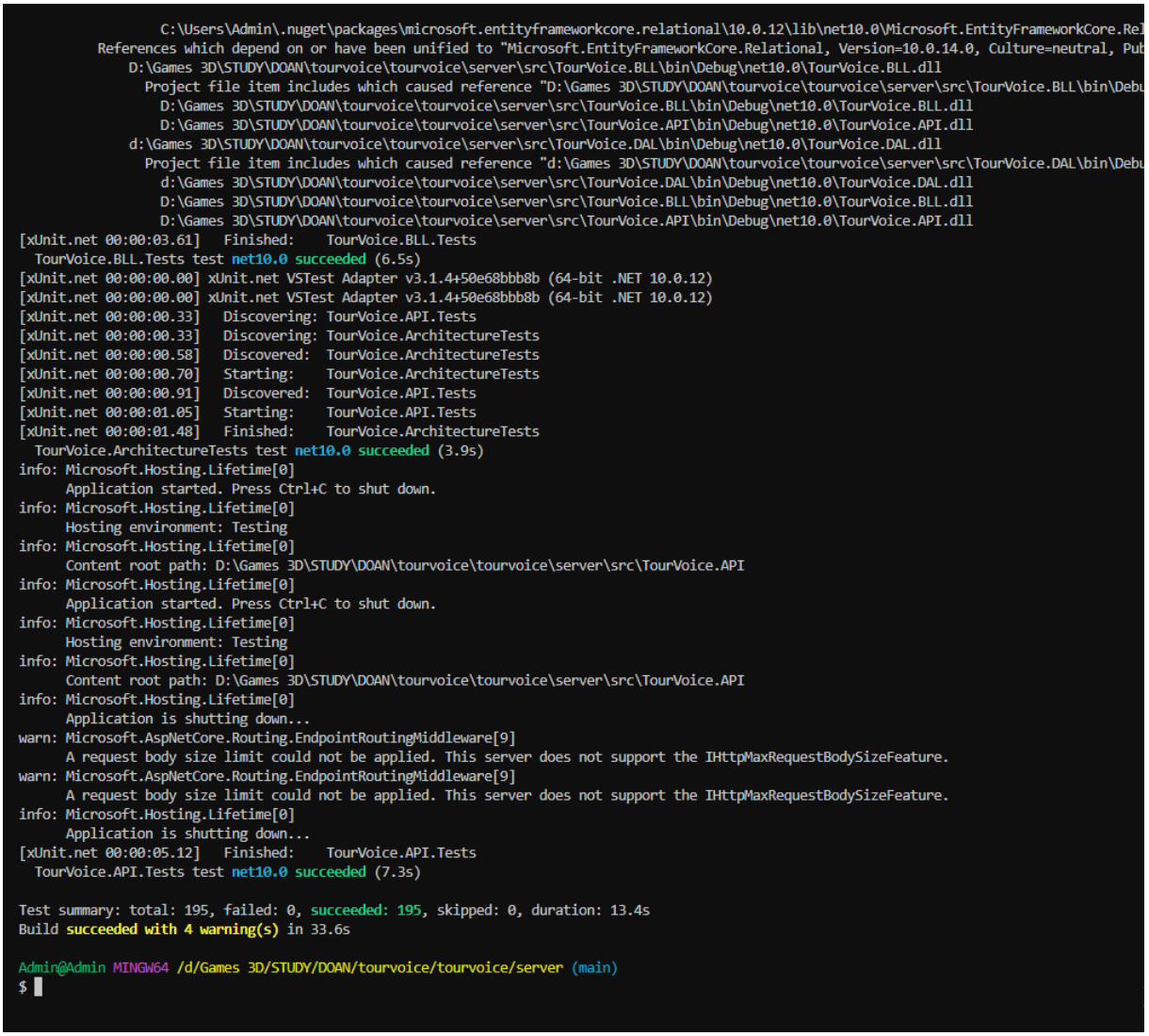
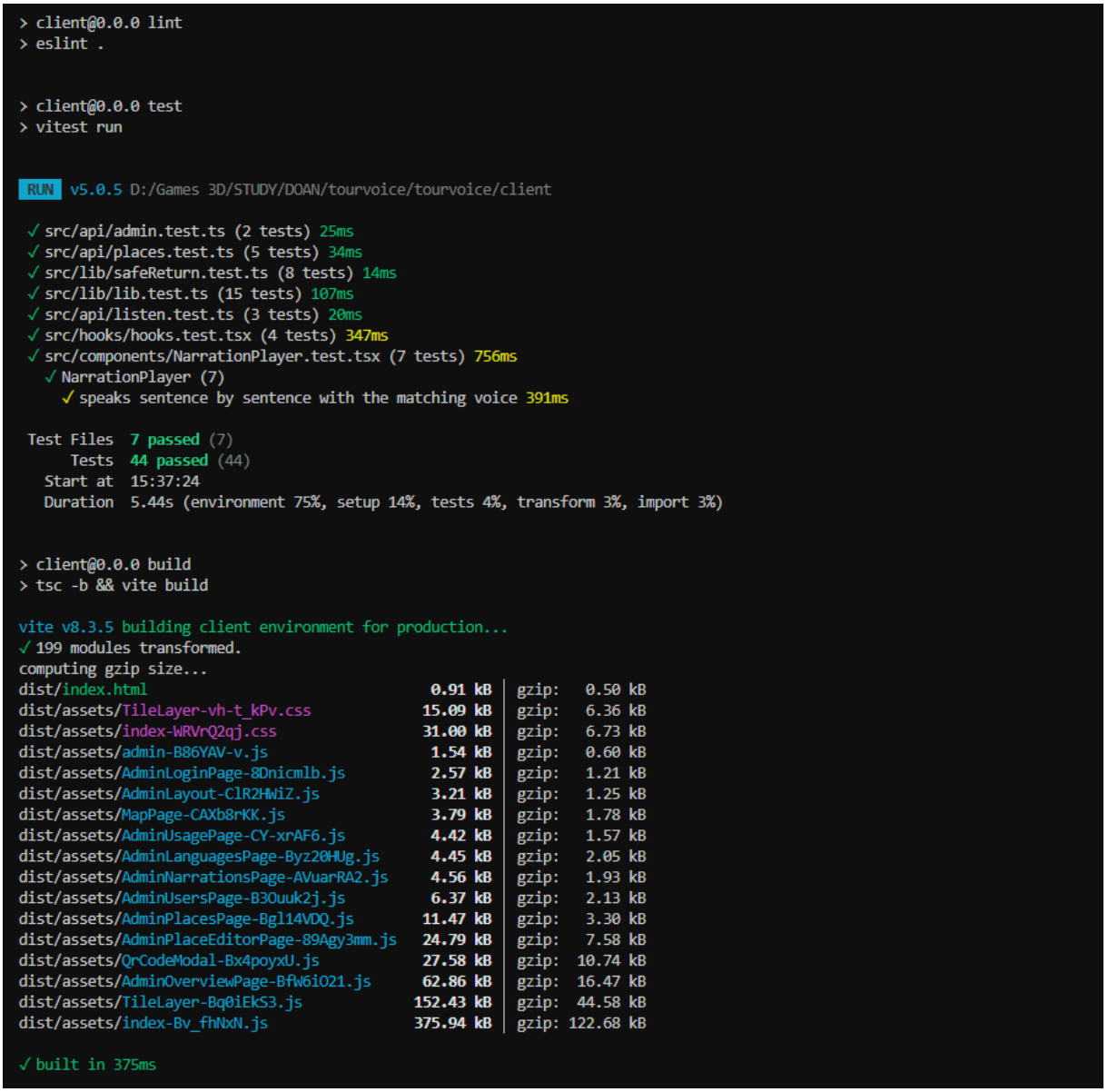



*Figure 5.4. Backend automated test results*



```
> client@0.0.0 lint
> eslint .


> client@0.0.0 test
> vitest run


 RUN  v5.0.5 D:/Games 3D/STUDY/DOAN/tourvoice/tourvoice/client

 ✓ src/api/admin.test.ts (2 tests) 25ms
 ✓ src/api/places.test.ts (5 tests) 34ms
 ✓ src/lib/safeReturn.test.ts (8 tests) 14ms
 ✓ src/lib/lib.test.ts (15 tests) 107ms
 ✓ src/api/listen.test.ts (3 tests) 20ms
 ✓ src/hooks/hooks.test.tsx (4 tests) 347ms
 ✓ src/components/NarrationPlayer.test.tsx (7 tests) 756ms
   ✓ NarrationPlayer (7)
     ✓ speaks sentence by sentence with the matching voice 391ms

 Test Files  7 passed (7)
      Tests  44 passed (44)
   Start at  15:37:24
   Duration  5.44s (environment 75%, setup 14%, tests 4%, transform 3%, import 3%)


> client@0.0.0 build
> tsc -b && vite build

vite v8.3.5 building client environment for production...
✓ 199 modules transformed.
computing gzip size...
dist/index.html                                0.91 kB | gzip:   0.50 kB
dist/assets/TileLayer-vh-t_kPv.css            15.09 kB | gzip:   6.36 kB
dist/assets/index-WRVrQ2qj.css                31.00 kB | gzip:   6.73 kB
dist/assets/admin-B86YAV-v.js                  1.54 kB | gzip:   0.60 kB
dist/assets/AdminLoginPage-8Dnicmlb.js         2.57 kB | gzip:   1.21 kB
dist/assets/AdminLayout-ClR2HWiZ.js            3.21 kB | gzip:   1.25 kB
dist/assets/MapPage-CAXb8rKK.js                3.79 kB | gzip:   1.78 kB
dist/assets/AdminUsagePage-CY-xrAF6.js         4.42 kB | gzip:   1.57 kB
dist/assets/AdminLanguagesPage-Byz20HUg.js     4.45 kB | gzip:   2.05 kB
dist/assets/AdminNarrationsPage-AVuarRA2.js    4.56 kB | gzip:   1.93 kB
dist/assets/AdminUsersPage-B3Ouuk2j.js         6.37 kB | gzip:   2.13 kB
dist/assets/AdminPlacesPage-Bgl14VDQ.js       11.47 kB | gzip:   3.30 kB
dist/assets/AdminPlaceEditorPage-89Agy3mm.js  24.79 kB | gzip:   7.58 kB
dist/assets/QrCodeModal-Bx4poyxU.js           27.58 kB | gzip:  10.74 kB
dist/assets/AdminOverviewPage-BfW6iO21.js     62.86 kB | gzip:  16.47 kB
dist/assets/TileLayer-Bq0iEkS3.js            152.43 kB | gzip:  44.58 kB
dist/assets/index-Bv_fhNxN.js                375.94 kB | gzip: 122.68 kB

✓ built in 375ms
```

*Figure 5.5. Frontend test and build results*

[Insert screenshot showing a tested endpoint, HTTP status, and response]

*Figure 5.6. API verification using Swagger*

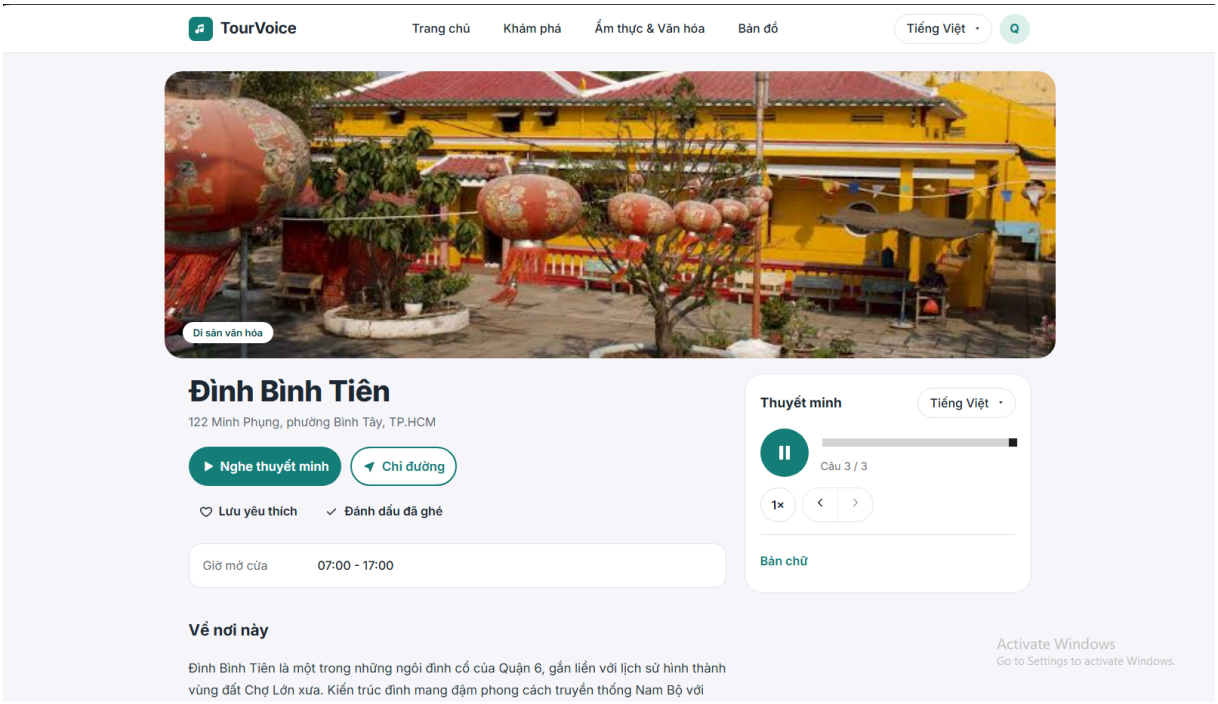



*Figure 5.7. Browser narration test*

# CHAPTER 6: CONCLUSION AND FUTURE WORK

## 6.1. Conclusion

The TourVoice project aims to improve the tourism experience by providing a web-based multilingual audio guide for destinations in District 6, Ho Chi Minh City. The system allows visitors to explore local attractions, food, and cultural information through a web interface without requiring a dedicated mobile application.

The implemented system provides several key features, including browsing and searching for destinations, viewing location details, exploring food and cultural content, and viewing locations on an interactive map. It supports five languages: Vietnamese, English, Chinese, Korean, and Japanese. The system also uses the browser's Web Speech API to provide audio narration and supports QR-code-based access to destination pages.

In addition, TourVoice includes customer authentication and personal features such as favorites, visited places, and browsing history. Its backend is developed using ASP.NET Core Web API, with a layered architecture and MySQL database. Supporting features include JWT-based authentication, translation management through MyMemory, and real-time visitor presence using SignalR.

Overall, TourVoice establishes a foundation for a multilingual digital tourism guide that makes local tourism information more accessible and convenient. However, some aspects still require further improvement, particularly the administrative web interface, translation quality, and systematic testing of the complete system. The actual reliability and performance of the system should be confirmed through test execution and evaluation.

## 6.2. Future Work

Although TourVoice provides the core features of a multilingual tourism guide, several improvements can be considered for future development.

#### *6.2.1. Complete the Administrative Web Interface*

Develop a dedicated administrative dashboard for managing destinations, food information, translations, and publication status. This interface would make content management easier and reduce the need for administrators to interact directly with API endpoints.

#### *6.2.2. Improve Multilingual Content and Audio Narration*

Improve the quality of translated content through review and approval workflows. Audio narration could also be enhanced by evaluating additional text-to-speech options, improving pronunciation of local names, and providing more consistent voice quality across supported languages.

#### *6.2.3. Expand Tourism Content and Navigation*

Add more destinations, local food, cultural information, images, and recommended routes. The interactive map could be extended with route guidance and location-based recommendations to help visitors plan their trips more effectively.

#### *6.2.4. Strengthen Testing, Performance, and Security*

Expand automated and integration tests to cover important visitor and administrative workflows. Conduct performance testing under different traffic levels and review authentication, authorization, input validation, and rate-limiting mechanisms to improve reliability and security.

#### *6.2.5. Improve User Experience and Accessibility*

Enhance the responsive interface for different screen sizes and improve accessibility for users with different needs. Potential improvements include clearer audio playback controls, adjustable narration speed where supported, and better handling of browsers that do not support the required speech features.

### 6.3. Final Remarks

TourVoice demonstrates how web technologies can be used to deliver multilingual tourism information through a combination of destination content,

interactive maps, QR-code access, and audio narration. With further development and systematic evaluation, the system could become a more complete digital guide that supports visitors in discovering the cultural and culinary attractions of District 6.

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